Chapter 5

Cardiac ICU

Managing the five commonest calls to ICU *128*
Definitions of common terms *130*
Transfer and admission to ICU *132*
Fast-track patients *134*
Postoperative pathophysiology *136*
Common ICU scenarios: CABG *138*
Common ICU scenarios: valves *140*
Common hemodynamic scenarios *142*
Bleeding *144*
Blood products, antifibrinolytics *146*
Transfusion reactions *148*
Re-exploration for bleeding *150*
Respiratory physiology *152*
Lung function after heart surgery *154*
Respiratory management *156*
Respiratory support modes *158*
Poor gases in ventilated patients *160*
Respiratory failure on ventilator *162*
Weaning from ventilation *164*
Tracheostomy *166*
Acid–base balance *168*
The acidotic patient *170*
Arterial cannulas *172*
Central venous catheters *174*
Pulmonary artery catheter *176*
Mixed venous oxygen saturation *178*
Other forms of monitoring *180*
Cardiac function after surgery *182*
Managing low cardiac output *184*
Optimizing preload and afterload *186*
Contractility *188*
Vasoplegia, SAM, and RV failure *190*
IABP insertion *192*
IABP management *194*
Pharmacological support *196*
Inotropes *198*
Pressors *200*
Vasodilators *202*
Antiarrhythmics *204*
Chronotropes *206*
Diuretics *208*
EKG monitoring *210*
Basic EKG interpretation *212*
Causes of dysrhythmias *214*
Lethal arrhythmias *216*
Dangerous atrial arrhythmias *218*
Other dangerous arrhythmias *220*
Benign arrhythmias *222*
Cardioversion *224*
Defibrillation *226*
Pacing overview *228*
Pacing *230*
Renal management *232*
The oliguric patient *234*
Management of renal failure *236*
Fluid management *238*
Sepsis, SIRS and MOD *239*
Gastrointestinal management *240*
Neurological management *242*
Placing arterial and venous lines *244*
PA catheter, pericardiocentesis *246*
Chest tube and cricothyroidotomy *248*
Management of chest drains *250*

Managing the five commonest calls to ICU

Atrial fibrillation

(📖 pp218–219)

- Give 10–20mmol K^+ via central line to get serum K^+ 4.5–5mmol/L
- Give empirical 20mmol Mg^{2+} via central line if none given post-op.
- Give 300mg amiodarone iv over 1h in patients with good LV, followed by 900mg amiodarone iv over 23h.
- In patient with poor LV function give digoxin in 125-microgram increments iv every 20min until rate control is obtained, up to a maximum of 1500 micrograms in 24h.
- Synchronized DC cardioversion for unstable patients (📖 pp224–225).

Bleeding

(📖 pp144–145)

- Get immediate help if bleeding is >400mL in 30min.
- Give colloid to get CVP 10–14 and systolic BP 80–100mmHg.
- Order 4 units of blood, 2 units FFP, and 2 pools platelets.
- Send coags and CBC, request a CXR stat.
- Transfuse to achieve Hb >8.0g/dL, platelets >100 × 10^9/L, APTT<40.
- Give empirical protamine 25mg iv, consider desmopression (DDAVP®).
- Emergency re-exploration is indicated for excessive bleeding.

Profound hypotension

(📖 pp142–143)

- Get immediate help. Check that reading is correct (📖 p173)
- Quickly assess pulse, rhythm, rate, CVP, O_2 sats and bleeding.
- Defibrillate VF or pulseless VT, treat AF as described.
- Treat bradycardia with atropine 0.3mg iv or pace (📖 p28).
- Give colloid to raise CVP to 12–16mmHg, place bed head-down.
- If you suspect cardiac tamponade (📖 p144) prepare for re-sternotomy.
- If patient warm and vasodilated draw up 10 micrograms of metaraminol into 10mL of saline and give *1mL* through a central line, and flush.
- If patient still very hypotensive repeat, or give 1mL 1:10,000 epinephrine iv and start infusion of presssor e.g., norepinephrine 0.1mcg/kg/min.

Poor gases

(📖 pp160–163)

- If sats <85% and falling get immediate help.
- Increase the FiO_2 to 100% temporarily, check the pulse oximeter.
- Look at expansion, auscultate the chest, check PaO_2.
- If you suspect a tension pneumothorax treat immediately (📖 p162).
- Suction the ET tube, check that the patient is not biting on it.
- Check that the drain tubing is patent, and drains are on suction.
- Treat bronchospasm with albuterol 2.5mg nebulizer.
- Disconnect from the ventilator and hand-ventilate the patient.
- Get a CXR: look for pneumothorax, hemothorax, atelectasis, ET tube position, and lobar collapse, and treat appropriately (📖 p162).

Poor urine output

(📖 pp234–235)

- Check that the Foley catheter is patent.
- If the patient is hypotensive treat this first (📖 pp142–143).
- Give a fluid challenge of colloid to raise the CVP to 14mmHg.
- If you are confident that the patient is not in a low cardiac output state (📖 pp142–143, p183), and is adequately filled, it is not unreasonable to treat profound oligura with 20mg furosemide iv (📖 pp208–209), and repeat if indicated.

Assessing cardiac output

- Cardiac output is defined on 📖 pp130–131, and is essentially a measure of cardiac function.
- When you're reviewing a patient it's always good to step back and ask yourself 'Is the cardiac output normal, high or low?' and treat any problems in that context. For example, if you are called to treat hypotension: the warm patient with bounding pulses and a high cardiac output, is going to need very different management to the patient that is cold, mottled, and oliguric, with a low cardiac output.
- There are many useful indicators of cardiac output (Table 5.1)—some may not fit (most are not isolated measures of cardiac output and can be affected by other issues including operator error), but you will usually get it right if you look at the whole picture, don't just treat one number, and go back and reassess the patient frequently (📖 p143).

Table 5.1 Assessment of cardiac output

Normal/high output state	Low cardiac output state
Peripheries warm	Peripheries cold
Brisk capillary refill	Extremities pale, mottled
Palpable/bounding distal pulses	Unable to feel distal pulses
If awake – patient appropriate	Patient obtunded
Wide pulse pressure, low CVP	Narrow pulse pressure, high CVP
Urine output >1ml/kg/hour	Urine output<1/2ml/kg/hour (📖 p234)
EKG normal	Dysrhythmias, ischemia on EKG (📖 p220)
CO>4l/min, CI>2.2l/min/m^2	CO<3l/min, CI<2.0l/min/m^2
SVO2>75% (📖 p178)	SVO2<65% (📖 p178)
Lactate <2.0	Lactate >2.5
Not acidotic	Metabolic acidosis (📖 pp168–171)
Creatinine, LFTs normal	Elevated creatinine, LFTs
Normal LV and RV on TEE/TTE	Poor LV or RV function on echo.
No effusion	Pericardial effusion (📖 p265)
No valve dysfunction	Valve dysfunction e.g. SAM (📖 p190)

Definitions of common terms

Basic concepts

- Instead of referring to systolic and diastolic BP, arterial pressure is frequently described as a single figure: the *mean arterial pressure* (MAP) which is calculated by adding 1/3 of the difference between diastolic and systolic pressures to the diastolic pressure.
- The MAP on its own does not adequately describe cardiac function: a number of other parameters are frequently used.
- *Cardiac output*: the volume of blood ejected by the heart per minute.
- *Stroke volume*: the volume of blood ejected by the heart per beat.
- Cardiac output equals heart rate × stroke volume.
- *Cardiac index* is the cardiac output adjusted to take into account the size of the patient, and is a more accurate reflection of cardiac function.

Preload

- Stroke volume depends on *preload, afterload*, and *contractility*.
- *Preload* is a measure of the wall tension in the LV at the end of diastole, but it is difficult to quantify directly.
- *Central venous pressure* (CVP), *pulmonary artery wedge pressure* (PAWP) and *left atrial pressure* (LAP) are indirect measures of preload: they are sometimes referred to as '*filling pressures*'.
- *Left ventricular end-diastolic volume* (LVEDV), and *left ventricular end-diastolic pressure* (LVEDP) are better measures of preload, but they are rarely measured in clinical practise.

Afterload

- *Afterload* is a measure of the wall tension of the LV during systole.
- It is determined by preload, which determines the maximum 'stretch,' and by the resistance against which the heart must eject, which is a function of *systemic vascular resistance* (SVR), *vascular compliance*, *mean arterial pressure*, and any LVOT pressure gradient.
- SVR, which usually reflects the amount of peripheral vasoconstriction, is a commonly used indirect measurement of afterload.

Contractility and compliance

- *Contractility* is a measure of the strength of myocardial contraction at a given preload and afterload.
- MAP and cardiac output are commonly used indirect measures of contractility.
- *Compliance* is a measure of the distensibility of the LV in diastole: stiff, hypertrophied ventricles have low compliance.
- Compliance is difficult to quantify in the clinical setting.
- LVEDP, which may be measured at preoperative cardiac catheterization or estimated by preoperative echo, is an indirect measure of compliance.

Important hemodynamic formulas

Cardiac output = SV x HR
Cardiac index = CO/BSA
Mean arterial pressure = DP + (SP – DP)/3
Systemic vascular resistance = ((MAP – CVP)/CO) x 80
Systemic vascular resistance index = SVR/BSA
Pulmonary vascular resistance = ((PAP – PAWP)/CO) x 80
Pulmonary vascular resistance index = PVR/BSA
Mixed venous = SaO_2 – (VO_2/(Hb ×1.39 x CO)) x 10

CO = cardiac output	4.5–8L/min
SV = stroke volume	60–100mL
BSA = body surface area	2–2.2 m^2
CI = cardiac index	2.0–4.0L/min/m^2
SVI = stroke volume index	33–47 mL/beat/m^2
MAP = mean arterial pressure	70–100mmHg
DP = diastolic pressure	60–80mmHg
SP = systolic pressure	110–150mmHg
SVR = systemic vascular resistance	800–1200 dyne/s/cm^5
CVP = central venous pressure	0–12mmHg
SVRI = systemic vascular resistance index	400–600 dynes/s/cm^5/m^2
PVR = pulmonary vascular resistance	50–250 dynes/s/cm^5
PAP = pulmonary artery pressure	20–30mmHg
PAWP = pulmonary artery wedge pressure	8–14mmHg
PVRI = pulmonary vascular resistance index	20–125 dynes/s/cm^5/m^2
SaO_2 = arterial saturation	95–100%
VO_2 = oxygen consumption	200–250mL/min

Oxygenation and ventilation

- *Oxygenation*, the amount of oxygen in arterial blood, is described in terms of the *partial pressure of* oxygen in arterial blood (PaO_2), and the *percentage saturation of arterial hemoglobin with oxygen* (SaO_2).
- *Ventilation*, the movement of air in and out of the lungs, is described in terms of minute volume, and assessed by measuring the partial pressure of carbon dioxide in arterial blood ($PaCO_2$).
- Oxygenation is independent of minute volumes until they are very low.
- In postoperative patients the primary cause of hypoxia is atelectasis and this must be reversed before the patient can benefit from increasing the fraction of oxygen in inspired air (FiO_2).
- *Positive end expiratory pressure* (PEEP) and *continuous positive airway pressure* (CPAP) (📖 pp158–159) treat and prevent atelectasis.
- The balance between oxygen consumption and delivery is assessed by measuring the *percentage saturation of mixed venous hemoglobin with oxygen* (SvO_2) or 'mixed venous', which is dependent on cardiac output, hematocrit, oxygenation, and oxygen consumption.
- Loss of functional residual capacity through atelectasis, supine position, lobar consolidation and collapse, effusions and obesity, results in hypoxia: CPAP, PEEP, and physiotherapy are aimed at limiting this loss.

Transfer and admission to ICU

It is easy for the surgeon to take this process for granted. Don't.

Patient requirements during transport to ICU

- Monitoring: EKG, MAP, and pulse oximetry.
- Respiratory support: E tank (most common portable size) of O_2 (check it is at least half full) connection tubing, 3L self-inflating reservoir bag, PEEP valve.
- Pharmacological support: all drugs infusing prior to transport: sedatives or anesthetics, inotropes, antihypertensives.
- Other support: IABP, other assist devices.
- Portable defibrillator.

Transfer

The postoperative patient can decompensate suddenly and catastrophically at any time during this process. The surgeon is responsible for the patient's safe arrival in ICU: surgery does not end with skin closure.

In the operating room

- Before unscrubbing, glance at the monitors to check that the patient remains in a stable rhythm, with isoelectric ST segments, satisfactory MAPs and oxygen saturations, and check the mediastinal drains: there should be no more than 100mL of blood.
- It will take several minutes for the anesthesiologist to disconnect, re-zero, and re-calibrate the monitors, and transfer the infusion pumps and ventilation to portable systems. At no stage should the patient be without electrocardiographic and hemodynamic monitoring. A surgeon should be present at all times.
- Lifting the patient can cause arrhythmias and hypotension in the volume-depleted patient, and hypertension in the waking patient.
- Make a note of what infusions are running.
- Ensure that the mediastinal drains do not become disconnected, and are placed below the level of the patient so that they continue to drain by gravity during transfer.
- Mediastinal drains should not be clamped during transfer, and if there is a significant air leak an underwater seal must be used in preference to cell salvage as tension pneumothorax is a risk.
- The patient must have an adequate rhythm, systemic pressure, and level of oxygenation before leaving the operating room.

Transport

- The anesthesiologist should have a clear view of the monitor at all times.
- During transport, manual ventilation of patients with a self-inflating bag and 100% O_2 is the anesthesiologist's responsibility: oxygenation is monitored with continuous pulse oximetry.
- All drug infusions should have been topped up and switched to battery-operated infusion pumps: running out of infusions with short half-lives such as norepinephrine during transfer can be catastrophic.

Admission to ICU

The key here is meticulous transfer of information to ensure continuity of care. The anesthesiologist, surgeon, and ICU staff ensure that the patient monitoring is transferred, re-zeroed, and re-calibrated, and transfer the portable infusion pumps and ventilation to ICU set-up. Check that hemodynamic and ventilatory parameters are acceptable, ensure that the mediastinal drains are placed on 10cmH_2O wall suction, and then outline preoperative status, operation, and the plan to the admitting team.

Typical postoperative handover

'Mr A is a 56-year-old man with good LV, three-vessel disease, smoker but no comorbidity, who has had an elective aortic valve replacement for tight aortic stenosis with a 23mm carbomedic mechanical valve and coronary artery bypass grafting times three under Dr A. Off bypass in sinus rhythm on dobutamine at 4 micrograms/kg/min. He has two atrial and two ventricular pacing wires, a mediastinal and right and left pleural drains, crossed. Aim for filling pressures of 14–16mmHg and MAPS 70–80mmHg. Wake, wean, and extubate as per protocol.'

Respiratory and cardiovascular monitoring

This is described in detail on 📖 pp156–157, and 📖 pp172–181 respectively.

Renal function

All cardiac surgical patients are catheterized. Suprapubic catheters should be used if strictures make insertion of urethral catheters difficult. Urine output is measured and recorded hourly, as are total crystalloid and colloid infusion volumes, surgical drainage, NGT aspirates, and infusions. Total fluid balance is calculated hourly. Serum electrolytes are checked on arrival in ICU and at 6–12h. Serum potassium is checked hourly.

Neurological function

Pupil size, afferent and efferent reflexes are usually checked after admission to the ICU and at 3-hourly intervals.

Temperature

Core temperature

This represents the temperature of the vital organs. The best estimate is obtained from the PA catheter thermistor. Nasopharyngeal temperature and tympanic temperature provide an accurate measure of brain temperature. Bladder temperature can be used but is inaccurate in oliguria.

Peripheral temperature

This lags behind core temperature during warming and cooling, and is affected by local factors, e.g., warmers and peripheral vasoconstriction.

Mediastinal drainage

Mediastinal and pleural drainage is recorded hourly until the drains are removed (📖 p254).

Fast-track patients

Patients undergoing CABG surgery with low preoperative risk scoring, no major comorbidity, and a straightforward operative course should be eligible for 'fast-tracking' (see Box 5.1).

Box 5.1 Exclusion criteria for 'fast-tracking'

Many units have minimal exclusion criteria, fast-tracking all but the most complex patients. Exclusion criteria used by other units include:

- Age >75 years.
- Poor LV.
- Previous history of CABG or valve surgery.
- Left bundle branch block.
- Previous problematic general anesthesia.
- Inotropic support preoperatively.
- MI within 3 weeks.
- Active congestive cardiac failure.
- IABP.
- Severe COPD.
- Pulmonary hypertension.
- Severe hepatic or renal insufficiency.
- Peripheral vascular disease or stroke.
- Ongoing bleeding.

Although patient selection and care varies widely between centers, the goal is rapid progress through hospital from operation to discharge. At the perioperative stage this mandates:

- Selection of short-acting anesthetic agents.
- Sustained normothermia.
- Effective control of postoperative pain, nausea, and vomiting.
- Early waking and extubating.
- Rapid weaning of any inotropic support.
- Early independence from invasive monitoring.
- Early mobilization.

These patients are frequently awake and extubated within 6h of arriving in ICU. Bleeding is minimal so mediastinal drains can be rapidly removed following extubation. They are hemodynamically stable with no need for inotropic support, with satisfactory respiratory, renal, and neurological function. They may be transferred to a high-dependency unit within 12h, to the floor within 24h, and discharged home within 4 or 5 days. There is often additional provision for early follow-up.

This page intentionally left blank

Postoperative pathophysiology

Key facts

- The majority of patients exhibit a broadly similar pattern of pathophysiological derangements in the hours following cardiac surgery. Intensive care protocols are designed to enable medical and nursing staff to avoid, recognize, and treat these changes.
- This section summarizes this process system by system. This should enable you to recognize and manage patients that fall outside these predictable response patterns.

Hemodynamic

The cardiovascular system is the system that undergoes the greatest physiological stresses during both bypass and beating heart surgery, even uncomplicated cardiac surgery.

- Myocardial compliance falls because of myocardial edema. Higher filling pressures are required to generate the same stroke volume.
- Contractility is impaired as a result of myocardial edema, metabolic derangements, myocardial stunning, and ischemia (which may be caused by coronary artery and arterial conduit spasm, air and particulate emboli, manipulation of the heart post bypass, periods of hypoperfusion during transfer, and factors affecting gas exchange).
- Myocardial function continues to decline for 6–8h postoperatively as a result of ischemia-reperfusion injury, before returning to baseline.
- Hypothermia is common as it is difficult to keep up with heat loss from an open chest in theatre, and it is detrimental as it predisposes to ventricular arrhythmias, coagulopathy, and metabolic acidosis, and raises afterload increasing myocardial oxygen demand.
- The high SVR helps, however, to maintain MAPs until the patient is rewarmed; it is vital to ensure that adequate volume replacement and if necessary pharmacologic support are given, to avoid hypotension during rewarming.
- Patients may 'overshoot' to temperatures above 37.5°C because of abnormalities in central thermoregulation or systemic inflammatory response syndrome (SIRS) resulting in peripheral vasodilatation and hypotension, and associated with a poorer neurological outcome.
- Vasodilatation and capillary leak mean that there is a progressive requirement for volume resuscitation to achieve satisfactory filling pressures that must be balanced against volume overloading the patient, which has a particularly deleterious effect on respiratory function.

Respiratory

Patients have a high preoperative incidence of obstructive airway disease: smoking is a common risk factor. Pulmonary function may be further compromised by pre-existing pulmonary edema, post-perfusion lung syndrome, administration of protamine, anaphylaxis, intrapulmonary shunts, mechanical factors such as pneumothorax, hemothorax, and obstruction of the tracheobronchial tree by blood or mucus plugs (📖 p163). Functional residual capacity is further reduced by supine position.

- Most patients arrive in the ICU heavily sedated, anesthetized, and paralysed. They require mechanical ventilation for a variable time.
- The anesthesiologist will frequently pre-oxygenate the patient prior to transfer with an FiO_2 of 100%. The FiO_2 is gradually weaned to 40% as long as the PaO_2 remains above 10kPa, and the rate of ventilation is reduced as the patient wakes.
- Fluid shift into the interstitial space, exacerbated by fluid overloading the vasodilated patient during rewarming, impairs oxygenation.
- Hypercapnia occurs in patients who are waking, shivering, or warming: increasing the tidal volume and/or the respiratory rate of ventilation is normally sufficient to treat this.
- The awake, neurologically appropriate patient who is stable hemodynamically and not bleeding is extubated if gases and respiratory rate are satisfactory after a period of pressure support ventilation.

Bleeding

Multiple perioperative factors predispose the patient to bleeding: preoperative antiplatelet therapy, residual heparin effect, platelet depletion and dysfunction post bypass, hypothermia, hypertension, clotting factor depletion, and surgical technique (📖 p144).

- Bleeding should taper quickly after a few hours.
- Prevention, recognition, and treatment of abnormal bleeding patterns and their sequelae is a key part of the recovery process (📖 p145).

Renal

Cardiopulmonary bypass results in major changes in fluid distribution and renal function, moderated by pharmacological agents employed primarily for hemodynamic support. The importance of correct fluid management cannot be overstated: it is the difference between a satisfactory postoperative course and one complicated by multisystem dysfunction.

- Most patients arrive in ICU in a state of total body sodium and water overload, which becomes more pronounced with the additional volume given as the patient warms and vasodilates, even though the patient may have a relatively depleted intravascular volume.
- The aim of fluid management is to ensure adequate intravascular volume to maintain perfusion. Volume resuscitation in isolation may not be enough to achieve this, and in excess will have a deleterious effect on both cardiac and respiratory function: low-dose vasoconstrictors are useful in this setting.
- Glomerular filtration rate and hence urine output is a sensitive measure of cardiac function.
- Patients with normal preoperative renal function should maintain a diuresis of at least 1ml/kg/h: renal dysfunction is uncommon in these patients in the absence of prolonged perioperative hypotension.

Neurological

Patients should be alert and oriented prior to extubation: grossly abnormal cognitive function at this stage requires resedation and continued ventilation so that the cause can be identified and treated. Agitated, confused patients commonly consume more oxygen than they will accept from a ventilator, and may remove critical lines and tubes.

Common ICU scenarios: CABG

Key facts

- In the immediate postoperative period most patients tend to follow one or more of several common clinical patterns. These are determined by the preoperative disease process, the nature of the surgery, and the body's response to the complex pathophysiological processes outlined earlier in this chapter. Common cardiac surgical intensive care scenarios are outlined in the following pages.
- Older patients, and those with more comorbidity or undergoing more complex surgery, often do not fit a single clinical pattern.

On-pump coronary artery bypass

Patients with good ventricular function and a full revascularization have an initial tendency to hypertension and tachycardia, and may also produce a large initial diuresis. Usually hypertension resolves quite quickly as the patient warms, vasodilates, and diureses.

- Early hypertension can be managed with:
  - Sedation: give 25–100 micrograms iv fentanyl bolus if the patient is agitated or in pain, and not otherwise ready to wean from the ventilator.
  - Nitrates: a continuous infusion of nitroglycerine (5–200 mcg/min iv) reduces coronary artery vasospasm, increases coronary flow to ischemic regions increasing contractility and the threshold for ventricular fibrillation, and decreases systemic vascular resistance and myocardial oxygen consumption (📖 p202). Especially useful in patients with borderline EKG changes.
  - Diltiazem: start infusion at 5mg/h to a maximum of 15mg/h (250mg in 250mL 5% dextrose) to provide similar effect on coronary blood flow, often reducing atrial tachycardias.
  - Hydralazine and β-blockers are useful later in the postoperative period in stable patients. They are not routinely used early on as they are difficult to titrate in patients with labile hemodynamics.
- The disadvantage of starting nitroglycerine routinely is that hypotension secondary to vasodilation may then be treated by giving volume to a patient who is already volume overloaded.
- Tachycardia is common, and heart rates of 90–100bpm achieve maximal cardiac output, but rates above this reduce cardiac output as diastolic filling times decrease.
  - Except in patients with LV hypertrophy (📖 p140) or SAM (📖 p190) who may be compromised by tachycardia there is no need to treat
  - Look for and address underlying causes—commonly hypovolemia, pain, anxiety, acidosis, hypercapnia, chronotropic agents (epinephrine, dobutamine), occasionally tamponade.
- 1/3 of patients go into AF, but not usually within the first postoperative day if K^+ is maintained at 4.5–5.0mmol/L (📖 p218).
- Many patients become hypotensive as a result of vasodilation in response to warming, or the long half-life of preoperative ACE inhibitors: if the patient is peripherally warm, with CI >2.2L/min/m^2

and SVO_2 >70, and CVP<14mmHg a small amount of norepinephrine 0.03–0.05 micrograms/kg/min) is usually sufficient to treat this.

- Patients with poor ventricular function or incomplete revascularization may also become increasingly hypotensive with poor urine outputs, as stunned myocardium is transiently further depressed in the first 6–12h post surgery. Look for:
  - Worsening lactic acidosis.
  - Rising filling pressures.
  - EKG changes, e.g., ST segment elevation.
- In this case dobutamine (250mg in 50mL 5% dextrose at 3–5 micrograms/kg/min) is the first-line choice, alternatively epinephrine (250mg in 250mL 5% dextrose at 0.03–0/05 micrograms/kg/min) will improve contractility.
- Ischemic EKG changes (ST segment elevation or depression, T-wave inversion, new bundle branch block) suggest poor myocardial protection, vasospasm of arterial grafts, incomplete revascularization, or acute graft occlusion: management is discussed on 📖 p185.
- Failure to respond to appropriate inotropic support mandates accurate assessment of cardiac indices with a pulmonary artery catheter if not already *in situ*, and echocardiography (📖 p185): an IABP or re-exploration may be indicated.

Early warning signs in CABG patients

Correct metabolic abnormalities, talk to the surgeon—do you need echo to evaluate regional wall motion abnormalities (ischemia) or collection (tamponade)?

- Frequent extrasystoles, ST segment changes.
- Hypotension, rising filling pressures, rising PAPs.
- Low cardiac output, low mixed venous, rising lactate, acidosis.
- Oliguria (do not empirically treat with diuretics).

Coronary artery bypass—off-pump

Patients with good ventricular function and a full revascularization are usually hypertensive and tachycardic. They do not produce a diuresis as they are often relatively underfilled. A significant metabolic acidosis frequently develops during the first 6h postoperatively, resolving within 12h. The cause of the acidosis is unclear but may reflect an intraoperative low cardiac output state. As bypass cannot be used to warm these patients they are often hypothermic.

- Ensure these patients are adequately filled.
- Use warming blankets and infusion warmers to achieve normothermia.
- These patients sometimes receive a reduced dose of intraoperative heparin depending on surgical preference, and should therefore receive early aspirin (81mg po) and low-molecular-weight heparin (5000U sc) to ensure graft patency and avoid thromboembolic complications.

Common ICU scenarios: valves

Aortic stenosis

These patients have a stiff, hypertrophied ventricle that is dependent on synchronized atrial contraction to provide 30% of stroke volume and therefore cardiac output. Shortened filling times reduce cardiac output.

- AF and tachyarrhythmias cause these patients to decompensate rapidly: have a low threshold for synchronized DC cardioversion if antiarrhythmic drugs do not achieve cardioversion quickly. It may be worth prophylactically loading these patients with 300mg iv amiodarone followed by 900mg over 24h.
- Sequential AV pacing achieves a much better cardiac output than ventricular pacing alone, and both atrial and ventricular epicardial pacing leads should be placed during surgery as debridement of heavily calcified valves frequently results in complete heart block, which may eventually require permanent pacing in 2–3% of all aortic valve replacements.
- Ensure adequate preload by filling to CVP 14–16mmHg: as the ventricle is small and non-compliant, very little volume may be required to achieve this.
- Hypertension usually develops after 6–8h as the heart recovers from bypass, and responds to the removal of the outflow gradient.
- In order to reduce myocardial oxygen demand, preserve suture lines and reduce the risk of hemorrhagic stroke, mean systemic pressures should be maintained at 70–80mmHg using diltiazem or nitroglycerine (5–200 mcg/min iv).
- Anticoagulate mechanical valves if chest tube output minimal, starting the second evening after surgery by commencing warfarin: the risk from thrombosis is negligible in the first 2–3 days post-op so there is no reason to start iv heparin at this stage.

Aortic regurgitation

These patients have dilated, compliant ventricles, and also rely on AV conduction.

- Compliant, dilated ventricles mean that large volumes are often required to raise filling pressures.
- Removal of the regurgitant fraction means LV performance improves compared to RV function: filling pressures are therefore lower postoperatively than they are in AS.
- Anticoagulate as described above.

Mitral stenosis and mitral regurgitation

Both types of patient are commonly in chronic AF and have a variable degree of pulmonary hypertension, which may be associated with RV dysfunction postoperatively. Patients with MS usually have small LVs and well-preserved function. Correcting MR, on the other hand, may unmask severe LV dysfunction as the low-resistance regurgitant pathway is removed: the preoperative ejection fraction usually overestimates LV function.

- Treatment is directed at reducing pulmonary hypertension and RV dysfunction, which may be exacerbated intraoperatively by poor

myocardial protection, and postoperatively by factors which raise pulmonary resistance: positive pressure ventilation, bronchospasm, pulmonary edema, reduction in functional residual capacity and vasoconstrictors such as norepinephrine.

- Higher filling pressures are required to ensure adequate cardiac output, but right-sided filling pressures may not correlate with left-sided pressure because of RV dysfunction and pulmonary hypertension.
- Large fluid volumes may be needed because of increased atrial or ventricular compliance, but giving volume may exacerbates RV dysfunction without achieving optimal LV filling pressures: inotropic support is commonly required.
- Systolic anterior motion may complicate mitral repair (📖 p190): the mainstay of medical management is avoiding tachycardia and hypovolemia, giving volume and β-blockade. SAM in the setting of significant RV dysfunction is a difficult problem, as the management of the two problems is completely opposed (📖 pp190–191).
- TEE or a PA catheter should be used to assess RV dysfunction.
- Respiratory dysfunction is common as a result of pulmonary hypertension and volume overload, and it must be managed aggressively with a diuresis and particularly careful ventilatory weaning.
- Dobutamine and milrinone result in pulmonary vasodilation and reduce pulmonary hypertension in addition to their inotropic effect: inhaled nitric oxide decreases PVR and helps in severe RV dysfunction.
- Some patients with chronic AF come off bypass in sinus or junctional rhythms, particularly if ablation has been performed: amiodarone should normally be started to maintain freedom from AF.
- Patients with mechanical valves do not normally need therapeutic anticoagulation in the first 48h.

Aortic surgery

Aortic dissection and aneurysm repair involves suturing Dacron graft to friable tissues. The proximal false lumen is obliterated, but distal extension of the dissection can occur with end-organ ischemia. Surgery often involves a period of hypothermic circulatory arrest. Postoperatively paraplegia can result from hypoperfusion of spinal cord (📖 pp436–437).

- Avoid extremes of BP: aim for MAPs 90–100mmHg using either antihypertensives that reduce the rate and force of contraction e.g. labetolol in hypertensive patients or vasoconstrictors and fluid in hypotensive patients.
- Bleeding may be a significant problem: long bypass times and hypothermia contribute to a significant coagulopathy. Fragile suture lines result in increased bleeding rates during periods of hypertension, and are often not improved by further surgery: there will be a higher threshold for re-sternotomy. Correct coagulopathy (📖 p145).
- Poor renal function, or worsening metabolic acidosis refractory to adequate perfusion pressures should be investigated by imaging the descending aorta; percutaneous fenestration of the dissection flap may salvage ischemic end-organs. Deteriorating lower extremity neurology may respond to increase perfusion pressure or spinal drainage (📖 pp436–437).

Common hemodynamic scenarios

Normotensive but low cardiac output

Peripheries are cold to touch. (If the patient is warm, with good peripheral pulses and normotensive, it is almost impossible for their cardiac output to be low.) Mean arterial and filling pressures are initially adequate. One of the conditions for terminating bypass is a core temperature of 36.6–37.5°C. After bypass the chest remains open while the surgeon establishes satisfactory hemostasis: although the anesthesiologist may make vigorous efforts to maintain core temperature using warming blankets and warming iv fluids, patients usually reach ICU hypothermic and vasoconstricted. Shivering increases filling pressures, and can double oxygen consumption, causing SVO_2 to fall in the hypovolemic patient.

- These patients should be managed with aggressive volume replacement to maintain preload which tends to fall as warming causes vasodilatation: as much as 3L of fluid are often required in the first few hours on ICU in response to capillary leak, diuresis, rewarming, and the fact the perfusionist may have hemoconcentrated the patient by filtering several litres of fluid off during bypass.
- Warming blankets correct core temperature slowly, mainly by minimizing peripheral heat loss. Avoid infusion of cold fluid.
- Nitroglycerine infusion reduces coronary artery vasospasm, increases coronary flow to ischemic regions increasing contractility and the threshold for ventricular fibrillation, and decreases systemic vascular resistance and myocardial oxygen consumption (📖 p202).
- The vasodilated patient equilibrates with external temperatures faster than the vasoconstricted patient.
- Stop shivering, which increases oxygen demand and compromises ventilation, by warming the patient and giving a muscle relaxant (vecuronium 0.1mg/kg iv) in the sedated, intubated patient.
- In the awake patient shivering can be treated with pethidine (25mg iv) which has an antispasmodic effect as well as its analgesic effect.

Hypotensive with low CVP despite giving volume

Warm patient with good cardiac output

If peripheries are warm to touch with good pulses, CVP is <10mmHg despite being >1–2L positive, cardiac index is >2.2L/min, although mixed venous saturation may be low, this suggests primary issue is vasodilation. As patients warm to 37°C they vasodilate. Falling filling pressures require volume to maintain systemic pressures.

- In a young patient, if urine >1mL/kg/h, mixed venous and pH are ok, and patient warm and well perfused, MAPs 55–60mmHg may be adequate for organ perfusion: less likely in elderly.
- Colloid is the first choice: blood to maintain the hemoglobin levels >8.0g/dL: Gelofusine®, Hespan®, or albumin where the hemoglobin is satisfactory, blood products in the bleeding patient.
- Fill to achieve RAPs of 12–14mmHg: MAPs should improve significantly and quickly.
- If the patient is mounting a large diuresis, "replace crystalloid with crystalloid": use normal saline in older patients who may have impaired

ability to conserve sodium, and are therefore at risk of hyponatremia, but look out for hyperchloremic acidosis in these patients (📖 p169).
- If the patient has low MAPs, low filling pressures, and poor urine output despite being over 2–3L positive, ***and is not bleeding***, this reflects abnormally high cardiac compliance and capillary permeability rather than a need for continued fluid challenges, which will adversely effect pulmonary function: start a small dose of a vasoconstrictor such as norepinephrine (4mg in 50mL 5% dextrose at 0.03–0.3 micrograms/kg/min)which should reduce the volume requirement.

Poor cardiac output, cool peripheries
- Recheck fluid balance—look for bleeding, which may be in the ***chest, groin, leg drains, gut, abdomen, or the bed, not just the chest tubes***. Is the hematocrit falling rapidly? Is the patient polyuric?
- Look for causes of low cardiac output while starting a low-dose epinephrine (4mg in 250mL 5% dextrose at 0.03–0.05 micrograms/kg/min) and continuing fluid resuscitation.

Hypotensive with high filling pressures

The causes of this scenario range from benign to acutely life-threatening.
- Fluids being run through CVP transducer port (most common).
- Cardiac tamponade (📖 p144).
- Other causes of pump failure (📖 pp184–185, 188–191).
- Patient waking: shivering and 'fighting' the ventilator (📖 p165).
- Massive PE (rare in patients without pre-op risk factors) (📖 p476).

Useful rules of thumb

Always look at the whole picture, and examine the patient—treating one number in isolation is almost always a mistake.
- Remember that the picture is dynamic—go back and reassess.
- If you are not sure what is going on, change one thing at a time, wait and reassess—making multiple changes at once can confuse the clinical picture. It's rare you don't have time to do this safely.
- If the numbers don't make sense, recheck them: check the arterial line pressure against a cuff pressure, check the CVP off a different port, check the cardiac output against clinical findings.
- If the patient is warm, with good peripheral pulses, and normotensive it is virtually impossible for their cardiac output to be low.
- If the patient is cold, with barely palpable peripheral pulses, their cardiac output is not adequate, whatever the number recorded.
- Remember that mixed venous saturation, is dependent on oxygen consumption (is the patient shivering, or agitated?), hematocrit and oxygenation—not just cardiac output. Treat the underlying causes, and do not just escalate inotropes without confirming cardiac output is inadequate.
- Low filling pressures mean hypovolemia: this is usually due to capillary leak, diuresis, rewarming, and vasodilatation, but don't forget to look for bleeding, which is not always evident in the chest tubes.
- High filling pressures are cause for concern: they may be due to a patient shivering or waking up, but can indicate RV or LV dysfunction, MR or TR, ischemia, tension pneumothorax or tamponade.

Bleeding

Key facts

- Excessive bleeding is bleeding that falls outside the 'normal' pattern of bleeding post cardiotomy (see Box 5.2), and except in rare cases should not be acutely life threatening if correctly managed.
- Every effort must be made to prevent excessive bleeding with good surgical technique, and careful correction of coagulopathy pre- and postoperatively, as bleeding and transfusion are associated with potentially major morbidity and mortality.

Cardiac tamponade

Cardiac tamponade is a surgical emergency. It is an indication for urgent re-sternotomy (📖 p150). It is not always an easy diagnosis to make, as severe LV or RV dysfunction may result in similar picture. Tamponade is suggested by:

- Hypotension *and* high filling pressures.
- Additional features may include:
  - Worsening urine output and acidosis despite diuretic therapy.
  - Equilibration of cardiac filling pressures.
  - Sudden cessation of bleeding.
  - Excessive widening of the mediastinum on CXR.
  - Decreased EKG voltage; pulseless electrical activity (PEA) is a very late sign.
  - TEE may show clot in pericardium, collapse of RV in diastole, and underfilled hyperdynamic LV, but is operator dependent and not 100% accurate: a negative echo does not rule out tamponade.

Box 5.2 'Normal' bleeding

Mediastinal bleeding should be greatest in the hours immediately after surgery, tailing off to near zero over the course of the following 6–12h. Acceptable rates of bleeding are: up to 2mL/kg/h for the first 2–3h, up to 1mL/kg/h for the next 3h, and <0.5mL/kg/h by 12h post closure. The trend should be downward.

Etiology

- *Preoperative:* medication, e.g., aspirin, clopidogrel, warfarin, thrombolytics, coagulopathy (ITP), age, previous sternotomy, infective endocarditis. Preoperative pleural effusions mimic bleeding if not drained in surgery.
- *Operative:* surgical technique, bypass causing platelet dysfunction, thrombocytopenia and fibrinolysis (📖 p106), residual heparin, open chest.
- *Postoperative:* dilutional, massive transfusion with red cells, hypothermia, hypertension, sepsis, DIC, ECMO, old collection.

Management

Prevent perioperative blood loss by identifying and correcting pre-existing coagulopathies (📖 p145). Some surgeons give prophylactic aminocaproic acid (Amicar®) or tranexamic acid (📖 p147). Meticulous operative technique is usually the most important factor.

Investigation

- If the chest looks wet on closing ask for TEG, heparin level, repeat ACT and platelet works in OR and treat. Check Hb, platelets, APTT, PT, and fibrinogen levels on arrival in ICU, and after each batch of blood products when treating bleeding.
- *Look* at the patient; are they hemodynamically unstable, is blood rising very rapidly up the drains, does drainage look arterial, are the conduit harvest sites bleeding, are line sites bleeding?
- Bleeding frequently needs treating empirically, before the results of clotting studies are known: replace blood loss with blood products.
- If bleeding does not decrease despite corrected APTT, PT, and platelets, consider re-exploration (📖 p150).
- A CXR will show pleural collections and may show excessive mediastinal widening (📖 p14, Fig. 1.2b).
- If you suspect that the drainage is bloodstained fluid (if the patient had pleural effusions preoperatively, or the chest was washed out at the end of the procedure) measure the hematocrit of the drain fluid (<10 suggests effusion rather than frank bleeding).

Treatment

- If bleeding approaches 500mL in 30min, or the patient is hemodynamically unstable, prepare for emergency resternotomy (📖 p150).
- Order 4–6 units of blood, FFP, platelets, and cryoprecipitate.
- Give blood, or colloid if no blood available, to maintain MAPs.
- Put the bed flat or head-down if patient very hypotensive.
- Control hypertension with adequate sedation and vasodilation (📖 p202), and control shivering with sedation and paralysis (📖 p142).
- Increasing PEEP can reduce bleeding, but also reduces cardiac filling causing hypotension.
- Check coagulation and treat the bleeding patient with platelets, FFP, blood, protamine, or cryoprecipitate as indicated (📖 p146): if there is no coagulopathy the bleeding patient needs to be re-explored
  - Give desmopressin(DDAVP®).
  - Give 2 units of platelets for functional platelet count <90 $\times 10^9$ /L.
  - Give 2–4 units of blood to maintain Hb >8.0g/dL.
  - Give 1 pool of cryoprecipitate to maintain fibrinogen >400.
  - Give 2 units FFP to maintain APTT 25–35s.
  - Give 25mg iv protamine to reduce residual heparin level to 0.
- Withhold postoperative aspirin and any anticoagulants.
- Ensure patency of chest drains by milking them regularly.
- Avoid giving large volumes of cold fluid: use infusion warmers.
- Warm the patient to 36°C: hypothermia suppresses coagulation mechanism and platelet function.
- If a drain appears blocked, clamp and disconnect it and quickly remove any clot with forceps: using a sterile fine-bore aspiration tube under wall suction may help but exposes patient to risk of sepsis and should be reserved for emergencies.

Blood products, antifibrinolytics

Blood

- 1 unit of blood increases Hb by about 1g/dL in a 70kg adult.
- Blood is normally provided as packed red cells (1 unit = ~350mL).
- Cross-matched blood can normally be provided within 20min: it contains blood from a single donor.
- In dire emergencies O −ve blood (universal donor) can be given to recipients of any ABO Rh group without incompatibility reaction.
- Some centers use autologous blood transfusion to avoid the complications described on 📖 p148: up to 2 units of blood are withdrawn from patients preoperatively, which may be stored for up to 5 weeks.
- Additionally *cell salvage* reduces the need for allogeneic blood. Shed blood is collected intraoperatively and postoperatively, 5000U of heparin are added to collected blood which is spun with normal saline to remove all material, including residual heparin, platelets, and clotting product, to produce RBCs suspended in saline.

Platelets

- 1 unit of platelets increases platelet count by 10^9/L in a 70kg adult.
- Platelets are provided as units (1 unit = ~50mL).
- Platelets do not need to be cross-matched but they should be ABO compatible (and rhesus matched in females of childbearing age).
- 1 unit contains platelets from a single donor.
- They are stored at 22°C, and have a shelf life of 5 days.

Fresh frozen plasma

- 1 unit of FFP contains all the coagulation factors except platelets.
- 1mL of FFP per kg will raise most clotting factors by 1% in a 70kg adult.
- 1 unit of FFP = 150–250mL, and 5–10mL/kg is normally given.
- 1 unit of FFP usually contains product from a single donor but is sometimes pooled, in which case it may be from several donors.
- FFP does not need to be cross-matched but should be ABO compatible (and Rhesus matched in females of childbearing age).
- FFP must be stored at <−18°C: it must be thawed, usually over 20min, before giving, and discarded if not used within 2h.
- FFP treats heparin resistance due to antithrombin III deficiency (📖 p93).

Cryoprecipitate

- I bag of cryoprecipitate contains 150–250mg fibrinogen, and factors VII and VIII, and von Willebrand factor.
- 10 bags of cryoprecipitate raises fibrinogen 0.6–0.7 mg/L in a 70kg adult.
- I bag of cryoprecipitate = 20mL: 5–10 pooled bags are normally given.
- ABO and rhesus compatibility is not relevant.

Recombinant activated factor VII

- Potent agent acting at site of tissue injury, binds to exposed tissue factor, activating platelets and leading to fibrin and clot formation.
- Few small studies showing significant reduction in blood loss, need for transfusion. Risk of adverse thrombotic sequelae and cost mean this

agent generally reserved for bleeding without a surgical solution that is refractory to two or more rounds of blood products.
- *Dose:* 50–90mcg/kg after discontinuation of bypass (discuss with hematologist).

Antifibrinolytics

There are two types of antifibrinolytic:
- Synthetic lysine analogues, e.g., tranexamic acid and aminocaproic acid (Amicar®).
- The non-specific serine protease inhibitor aprotinin (Trasylol®) which was withdrawn from world markets in May 2008 after several non-randomized studies raised safety concerns.

Aminocaproic acid (Amicar®)
- *Action*: inhibits fibrinolysis by blocking lysine binding sites on plasminogen, inhibiting formation of plasmin
- *Pharmacodynamics*: renal elimination.
- *Dose*: loading dose ranges from 75–150mg/kg and maintenance dose range is 12.5mg/kg/h to 30mg/kg/h over varying timeframe.
- *Indication*: prophylactic prevention of bleeding.
- Randomized controlled trials and meta-analyses show blood loss reduced on average by 200mL per patient, and blood transfused by 1.3 units per patient, but no impact on reoperation for bleeding.
- No apparent increase in death, stroke, myocardial infarction, renal failure, or pulmonary embolism

Tranexamic acid (Cyklokapron®)
- *Action:* binds to plasminogen and plasmin to inhibit fibrinolysis. It promotes clot stability
- *Pharmacodynamics*: renal elimination.
- *Indication*: prophylactic prevention of bleeding.
- *Dose*: normally given as an iv infusion 1–2g on induction, or up to 1g an hour during operation. Can be given postoperatively as a single dose of 1–2g iv.
- RCTs and meta-analyses show blood loss reduced on average by 300mL per patient, and blood transfused by just under 1 unit per patient, but no impact on reoperation for bleeding.
- No apparent increase in death, stroke, myocardial infarction, renal failure, or PE.

Desmopressin (DDAVP)

- *Action*: releases von Willebrand factor from endothelial cells, increasing vWF and factor VIII: main indication in hemophilia A and von Willebrand disease, but also used after cardiac surgery.
- *Dose*: slow infusion 0.3g/kg.
- *Indication*: prophylactic prevention of bleeding.
- One early study showed 30% reduction in blood loss and transfusion requirements, subsequent studies show blood loss reduced on average by 115mL per patient, but no impact on transfusion requirements, or reoperation for bleeding.
- No significant increase in death, stroke, myocardial infarction, renal failure, or PE.

Transfusion reactions

Acute hemolytic reaction

ABO incompatibility as a result of clerical, bedside, sampling or laboratory error is the most common cause. May also be caused by incompatibility within other antigen systems (Duffy/Kidd). ***Donor erythrocyte A or B antigens*** bind to the recipient's anti-A or anti-B antibodies, resulting in complement formation, membrane attack complex, and immediate hemolysis. Cytokine and chemokine release mediates sympathetic inflammatory response characterized by sudden onset of hypotension, tachycardia, pyrexia, breathlessness, tachypnea and back pain. Bilirubinemia, anemia, and hemoglobinuria as a result of hemolysis ensue.

- Stop the transfusion immediately and give basic life support if required.
- Keep the bag and giving set for analysis, inform haematology.
- Give crystalloid and furosemide to encourage a diuresis.
- Dialysis may be required.

Anaphylaxis and allergic reactions

Normally *IgE-mediated* histamine release reactions to plasma, platelets and red blood cells. Mild allergic reactions are relatively common, and are characterized by erythematous papular rashes, wheals, pruritus and pyrexia. These are treated by stopping the transfusion and administering chlorphenamine (piriton) (10mg iv). Anaphylaxis characterized by hypotension, bronchospasm and angioedema occasionally occurs.

- Stop the transfusion immediately and disconnect connection tubing.
- Basic life support may be required.
- Treat bronchospasm and angioedema with epinephrine (1mL of 1:10,000 iv), chlorpheniramine (10mg iv), and hydrocortisone (100mg iv).

Non-hemolytic febrile reaction

These common and normally mild reactions are caused by recipient antibodies directed against ***donor HLA and leukocyte specific antigen on leukocytes and platelets.*** Cytokine release mediates a mild pyrexia, typically over an hour after transfusion is started. Antipyrogens such as acetaminophen 500mg–1g po/pr limit pyrexia, but antihistamines are not helpful. Severe reactions feature high grade fever, rigors, nausea, and vomiting. The severity of symptoms is proportional to the number of leukocytes in the transfused blood and the rate of transfusion. Leukocyte-depleted blood helps prevent these reactions.

Delayed extravascular hemolytic reaction

Although pretransfusion antibody testing is negative (a satisfactory XM) these patients experience accelerated destruction of transfused red blood cells 7–10 days following transfusion. This is an antibody-mediated reaction, usually by a patient antibody (commonly rhesus E, Kell, Duffy, and Kidd), present in levels too low to be detected clinically until produced in larger amounts on exposure to ***circulating donor antigen***. As hemolysis is extravascular, hemoglobinaemia and hemoglobinuria are uncommon: it is characterized by an unexpected fall in haematocrit a few days post transfusion, hyperbilirubinemia and positive Coombs test.

Transfusion-related acute lung injury (TRALI)

Non-cardiogenic pulmonary edema typically within 6h of transfusion is mediated by recipient antibodies against ***donor HLA***. Activated recipient leukocytes migrate to the lung, releasing proteolytic enzymes that cause a localized capillary leak syndrome and pulmonary edema (📖 p163).

Infection

Bacterial

Serious bacterial contamination of stored blood may occur, although platelets, which are usually stored at room temperature, are at greater risk. Common organisms are *Staphylococcus* spp., *Enterobacter*, *Yersinia*, and *Pseudomonas* spp. Contamination is difficult to detect. The recipient is pyrexial >40°C, and hypotensive. This may occur during or hours after transfusion, and unlike febrile transfusion reactions is not self-limiting.

- Volume resuscitation.
- Culture the patient and send bag and giving sets to microbiology.
- Start empirical broad-spectrum antibiotics.

Non-bacterial

- Pretransfusion testing includes screening for Hep B (HbsAg, anti HBc), hepatitis C (anti HCV), HIV (anti HIV-1/2, HIV-1 p24 antigen), HTLV (anti TLC-1/2) and syphilis. HIV can be transmitted by an infective but seronegative donor for about 15 days after infection. The HCV window is 20 days. CMV is common in the donor population (40–60%) and immunocompromised recipients must receive leukocyte-depleted or CMV negative blood. Malaria may be transmitted by blood transfusion as may nvCJD.

Fluid overload

Characterized by hypotension, high filling pressures, and poor gases.

- Stop the transfusion.
- Give high-flow O_2 and loop diuretics (40mg furosemide iv).

Massive transfusion

Massive transfusion is replacement of the patient's circulating volume within 24h. Blood is stored at 4°C for up to 5 weeks, but this is associated with a variety of abnormalities including depletion of ATP, 2,3 DPG and leakage of potassium from RBCs. $\downarrow Ca^{2+}$ (citrate used to store blood binds Ca^{2+}): $\uparrow K^+$, hypothermia (if blood is not warmed), and dilutional coagulopathy are complications of massive transfusion.

- Use infusion warmers and a warming blanket.
- Monitor central circulation and respiratory function closely.
- Give CaCl iv slowly (10mL of 10%) after 8–10 units of blood: calcium gluconate provides 1/3 less ionized calcium.
- Check platelets, APTT and fibrinogen: replace empirically if needed.
- Check K^+ regularly.

Re-exploration for bleeding

Key facts

- The re-sternotomy rate for bleeding or tamponade following first pericardiotomy is ~1–2%.
- Resternotomy is best performed in OR (sterility, light, equipment), but if patient too unstable to transfer can be performed at bedside on ICU.

Indications for emergency re-exploration

1. ***Incipient or actual cardiac arrest*** due to cardiac tamponade, or torrential hemorrhage.
2. ***Excessive bleeding***: re-exploration is indicated for grossly excessive bleeding. The following rates are given for guidance only. The decision is a subjective one as the factors listed below must also be taken into account. Consider emergency resternotomy for bleeding rates of:

- >500 mL in <30min.
- >7 mL/kg for 1h.
- >5 mL/kg for 2 consecutive hours.
- >3 mL/kg for 3 consecutive hours.
- >2 mL/kg for 5 consecutive hours.

Re-exploration versus medical management

Higher bleeding rates are accepted where there is a correctable coagulopathy, where there are known risk factors for bleeding (📖 p144), where the surgeon is particularly sure that there is no surgical source for bleeding (e.g., where prolonged efforts were made to secure hemostasis in theatre), or where the surgeon feels that the source of bleeding is not surgically correctable (raw surface, needle holes). If bleeding is excessive but the trend is reducing, it may be acceptable to delay the decision to reopen. The following factors indicate that re-exploration is needed:

- Torrential hemorrhage or increasing rate of bleeding, tamponade.
- Hemodynamic instability (e.g., escalating pressor requirement).
- Large hemothorax.
- Minimal coagulopathy.

Technique of re-exploration

Be familiar with the location and contents of the emergency re-exploration set. If you can stabilize the patient and transfer to the OR, do so. It is much more difficult operating over an ICU bed, with poor light and poor sterility, and no scrub nurse, and results reflect this.

Items that are often NOT on the re-exploration set in ICU:

- Prep: pour over patient before putting on gown and gloves.
- Scalpel blades: open up a disposable pre-mounted one first.
- Suction: pass out the tubing so it can be connected to wall suction.
- Light: send someone to get a portable overhead light.
- Defibrillator: don't forget to connect the internal paddles.

Think! Do you need a perfusionist? Ask someone to phone the senior surgeon responsible for the patient.

- Patient must be intubated and sedated (📖 p90) as you prep and drape.
- Open the wound down to the sternum with a knife. Suction ready.
- Cut the sternal wires with a wire cutter: if there is no wire cutter use a heavy needle holder to grab and twist the wires until they fracture.
- Pull ***all*** the wires out: leaving fragments of wire risks lacerating the right ventricle and makes it more difficult to get good exposure.
- Lever the sternum open with your fingers, and check edges well clear of any grafts, pacing wires, etc.
- Place the sternal retractor (use the Finochietto if you are on your own as this needs least assembling) and carefully expose the heart.
- You should see an immediate improvement in perfusion pressures.
- The commonest finding is a mass of clot with a liquid component: suction this, and remove the clots carefully: filling pressures should fall.
- If there is a source of major bleeding, put a finger over it if you cannot rapidly gain control, and get more senior help.
- If no source of bleeding is immediately apparent, wash the chest out several times with saline (this physically reduces the bacterial load, and helps highlight continued bleeding points) and pack, systematically looking for bleeding at the surgical sites listed (see Box 5.3), taking great care not to avulse grafts while manipulating the heart: inspection of the inferior aspect of the heart is difficult, and it must be swift as it causes significant hemodynamic instability.
- The suprasternal veins sometimes only bleed as the retractor is closed, releasing tension from the surrounding tissues.
- Remove the packs, noting whether they come out 'red, pink or white'.
- Go back and concentrate the search where swabs are 'red': if bleeding is from needle holes or you cannot see an obvious source of bleeding, pack again with dry swabs. Wait. Pack again.
- If 'non-surgical' bleeding persists despite appropriate clotting factors try a topical procoagulant such as Surgicel®, or a tissue glue such as Floseal®.
- Before closing flush the chest drains through with a 60mL 'bladder' syringe filled with saline to remove clot, and replace them in the chest.
- On the rare occasion that significant 'non-surgical' bleeding persists one option is to pack the chest and close the wound with a 'Bogota' bag (the sterile inside bag of iv fluid, or Goretex® membrane), cut to shape and stitched over the open wound using running 3/0 Prolene. The sternum can be splinted open using a 10mL syringe, with the ends cut off and shaped to fit using eye-cautery.

Box 5.3 Sites of bleeding

- *Venous:* atrial cannulation sites, innominate vein, thymic vein, suprasternal veins, cardiac veins, mammary bed, bone marrow, left pulmonary artery if aorta 'taped', pacing wire sites, 'raw' surface of myocardium, right ventricle, right atriotomy.
- *Arterial*: aortic cannulation site, aortotomy, top-ends, left atriotomy, vent sites (aortic, right superior pulmonary vein, LV apex), vein/IMA side-branch, IMA stump, coronary anastomosis, periosteal arteries.

Respiratory physiology

Normal values

- PaO_2 >10.5kPa (75mmHg)
- Pa CO_2 4.7–6.0kPa (35–54mmHg)
- SaO_2 >97%
- SvO_2 65–80%
- O_2 delivery = CO × [(Hb × SaO_2 × 1.39 × 10) + (PaO_2 × 0.0225 × 10)]

(Hb is in g/dL, and PaO_2 is in kPa.)

Basic concepts of oxygen carriage

Definitions of common terms are listed on 📖 pp130–131.

- Oxygen consumption is about 250mL/min O_2 in the adult at rest, rising to over 4L/min during exercise.
- Oxygen diffuses from alveolus to pulmonary capillary until arterial PO_2 (PaO_2) = alveolar PO_2 (P_AO_2).
- The solubility of oxygen in blood is low (0.000225mL O_2 per mL of blood per kPa): at a normal PaO_2 of 13kPa, there is only 0.3mL O_2 dissolved in 100mL of blood.
- Each gram of hemoglobin combines with 1.34mL O_2: at a normal Hb 13g/dL there is 20mL O_2 bound to hemoglobin in 100mL of blood.
- Hence the percentage of arterial hemoglobin binding sites bound to oxygen (SaO_2) or 'oxygen sats' is the main determinant of the O_2 content of arterial blood.
- A fall in pH, a rise in $PaCO_2$ or increased temperature cause a right shift in the oxygen dissociation curve, encouraging oxygen release in metabolically active tissues (the Bohr effect).

The relationship between PaO_2 and SaO_2 is non-linear. As each O_2 molecule binds to 1 of the 4 heme groups on the hemoglobin molecule, the affinity of the remaining heme molecules for oxygen increases, increasing the steepness of the *oxygen-hemoglobin dissociation curve* (Fig. 5.1). The curve flattens out at partial pressures above 8kPa because there are few remaining sites. Above 8kPa (60mmHg) rises and falls in PaO_2 make very little difference to O_2 saturation. Below 8kPa small drops in PaO_2 result in dramatic drops in SaO_2. PaO_2 is an effective measure of changes in oxygenation where SaO_2 plateaus, which is why it is used as a measure of oxygenation. Below this falling PaO_2 heralds profound desaturation, reflecting a dangerous drop in O_2 content of arterial blood.

Other points about oxygen carriage

- Increasing hemoglobin has a major impact on oxygen content.
- Maintaining the PaO_2 well above 10kPa (75mmHg) maintains a slight safety margin for oxygenation in the event of a drop in cardiac output, acute respiratory compromise, or bleeding.
- In non-cardiac patients, a rise in cardiac output can compensate for SaO_2 >80%, so SvO_2 remains at normal levels. In cardiac patients this does not happen to the same extent. Using inotropes to increase cardiac output to compensate for hypoxemia risks myocardial ischemia.

Basic concepts of carbon dioxide transport

Ventilation has two goals: (1) achieving satisfactory PaO_2 and (2) regulation of $PaCO_2$. It has a major bearing on acid–base balance (📖 pp168–169). CO_2 is transported in the blood as bicarbonate ions (HCO_3^-), and carbamino compounds combined with proteins and dissolved in plasma. As can be seen from the CO_2 dissociation curve (Fig. 5.1) the amount of CO_2 that can be carried in blood is much more than the amount of O_2, and it does not plateau. This is because CO_2 is highly soluble. 60% of CO_2 is transported in the form of bicarbonate:

$$CO_2 + H_20 \Leftrightarrow H_2CO_3 \Leftrightarrow H^+ + HCO_3^-$$

The left-hand side of the equation proceeds slowly in plasma. Carbonic anhydrase, an enzyme present in red cells, accelerates the reaction. The right-hand side of the equation is rapid in any phase. HCO_3^- diffuses out into plasma, but the red cell membrane is impermeable to H^+. To maintain electrical neutrality Cl^- ions diffuse into the cell: the *chloride shift*. H^+ binds avidly to deoxygenated hemoglobin, which acts as a buffer. Oxygenated hemoglobin does not bind H^+ so well.

- The *Haldane effect* means that for any given $PaCO_2$ the CO_2 content of deoxygenated blood is greater than that of oxygenated blood: CO_2 is appropriately taken up in respiring deoxygenated tissues, and released in the oxygenated blood of the lungs.
- Treating metabolic acidosis by administering *sodium bicarbonate* drives the equation to the right: it is important not to give excessive sodium bicarbonate, and to ensure that the patient is adequately ventilating otherwise a paradoxical intracellular acidosis results as CO_2 accumulates.

Other points about carbon dioxide carriage

- CO_2 is 20 times more soluble than O_2.
- At any given metabolic rate doubling the alveolar ventilation halves alveolar and arterial PCO_2, and vice versa, whereas the relationship between ventilation and PaO_2 is more complicated: hypo- and hyperventilation are therefore defined in terms of $PaCO_2$.

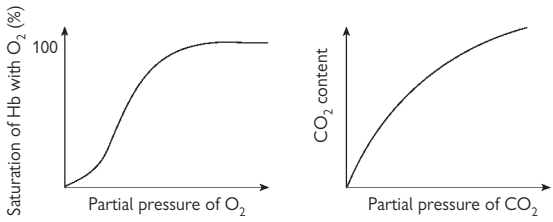


Fig. 5.1 The oxygen and carbon dioxide dissociation curves.

Lung function after heart surgery

Key facts

- Pulmonary dysfunction is maximal for 24h postoperatively, and normally improves over the week following surgery.
- Factors causing changes in pulmonary function can be classified as preoperative, intraoperative, and postoperative, resulting in ventilation-perfusion mismatch and intrapulmonary shunting.

Preoperative factors and respiratory function

- The vast majority of coronary patients have a history of ***smoking***.
- Chronic smokers almost invariably have evidence of chronic obstructive airway disease: exertional dyspnea, sputum production, or wheeze is associated with over 5× the rate of pulmonary complication, and 1/3 of these patients will have pulmonary complications requiring additional therapy.
- Stopping smoking 8 weeks or more before surgery has been shown to reduce some of the chronic changes in the bronchopulmonary tree and decrease the rate of pulmonary complications to <15%, but pulmonary complications are probably *increased* in patients who stop smoking <8 weeks before surgery.
- A ***recent respiratory infection*** is an indication for cancelling non-emergency cardiac surgery, as increased or purulent secretions can lead to atelectasis, pneumonia, bronchial plugging, and eventual occlusion of smaller endotracheal tubes requiring reintubation.
- Resting PaO_2 decreases linearly with age: ***older patients*** >75 years of age are at greater risk of pulmonary problems.
- ***Obesity*** increases the risk of postoperative atelectasis: in the morbidly obese sleep apnea and chronic atelectasis may lead to pulmonary hypertension.
- ***Pulmonary hypertension*** contributes to respiratory insufficiency post-op.

Intraoperative factors and respiratory function

- Both median sternotomy and thoracotomy are associated with ***decreased lung capacities*** postoperatively: ***left lower lobe collapse*** occurs in over 50% of on-pump coronary bypass patients.
- ***Selective lung ventilation*** on bypass for thoracotomy approaches appears to be associated with increased lung injury in the deflated lung.
- The increased ***pain*** associated with a thoracotomy further predisposes patients to ***atelectasis***: an epidural is a key part of improving respiratory function in the postoperative period.
- Either pleura may be opened during surgery: the evidence is mixed as to whether this increases the risk of postoperative atelectasis and ***effusions***, but increased handling of the lung is probably detrimental.
- Cardiopulmonary bypass results in increased intra and extravascular fluid and ***pulmonary edema***: pulmonary vascular resistance is increased, ***lung compliance is decreased***, atelectasis is increased, and ***intrapulmonary shunting*** is frequently increased as a result.
- ***Transfusion-related acute lung injury*** (TRALI) 📖 p149: blood and FFP post a higher risk than platelets and cryoprecipitate.
- Damage to either phrenic nerve when opening the chest, harvesting the internal mammary artery, creating a window in the pleura, or

using topical cooling results in ***phrenic nerve paresis*** which may be temporary or permanent (📖 p269).

Postoperative factors

72h postoperatively ***vital capacity may be less than 50%*** of preoperative values: it takes 2 weeks to return to normal. ***Total lung compliance*** falls by 25%, and ***dead space is increased***. Atelectasis is present in >50% of patients and is associated with increased intrapulmonary shunts, and raised PVR. Over 50% of patients consequently have a PaO_2 on room air of <7.5kPa (55mmHg).

- ***Narcotics, opioid analgesics and sedatives*** depress ventilation.
- ***Anticholinergics*** dilate the airways, increasing dead space.
- ***Inotropes and catecholamines*** with alpha activity increase PVR.
- ***Beta-blockers*** can cause bronchoconstriction.

Hemodynamic effects of ventilation

Reduced cardiac output

Increased intrathoracic pressure increases RAP, decreasing venous return and therefore preload. Right ventricular and therefore LV stroke volume and cardiac output are decreased. Induction and sedative agents (fentanyl, propofol) cause vasodilatation which reduces preload and cardiac output. The interventricular septum is displaced leftwards, decreasing LV compliance, LVEDV, and stroke volume.

Increased peripheral vascular resistance (PVR)

Air space dilatation and blood vessel compression caused by PEEP increases PVR by approximately 10%, most of which is at end-inspiration.

Decreased O_2 requirement

Although the work of breathing in normal circumstances utilizes <5% of the body's total O_2 requirements, in respiratory failure this may increase to 50%. Ventilation reduces the total O_2 requirement.

Negative inotropic effect

Hyperinflation occasionally causes reflex vasodilatation, bradycardia and negative inotropic effect.

Intubating hemodynamically unstable patients

As a result of the hemodynamic effects of induction and sedative agents, and ventilation described, intubating patients who are already hemodynamically compromised can precipitate cardiac arrest. This sometimes occurs postoperatively, but must also be anticipated in certain patients being anesthetized for surgery:

- Cardiac tamponade.
- Critical AS.
- Critical left main stenosis, emergency CABG.
- Pulmonary edema.
- Hypotension (hemorrhage, sepsis).

Initial ventilator settings for post-op patients

- FiO_2: 70–100%, mode: SIMV or pressure control.
- Tidal volume: 8–10mL/kg, respiratory rate: 8–14 breaths/min.
- Minute volume: 100–120mL/kg/min.
- PEEP: 5–7.5cmH_2O, pressure support: 8–10 cmH_2O.
- Inspiratory–expiratory ratio: 1:2, inspiratory flow rate: 30–60L/min.

Respiratory management

Key facts

- Pulmonary dysfunction is maximal for 24h postoperatively, and normally improves over the week following surgery.
- Most patients need mechanical ventilation for at least 4–6h.

Standard ventilator settings

Ventilation should improve the efficiency of gas exchange and reduce the work of breathing in the sedated, paralysed patient who is pre-oxygenated prior to leaving the OR with a FiO_2 of 100% to ensure oxygenation is adequate during any problems in transfer. The FiO_2 is slowly weaned to 30–40% as long as the PaO_2 stays >10.0kPa (75mmHg).

- Any patient can be ventilated safely using SIMV, but pressure control may be used initially: modes of ventilation are described on 📖 p158.
- A tidal volume of 8–10mL/kg helps prevent atelectasis: but excessive tidal volumes risk barotrauma, and can place a borderline LIMA graft under tension as it crosses the upper lobe of the left lung to the LAD.
- A respiratory rate of 8–14 breaths per minute is adequate for most patients, but patients with higher metabolic demands (shivering, sepsis) may benefit from higher rates which help prevent respiratory acidosis.
- 5 cmH_2O of PEEP is added to prevent atelectasis, dramatically reducing FiO_2 requirements.
- Patients with COPD benefit from lower respiratory rates and higher tidal volumes, allowing time for the longer expiratory phase, avoiding air trapping ('autoPEEP') which can cause high intrathoracic pressures, barotrauma, and reduced venous return causing hypotension.
- Increasing PEEP to 10cmH_2O helps to reverse atelectasis improving oxygenation, but can reduce venous return and MAPs in patients.
- Higher levels of PEEP are sometimes used in the hemodynamically stable, bleeding patient to tamponade chest wall bleeding: this probably works primarily by reducing MAPs.
- The inspiratory–expiratory (I:E) ratio is 1:2., allowing optimal peak inspiratory pressures and reducing the risk of air trapping: in selected patients that are difficult to oxygenate the intensivist or anesthesiologist will change or even reverse these ratios.
- Peak inflation pressures in the adult are usually <30 cmH_2O, and the usual inspiratory flow rate is 30L/min, higher peak inflation pressures occur in obesity, active resistance from the patient, or a mechanical obstruction to airflow such as bronchospasm, endobronchial intubation or mucus plugging with lobar collapse.

Pulse oximetry

Pulse oximetry allows non-invasive continuous measurement of SaO_2. Only available since 1990, it is now mandatory in every ventilated patient.

Principles

- Oxygenated (HbO_2) and deoxygenated hemoglobin (Hb) have different absorption spectra. HbO_2 absorbs less red light (wavelength 660nm), and more infrared light (wavelength 980nm) than Hb.

- The pulse oximeter probe consists of two light-emitting diodes (LEDs) emitting red and infrared light pulses and a photocell detector.
- The probe is placed on a well-perfused extremity, e.g., digit or pinna.
- It detects maximal (systole) and minimum absorption (diastole), and subtracts the constant, non-arterial component from the pulsatile (arterial) component: the ratio of the two wavelengths passing through the tissue is compared to an algorithm derived from experimental data to give a measure of the relative amounts of HbO and Hb (SaO_2).
- A pulse waveform is displayed with the SaO_2.

Sources of error

- Inaccurate at SaO_2 <70% as not calibrated in volunteers at these levels.
- Loss of the pulsatile component in hypoperfusion, peripheral vasoconstriction, narrow pulse pressure, non-pulsatile cardiopulmonary bypass and venous congestion leads to loss of readings.
- Carboxyhemoglobin (in heavy smokers and carbon monoxide poisoning) gives falsely high readings. Jaundiced patients, and patients with systemic methylene blue, give falsely low readings.
- Movement and electrical artifact, interference from direct, bright ambient light, and absorption by substances such as nail varnish and nicotine staining can distort measurements.

Capnography

Capnography is also mandatory in ventilated patients.

Principles

- Similar principles to pulse oximetry: different gases absorb different amounts of infrared light. Systems can be in-line (in the ET tube) or sampling (withdraw gas from the ventilator circuit and sample it at a remote site), and consist of an infrared LED and a photocell detector.
- The fraction of infrared light absorbed is compared against reference values before the trace is displayed: regular calibration against known CO_2 concentrations is essential.

Interpreting the trace

- Adequate ventilation is indicated by normal end-tidal CO_2. *No CO_2 trace* suggests ET tube displacement, tubing disconnection, CO_2 sampling tubing occlusion, esophageal intubation, or circulatory arrest.
- Low or falling end-tidal CO_2 indicates hyperventilation, PE or low cardiac output; high or rising end-tidal CO_2 indicates inadequate ventilation, and hypermetabolic states in which CO_2 production increases (sepsis, shivering) and rarely malignant hyperpyrexia.
- A slow upstroke and slowly rising plateau is seen in COPD.
- Inspiratory dips indicate that the patient is initiating breaths.

Respiratory support modes

Invasive methods

Intermittent positive pressure ventilation (IPPV); controlled mechanical ventilation (CMV); pressure control ventilation

IPPV or *CMV* is the mode commonly used during routine surgery. Raising airway pressure forces air into the lungs via an endotracheal or tracheostomy tube. Expiration occurs when airway pressures are allowed to fall to zero. This mode is poorly tolerated by the awake patient.

Positive end-expiratory pressure (PEEP)

Instead of allowing airway pressures to fall to zero, a small positive airway pressure is maintained throughout expiration (*PEEP*), preventing collapse of small airways and alveoli that occurs at the end of expiration. Functional residual capacity, shunts, lung compliance and PaO_2 are improved, and the work of breathing is reduced. High levels of PEEP (>10cmH_2O) may be necessary to reverse established atelectasis, but increase intrathoracic pressure, reducing venous return and cardiac output, which may reduce PaO_2. Barotrauma is a complication of high PEEP. PEEP as low as 5cmH_2O may cause hemodynamic compromise in the patient with poor LV. Physiological PEEP is normally provided by an intact glottis: patients with COPD sometimes purse their lips during expiration to increase physiological PEEP.

Synchronized intermittent mandatory ventilation (SIMV)

One variation of IPPV is SIMV, where positive airway pressure may be synchronized with patient-initiated breaths. Mandatory (machine initiated) breaths are given if no spontaneous breaths occur in a preset time, but do not occur at the same time as patient-initiated breaths. This mode is used in patients that are awake, but any patient may safely be ventilated on SIMV.

Mandatory minute ventilation (MMV)

Another variation of IPPV is MMV where the ventilator initiates breaths only if patient-initiated ventilation falls below a preset minute volume.

Pressure support (PS); assisted spontaneous breathing (ASB)

Patient-initiated breaths can be supported with a preset positive airway pressure of 6–20cmH_2O. The ventilator detects the drop in airway pressures as the patient generates negative intrathoracic pressures at the beginning of inspiration and assists air inflow with a positive airway pressure. This is lower than the preset inflationary pressures in IPPV, as the patient is making some inspiratory effort. At high levels of pressure support (>20cmH_2O) the patient, although controlling the timing and frequency of respiration, is effectively on IPPV. Pressure support of 8cmH_2O is usually the last stage of weaning prior to extubation.

Continuous positive airway pressure (CPAP)

In CPAP a standing airway pressure, continuous throughout all phases of respiration is applied via the ET tube. It is more commonly used in the extubated patient via a tight-fitting face mask (see 📖 p159).

Non-invasive methods

Intermittent negative pressure ventilation (INPV)

Virtually never used today, INPV involves placing the patient inside a tank ventilator sealed at the neck. Tank pressure is intermittently lowered, expanding the chest and lowering intrapleural pressure.

Non-invasive intermittent positive pressure ventilation (NIPPV)

This is IPPV delivered by face mask, or more commonly, nasal mask. The patient must be cooperative in order to understand how to synchronize their breathing with the ventilator. It is sometimes used in the tiring COPD patient where the primary concern is not recruitment of alveoli, but reducing the work of breathing, without resorting to intubation.

Continuous positive airway pressure (CPAP)

In CPAP a standing airway pressure, continuous throughout all phases of respiration, is applied via an endotracheal tube, tracheostomy, nasal mask or, more commonly, face mask to a spontaneously breathing patient. This results in a kind of non-invasive PEEP, where additional alveoli are recruited with the benefits described on 📖 p158. CPAP is a useful adjunct in the extubated patient with COPD, atelectasis, pulmonary edema or ARDS. It cannot produce ventilation by itself.

Biphasic positive pressure ventilation (BiPAP)

BiPAP is a solution to the problem of air trapping that can occur in patients, particularly those with COPD, on CPAP. Airway pressure is cycled at preset rates between high and low levels.

High-flow (Venturi) face mask (fixed performance)

The key part of the Venturi facemask is the Venturi valve which draws in an amount of air through calibrated inlets, which is mixed with oxygen flowing into the valve, before entering the mask. More mixed air (up to 30L/min) is delivered to the mask than the patient can use: the excess escapes though holes in the mask. The FiO_2 is set by the choice of valve, not by the patient's breathing pattern (hence the term fixed performance). The maximum FiO_2 that can be delivered by a Venturi mask is about 60%. There is a minimum flow rate of O_2 for each Venturi.

Low flow (Hudson) face mask (variable performance)

Oxygen flows at a set rate (e.g., 2L/min) into the mask. It is diluted by air drawn into the mask, which depends on the patient's minute volumes which can range from 5–30L/min. The FiO_2 achieved thus depends primarily on the patient, and the delivery system should therefore not be used when accurate control of FiO_2 is required. The maximum FiO_2 that can reliably be delivered in this fashion is about 30%. Use of a non-rebreathing mask and reservoir bag, into which 100% O_2 is drawn during expiration and inhaled during inspiration, increases FiO_2 to up to 60%. The reservoir bag must be filled with O_2 before the patient uses it.

Nasal prongs

Nasal prongs deliver a FiO_2 that is determined primarily by the patient, as in a low-flow mask, but they are less obtrusive, allowing the patient to expectorate and eat. They increase tracheal FiO_2 to barely more than room air levels, particularly if the patient breathes through their mouth.

Poor gases in ventilated patients

Key facts

- Most patients recover respiratory function and are weaned from ventilatory support within 24h.
- 5% require additional and prolonged support, often due to multiple risk factors for postoperative respiratory dysfunction (📖 p155).
- Common causes of respiratory insufficiency in ventilated patients are listed on 📖 pp162–163 together with management.

'Poor gases' or respiratory insufficiency

These depend on the context: on FiO_2 of 100% O_2 a 'normal' patient should be able to achieve PaO_2 >400mmHg. Generally the following definitions apply:

- ***Hypoxia:*** PaO_2 <10.5kPa (80mmHg). SaO_2 >90%.
- ***Hypercapnia:*** $PaCO_2$ >6.5kPa (50mmHg).
- ***Hypocapnia:*** $PaCO_2$ <3.5kPa (25mmHg).

Basic assessment and management

- *Look at the patient*: is the ET tube obviously displaced, is the patient biting the ET tube or fighting the ventilator, is chest expansion asymmetrical, can you see or feel subcutaneous emphysema? What is the hemodynamic status of the patient? Poor pulse oximeter readings may reflect low cardiac output and peripheral vasoconstriction.
- *Auscultate the chest* for bilateral breath sounds, and the stomach: is the ET tube advanced too far past the carina resulting, in single lung ventilation, or misplaced in the esophagus?
- Compare SaO_2 from pulse oximeter with SaO_2 from new blood gas, and PaO_2: pulse oximetry may give falsely low reading (📖 p157).
- *Increase the FiO_2* to 100% O_2 which can be given for short periods without causing oxygen toxicity, and usually reverses hypoxia.
- If the O_2 sats do not start quickly rising to >95%, *disconnect the patient from the ventilator and manually ventilate* with 100% O_2. This allows you to:
  - Identify if the ventilator is the source of the problem.
  - Perform recruitment maneuvers to treat atelectasis.
  - Reassess whether both sides of chest moving symmetrically.
  - Assess the patient's airway compliance directly: reasons a patient may be 'difficult to bag' include kinked or blocked tubing, including biting on ET tube, ET tube up against carina or in one main bronchus, bronchospasm, pulmonary edema, tension pneumothorax.
  - Suction the patient to look for and treat mucus plugs or bleeding: suctioning can cause rapid desaturation in patient dependent on PEEP, as well as coughing and agitation.
- Is the patient biting on the ET tube? If they are uncooperative they are not ready for weaning and should be resedated: *sedation or paralysis* improves gas exchange by improving efficiency of ventilation and reducing the energy demands of spontaneous ventilation.

- *Treat bronchospasm with* nebulized albuterol 5mg.
- *Get a CXR.* Assess the position of the ET tube. It should be at least 2cm above the carina—pull it back 2cm if this is not the case.
- Look for *pneumothoraces* or *pleural collections* which should be treated with a formal chest drain, and *atelectasis* or *mucus plug* (complete whiteout lobe or hemithorax—can look similar to hemothorax 📖 p14, Fig. 1.2c) which should be managed with intermittent ET tube suction and increased PEEP.
- Change the *ventilator settings* (see Box 5.4).
- If you suspect mucus plugging (copious secretions, CXR showing lobar collapse or complete whiteout hemithorax) perform fiberoptic bronchoscopy at bedside (📖 pp654–655).

Box 5.4 Impact of ventilator settings on PaO_2 and $PaCO_2$

- FiO_2: 0.7–1.0 (↑ to increase PaO_2).
- Tidal volume: 8–12mL/kg (↑ to reduce $PaCO_2$).
- Respiratory rate: 8–14 breaths/min (↑ to reduce $PaCO_2$).
- Minute volume: 100–120mL/kg/min (↑ to reduce $PaCO_2$).
- PEEP: 5–10cmH_2O (↑ to increase PaO_2).
- Pressure support: 8–10cmH_2O (↑ to increase PaO_2).
- Inspiratory–expiratory ratio: 1:2.
- Inspiratory flow rate: 30–60L/min (↑ to increase PaO_2).

Respiratory failure on ventilator

Management

Basic assessment and management of acute respiratory insufficiency is described on 📖 pp160–161. Treatments of the various causes of respiratory failure listed in Box 5.5 are outlined here.

Box 5.5 Etiology of respiratory failure

- ***Inadequate O_2 delivery to patient:*** ventilator malfunction, incorrect ventilator settings, endotracheal tube problems.
- ***Pulmonary dysfunction:*** pneumothorax, hemothorax, pulmonary edema, atelectasis, consolidation, bronchospasm, COPD, ARDS.
- ***Inadequate O_2 delivery to tissues:*** low cardiac output, anemia.
- ↑ *O_2 demand*: Shivering, sepsis, pyrexia, hyperalimentation, drugs.

Ventilator malfunction: manually ventilate with 100% O_2, which should improve oxygenation immediately and also give an indication of pulmonary compliance. If oxygenation does not improve seek other causes.

Incorrect ventilator settings: inadequate FiO_2, PEEP or pressure support will result in low PaO_2. High $PaCO_2$ may be caused by too low a tidal volume and/or respiratory rate. Check inspiratory flow rate, and increase in COPD. Standard ventilator settings are described on 📖 p155.

ET tube problem: is the tube occluded by sputum plugs, kinking or patient biting, or misplaced in the esophagus, larynx or main bronchus; is there a cuff leak? Listening to both lungs and the epigastrium should confirm placement. A CXR should show the end of the ET tube 2cm above the carina. Suction the pharynx, deflate the cuff, withdraw the tube as appropriate, re-inflate, and look for symmetrical chest expansion. Suction the tube. Be aware in long-term intubated patients that the lumen of the tube can be reduced to little more than the diameter of the fine-bore suction tube by solidified secretions. A Guedel airway stops the patient biting the ET tube.

Low cardiac output: optimize cardiac output (📖 pp184–185).

Anaemia: an Hb >8.0g/dl is adequate for most patients post cardiac surgery. Reduced blood viscosity improves perfusion of microvasculature and leads to reduced risk of graft occlusion. In patients with borderline SaO_2 such as those with COPD, increasing Hb has a large impact in oxygen delivery, seen in increased mixed venous O_2 saturations.

Pneumothorax: increasing $PaCO_2$ and decreasing compliance suggest a pneumothorax, which occurs in approximately 1% of ventilated patients post cardiac surgery. Causes include trauma during central line insertion, median sternotomy, or barotrauma. Breath sounds are audible on the side of the pneumothorax until it is >50%. Get a CXR. If you suspect tension pneumothorax, place an immediate intercostal drain (📖 p248). A pneumothorax is unlikely if there is a patent pleural drain.

Hemothorax: increasing $PaCO_2$, falling Hb, falling CVP and pleural drainage suggests hemothorax which can be confirmed by CXR. Even if the pleura was not opened during surgery, trauma to neck vessels during central venous catheterization or to internal mammary vessels at chest closure may cause a hemothorax. Insert a chest drain (📖 p248).

Pulmonary edema: cardiogenic or non-cardiogenic. Treat the cause. Loop diuretics are an important temporizing measure (📖 p209). Failure of mitral valve repair, paraprosthetic leaks, conduit occlusion, cardiac tamponade and endocarditis are causes of acute postoperative pulmonary edema. Non-cardiogenic causes include fluid overload and renal failure.

Atelectasis: prevent with PEEP or CPAP, and treat with physiotherapy.

Consolidation: prevent with early mobilization, prophylactic antibiotics. Do not deflate the cuff on a tracheostomy tube until speech and language assessment is satisfactory. Treat with organism-specific antibiotics. Aspiration pneumonia, suggested by a history of aspiration of vomitus or NG feed while the ET cuff deflated, and bibasal shadowing on CXR, should be treated with cefuroxime 750mg and metronidazole 500mg iv 8-hourly. Prevent mucus plugging by giving humidified oxygen or saline nebulizers. Treat mucus plugs with physiotherapy, suctioning and if necessary bronchial toilet via bronchoscopy.

Bronchospasm and COPD: the ET tube touching the carina, and suctioning can precipitate bronchospasm, as can β-blockers. It is characterized by wheeze, high inflationary pressures, low PaO_2 and high $PaCO_2$. Treat with nebulized albuterol 5mg prn (📖 p161). Aminophylline infusions are used in refractory cases. Withdraw the ET tube if indicated.

ARDS: diagnostic criteria: (1) progressive hypoxemia (PaO_2 (mmHg): FiO_2 (fraction not percentage) ratio <200). (2) Predisposing cause (cardiopulmonary bypass, pancreatitis, SIRS, aspiration of gastric contents, pulmonary trauma, fat embolism, smoke inhalation), without pre-existing lung disease. (3) Bilateral infiltrates on CXR. (4) Reduced pulmonary compliance. (5) Normal LVEDP—excluding cardiogenic pulmonary edema. Treatment: There is no specific therapy for ARDS. Management is supportive. PEEP maintains lung volumes, decreases shunt and reduces V/Q mismatch. Lowest FiO_2 consistent with O_2 delivery. Nitric oxide and ventilation prone are occasionally used.

Shivering: increases O_2 demand and CO_2 production. Warm patient with warming blanket and use a warmer for iv infusions. Paralyze, and consider improving peripheral perfusion with vasoconstrictors (📖 p143). Pethidine 25mg iv has an effective antispasmodic effect.

Sepsis: increases O_2 demand and CO_2 production. Culture everything: blood, sputum, urine, wounds, stools, line tips. Treat the cause. Consider hemofiltration to remove cytokines in hyperpyrexia.

Drugs: drugs which exacerbate pulmonary shunting by preventing hypoxic pulmonary vasoconstriction (NTG, SNP) reduce PaO_2. Excessive or rapid administration of sodium bicarbonate leads to ↑CO_2 production (📖 p153).

Weaning from ventilation

Key facts

- Weaning is normally completed within 6–12h of surgery.
- Patients with poor ventricular function or pulmonary hypertension may take longer as will patients with surgical problems, e.g., bleeding.
- Waking causes vasoconstriction, increased afterload, and tachycardia, thus increasing myocardial oxygen demand.

The weaning process

Short-term ventilated patients

Withhold or minimize narcotics. Reduce the ventilator rate by 2 breaths per hour. At a ventilator rate of 4 with FiO_2 <0.6,check ABGs. If these remain satisfactory the patient can be weaned onto CPAP of $5cmH_2O$, or onto pressure support ventilation of $8cmH_2O$ which is gradually reduced, and then onto a T-piece. ABGs are checked after 45min. Proceed to extubation if the patient meets the extubation criteria.

Long-term ventilated patients

The same process described for short-term ventilated patients occurs over days to weeks (📖 p281). CPAP may be delivered via the tracheostomy, for several hours in between shortening periods of SIMV.

Extubation

All the equipment necessary for emergency re-intubation should be to hand and checked. Have post-extubation oxygen mask and connection tubing to hand, checked and ready. Grade III or IV intubation patients (check the anesthetic record) should be extubated with an anesthesiologist present. The patient should satisfy the criteria in Box 5.6.

- Position the patient upright at 45°.
- Suction the ET tube, mouth, and pharynx. Treat any resultant hypoxia.
- Deflate the ET tube cuff, remove the tube and suction mouth again.
- Apply face mask with 8L/min O_2 and 4L/min via nasal cannulas.
- Patient should be carefully observed for 20min following extubation with continuous pulse oximetry, and ABGs after 10–15min.
- If the post-extubation gas is good, begin to titrate FiO_2 downwards.
- Keep patient nil by mouth for 4h to allow any vocal cord irritation to settle, then start oral hydration with sips of water, and monitor.

Postextubation management

Oxygen delivery may be compromised by respiratory dysfunction, fluid overload and hemodynamic compromise (📖 p162). ***Give 4–6L humidified O_2 by face mask for several hours***. Most patients rapidly progress to lower FiO_2, O_2 delivery via nasal cannulas and then room air. Breathing is compromised by pain and decreased chest wall compliance. Shallow breathing, splinting, and poor cough predispose to atelectasis. ***Ensure adequate analgesia, and mobilization, and educate the patient***. A cough cushion held against the chest when coughing helps reduce pain and sternal movement (the abstraction force of pectoralis major is reduced).

Box 5.6 Criteria for weaning and extubation

Respiratory criteria

Ventilation and gas exchange

- *Criteria for weaning*: PaO_2 >10kPa on FiO_2 30%, $PaCO_2$ <7kPa; arterial pH >7.35 unless cause of acidosis known and improving; PEEP <5cmH_2O; awake and initiating breaths.
- *Criteria for extubation*: Negative inspiratory pressure >−20cmH_2O, vital capacity >10mL/kg, tidal volumes >5mL/kg, resting minute volume >8L, respiratory rate 10–25, PEEP < 5cmH_2O.

Safe airway: adequate cough for sputum and secretion clearance.
Lung compliance: compliance should be > 25ml/cmH_2O.
Rapid shallow breathing index (frequency/tidal volume (L) <100: patients who fail tend to breathe rapidly and shallowly, so RSBI >100.

Hemodynamic criteria

The EKG should be satisfactory. Cardiac failure with pulmonary edema will impair gas exchange, and may worsen with extubation. High or escalating inotropic requirements mean that myocardial function may be compromised by weaning too soon.

Neurological criteria

Patient should be awake, cooperative, with an adequate gag reflex.

Surgical criteria

Adequate hemostasis: mediastinal drainage <1mL/kg/h. Adequate pain relief. No further surgical procedures planned.

Additional factors

Multiorgan failure, renal failure with fluid overload, ARDS are relative contraindications to weaning. Patients should be normothermic. A base deficit is not a contraindication if cause known and treatable.

Common weaning problems

Drowsiness

Patients may show good respiratory parameters when awake but become apneic, bradycardic, and hypotensive when they sleep. Titrate opioids carefully. Naloxone 0.2–0.4mg iv prn provides emergency reversal.

'Fighting the ventilator'

These patients are unable to breath in synchrony with delivered breaths. Awake patients may bite the ET tube, causing hypoxia. Coughing, gagging, and fighting the ventilator result in raised intrathoracic pressures, which lead to sudden rises in CVP and falls in MAPs which mimic cardiac tamponade. If the patient is confused, with very poor cardiopulmonary parameters, resedate, ventilate, and paralyze if no improvement. In uncooperative but otherwise stable patients early extubation may be beneficial.

Failure to wean from ventilation

In addition to the causes of respiratory failure listed on 📖 p162, late failure to wean from ventilation is caused by myocardial ischemia, valvular heart disease, sternal non-union, stroke, and critical illness neuropathy.

Tracheostomy

Key facts

- A tracheostomy is normally performed electively, if the patient proves difficult to wean from ventilation after 7–10 days.
- It can be done as a formal surgical procedure (📖 p664), as a percutaneous ICU procedure (see 📖 p166), or on the HDU as a minitracheostomy (📖 p249) for tracheal toilet (not used for ventilation).

Indications

- Airway protection.
- Prolonged ventilatory support likely—facilitates gradual weaning.
- Tracheobronchial toilet (particularly in pneumonectomy patients).

Contraindications

- Local sepsis and hemodynamic instability are relative contraindications.

Technique of percutaneous tracheostomy

Percutaneous tracheostomy is based on a modified Seldinger technique. It is usually performed in selected patients (thin patients with long necks and good extension) by an intensivist. In about 50% of units a second intensivist routinely monitors the procedure via a fiberoptic bronchoscope as it is easy to perforate the posterior wall of the trachea.

- The patient is preoxygenated.
- With the patient positioned with the neck straight and extended, and the cuff on the ET tube deflated, the midline is carefully identified: this is critical to prevent malpositioning of the tracheostomy tube in the soft tissues adjacent to the trachea.
- The isthmus of the thyroid overlies the 2nd–4th tracheal rings: an approach higher than this avoids the isthmus but is associated with greater incidence of tracheal stenosis.
- Most practitioners elect to enter the trachea below the 2nd or 3rd ring.
- After infiltrating carefully with 10mL 1% lidocaine a skin incision is made over the 2nd ring in the midline.
- A hollow needle is inserted into the trachea, and the guidewire passed through it.
- Progressively larger dilators enlarge the diameter of the hole until it will accept a definitive tracheostomy tube; alternatively, a specially designed dilating forceps can be passed over the guidewire.
- At this point the indwelling ET tube is withdrawn slowly until the tracheostomy tube can be fed over the guidewire into position.
- Secure the tube by suturing it to the skin as well as tying it in place.
- The cuff is inflated, the tube is connected to a bag reservoir and the chest checked for symmetrical inflation while the patient is manually ventilated and suctioned if necessary.

Complications

Percutaneous tracheostomy has a complication rate of 5–7%, lower than that for surgical tracheostomy. Use of fiber optic bronchoscopy does not

affect the rate of complications, but appears to protect against the more serious complications. Complications are immediate, early and late.

Hemodynamic instability: this is common due to wide variation in the level of autonomic stimulation during the procedure.

Bleeding: bleeding during or immediately after insertion of the tracheostomy tube is usually due to bleeding from thyroid veins. Correct any coagulopathy. If bleeding does not settle with pressure, surgical exploration is indicated. Late bleeding from around the tube suggests erosion of the tube into the thyroid gland or neck vessels.

Failure to intubate the trachea: creation of a false passage can be avoided by careful positioning of the patient and constant reference to the surface landmarks of the midline. Difficulty ventilating the patient, absence of an end-tidal CO_2 trace, and surgical emphysema all suggest failure to intubate the trachea. Loss of an airway can be avoided by withdrawing the ET tube only as far as necessary to allow insertion of the tracheostomy tube, and leaving the ET tube in place until the tracheostomy tube has been checked and secured.

Esophageal perforation: simultaneous fibreoptic bronchoscopy is designed to avoid this by preventing laceration of the posterior wall of the trachea. Leaving the indwelling ET tube in situ while performing tracheostomy also helps protect against inadvertent damage.

Barotrauma: pneumothorax, pneumomediastinum, and surgical emphysema result from direct injury to the apex of the lung, excessive negative intrathoracic pressures in awake distressed patients, or excessive positive pressure manual ventilation. Management of pneumothorax is described on 📖 p162, pneumomediastinum on 📖 pp731, 790.

Tracheoinnominate fistula (TIF): Catastrophic late bleeding suggests formation of a tracheoinnominate fistula. It is often preceded by a small herald bleed, and may be associated with a pulsating tracheostomy. To gain emergency control compress the innominate artery against the sternum by removing the tracheostomy and placing a finger inside the stoma. ET intubation must be carried out with a cuffed tube to achieve airway control. Repair is via median sternotomy (📖 p665).

Tracheoesophageal fistula (TEF): erosion of the tube cuff through the membranous trachea is less common now that low-pressure cuffs and lighter weight connection tubing are in wide use. Repair is usually delayed until the patient no longer requires ventilatory support. Placing the tracheostomy tube so that the cuff rests below the fistula provides airway protection from gastric contents.

Infection: positive cultures from a tracheostomy tube represent clinical infection and should be treated. Treat cellulitis with antibiotics.

Inadvertent extubation and loss of airway control: in tracheostomies over 7 days old a tract makes re-insertion of the tracheostomy tube relatively easy. Where there is no tract get anesthetic help to attempt oral intubation. If this fails, cricothyroidotomy should be performed.

Tracheal stenosis and granulomas: these can occur at the site of the stoma or the site of the cuff. Management is described on 📖 p751.

Acid–base balance

Key facts

- The pH of arterial blood is maintained at 7.35–7.45. Normal functioning of the body's complex enzyme systems depends on this stability.
- Derangements are due to *respiratory and/or metabolic* dysfunction. Compensatory mechanisms are also divided into metabolic and respiratory. The true clinical picture is usually mixed.

Diagnosing acid–base abnormalities

pH <7.35 is an acidosis: pH>7.45 is an alkalosis

- Look at the pH—is there an acidosis or an alkalosis?
- Look at the $PaCO_2$—is there a change *in keeping with* the pH derangement? If so, the derangement is a respiratory one.
- Look at the base deficit (or anion gap). This will tell you if there is a metabolic derangement: if pH is normal it is fully compensated.
- The *Flenley nomogram* (Fig. 5.2, 📖 p171) is a useful diagnostic aid where mixed metabolic and respiratory derangements are present.

Acid–base physiology

Acids

An *acid* is a substance that can donate a hydrogen ion or proton (H^+) to another substance. Some molecules are stronger donors than others. Acids produced by the body are conventionally divided into *metabolic or fixed acids*, e.g., lactic acid, and *respiratory or volatile acids*, i.e., CO_2 which forms carbonic acid in solution(📖 p153). The production of respiratory acids is many times greater than the production of metabolic acid. Excretion of respiratory acid occurs rapidly through the lungs, whereas excretion of metabolic acid occurs more slowly through the kidneys.

Bases

A *base* can be loosely defined as a substance that accepts a hydrogen ion. Some molecules have higher affinity for H^+ and are stronger bases than others. The most important base is sodium bicarbonate ($NaHCO_3$).

Buffers

Buffers are usually made up of a weak acid, the salt of that acid, and a strong base, and can bind or release H^+ according to the pH, helping to maintain pH stability. The body has an enormous and immediate buffering capacity, which conceals huge changes in H^+ production. This is vital because small changes in pH lead to big changes in the amounts of ionized substances in solution, and speed of enzyme-catalysed reactions. The most important buffers in blood are hemoglobin (Hb) and bicarbonate (HCO_3^-), which is produced and excreted by the kidneys. CO_2 carriage in blood and its removal in the lungs is the most significant factor in acid–base homeostasis. The strength of a buffer or pKa (in this case for HCO_3^-) is described by the *Henderson–Hasselbach equation*:

$$pH = pK_A + \log([HCO_3^-]/[H_2CO_3])$$

Base excess, base deficit, and anion gap

These are derived numbers, calculated by blood-gas analysers, quantifying changes in metabolic or fixed acids but, because they depend on several assumptions, they do not always reflect the true acid–base balance.

- ***Base excess*** is defined as the millimoles per litre of *acid* required to titrate the blood pH back to pH 7.4, if the pCO_2 were normal.
- ***Base deficit (negative base excess)*** is defined as the millimoles per litre of *base* to titrate the blood pH back to pH 7.4, if the pCO_2 were normal.
- A base deficit is negative and a base excess is positive by convention. Normal values are −2mmol/L to 2mmol/L. A base deficit greater than this (e.g., −6mmol) indicates a metabolic acidosis.
- The ***anion gap*** is the difference between measured cations and measured anions ($= [K^+ + Na^+] - [Cl^- + HCO_3^-]$). This is made up of metabolic acids: ketones, lactate and phosphates. The anion gap is normally 8–16mmol/L: an increase in anion gap indicates a metabolic acidosis.

Metabolic acidosis

Uncompensated: ↓pH, ↔pCO_2, ↓HCO_3^-,
Compensated: ↓ ↔pH, ↓pCO_2, ↓HCO_3^-.

Metabolic acidosis due to increased metabolic acids (increased anion gap)

- Lactic acid (global and/or regional hypoperfusion, hypoxia, sepsis, hepatic failure as the liver normally metabolizes lactate).
- Uric acid (renal failure).
- Ketones (diabetic ketoacidosis, alcoholic and starvation ketoacidosis).
- Drugs/toxins (salicylates, SNP overdose).

Due to loss of bicarbonate or hyperchloremia (normal anion gap)

- Renal tubular acidosis (loss of bicarbonate).
- Diarrhea, high-output ileostomy (loss of bicarbonate).
- Pancreatic fistulae (loss of bicarbonate).
- Hyperchloremic acidosis (excessive saline administration).

Metabolic alkalosis

- Uncompensated: ↑ pH, ↔pCO_2, ↑HCO_3^-.
- Compensated: ↑ ↔pH, ↑pCO_2, ↑HCO_3^-.
- Loss of H^+ from gut (vomiting, NG tube suction).
- Renal loss of H^+ (diuretics), ↑reabsorption of HCO_3 (hypochloremia).
- Administration of base ($NaHCO_3$, citrate in blood transfusions).

Respiratory acidosis

- Uncompensated: ↓pH, ↑pCO_2, ↔HCO_3^-.
- Compensated: ↓ ↔pH, ↑pCO_2, ↑HCO_3^-.
- Any cause of respiratory failure, or hypoventilation (📖 pp162–163).
- Increased production of CO_2, e.g., sepsis, malignant hyperpyrexia.
- Rebreathing CO_2 (circuit misconnections, soda lime exhaustion).

Respiratory alkalosis

- Uncompensated: ↑pH, ↓pCO_2, ↔HCO_3^-.
- Compensated: ↔↑pH, ↓pCO_2, ↓HCO_3^-.
- Hyperventilation: deliberate, inadvertent, or in non-ventilated patients caused by stroke, anxiety, PE, pneumonia, asthma, pulmonary edema.

The acidotic patient

Key facts

- Acid–base disturbances not only indicate profound hemodynamic, respiratory, and metabolic dysfunction, but also exacerbate myocardial and pulmonary dysfunction.
- It is usually easy and important to distinguish between a metabolic and a respiratory acidosis (📖 pp168–169).

Common causes of acidosis after cardiac surgery

See 📖 p169. Low cardiac output states predispose to poor peripheral perfusion, which may result in lactic acidosis. This is exacerbated by vasoconstriction caused by hypothermia, reflex response to hypotension, or vasoconstrictors such as norepinephrine. Respiratory acidosis is common in obtunded patients without ventilatory support (📖 p281). Other important causes of metabolic acidosis after surgery include diabetic ketoacidosis, regional ischemia including mesenteric and peripheral limb ischemia, sepsis, renal failure, and hyperchloremic acidosis.

Basic investigation and management

Review the diagnostic criteria outlined on 📖 pp168–169.

- In addition to pH, you need to know the pCO_2.
- Base excess or anion gap (📖 p169), and blood glucose and lactate will help you diagnose and treat the causes more accurately.
- Treat respiratory acidosis by increasing respiratory rate and tidal volumes of the ventilator settings: aim for a minute volume of 100mL/kg/min.
- The weaning process should be reversed in the patient that develops a respiratory acidosis while being weaned from ventilation.
- It is sometimes reasonable to support hypoxic extubated patients with CPAP to reduce the work of breathing, but careful attention should be paid to ruling out other reversible causes of hypoxia in addition to atelectasis, and deciding whether in fact they would benefit more from elective rather than emergency reintubation.
- Check the *blood sugar*, and dipstick urine: start an iv insulin sliding scale if serum glucose >140g/dL (8.0mmol/L), or if there are ketones in the urine: many units routinely run patients on sliding scales to achieve tight glycemic control (blood glucose between 80–110mg/dL).
- *Treat peripheral hypoperfusion* by using a warming blanket if necessary to achieve core temperatures 36.5–37.5°C.
- Optimize preload by filling to maintain CVP 14–16mmHg.
- Optimize cardiac output: dopamine 250mcg in 250mL 5% dextrose at 3–6.0 micrograms/kg/min is a good first choice.
- Dopexamine and dobutamine cause peripheral and splanchnic vasodilatation which helps prevent end organ hypoperfusion.
- If the patient is still vasoconstricted with cold peripheries start a NTG infusion (5–200mcg/min).
- *Optimize urine output* by giving loop diuretics in the adequately filled patient to maintain a diuresis >1mL/kg/h, and inotropes if indicated.

- *Give sodium bicarbonate* 25–50mL 8.4% iv (📖 p163). There are situations where the underlying cause of acidosis cannot immediately be treated (off-pump patients who develop a metabolic acidosis possibly as a result of hypoperfusion in theatre, patients requiring high doses of epinephrine or norepinephrine, and patients with sepsis, renal failure and end organ ischemia. Sodium bicarbonate helps to normalize pH temporarily but will increase $PaCO_2$: ventilation must be adequate.
- A deteriorating base deficit, refractory to medical management, in the patient with renal failure is an indication for urgent *hemodialysis*.
- Look for and investigate signs of regional ischemia (📖 p185).

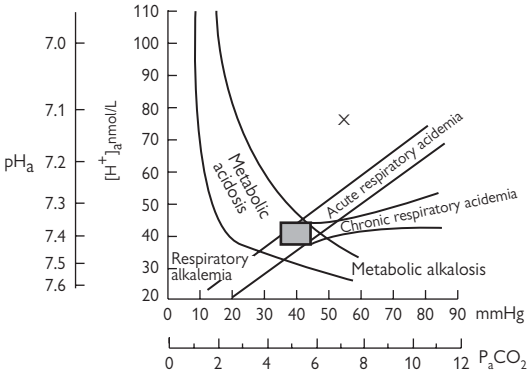


Fig. 5.2 The Flenley nomogram.

Arterial cannulas

Key facts

- All cardiac surgical patients should have an arterial catheter intraoperatively and for the first 12–24h postoperatively.
- Anesthesiologists insert a radial or axillary artery cannula in the non-dominant arm prior to induction.
- Intra-arterial pressure monitoring allows continuous and more precise measurement and control of BP than non-invasive cuff measurement in an environment where rapid changes are commonplace. It also facilitates frequent sampling of ABGs.

Contraindications

Absolute: infection at the site of insertion, distal limb ischemia.
Relative: coagulopathy, proximal obstruction, surgical considerations.

Allen's test demonstrates a patent palmar collateral circulation: the patient clenches a fist to exclude blood from the palm, while you firmly compress both ulnar and radial pulses. The patient then opens his palm, which should be blanched. You release the ulnar compression while still occluding the radial pulse: the palm becomes pink in <5s if there is good collateral supply from the ulnar artery (📖 p7). About 3% of people do not have a collateral palmar supply, and hand ischemia is a potential risk if the radial artery is cannulated. Although the radial artery is most commonly cannulated because of accessibility, pressure readings are more accurate from axillary and femoral artery, especially after long cases when radial line becomes damped by peripheral vasoconstriction.

The arterial waveform

(See Fig. 5.3.)

- The MAP is the average pressure over 1 cardiac cycle. As you can see from the waveform, systole is shorter than diastole, hence the equation MAP = DP + (SP − DP)/3.
- The slope of the upstroke reflects contractility (dP/dt).
- The slope of the diastolic decay reflects resistance to outflow: a long gentle slope is seen in vasoconstriction.
- The stroke volume and CO can be calculated by measuring the area from the beginning of the upstroke to the dicrotic notch.

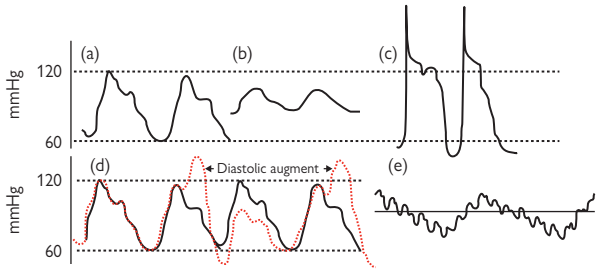


Fig. 5.3 The arterial waveform. (a) Normal. (b) Underdamped. (c) Overdamped. (d) IABP 1:2. (e) Respiratory variation.

Complications

Ischemia, thrombosis, distal embolism, bleeding, damage to radial and median nerve. Inadvertent intra-arterial injection of drugs.

Interpretation and common problems

Low-amplitude trace

This may be due to *poor cardiac output*: palpate a major pulse and ensure a non-invasive cuff measurement is obtained at 30s intervals from the opposite arm. If cardiac output is satisfactory exclude *occlusion* by flushing with 5mL heparinized saline and check that it is easy to aspirate. Ensure the wrist is extended (altering the position of the wrist may help get a good trace if the cannula is *against the vessel wall*). Check cannula and connecting tubing are not *kinked* and that all three-way taps between the patient and transducer are open. *Damping* is caused by cannulas that are too small, excessive lengths of compliant connection tubing, excessive use of three-way taps, air bubbles, and blood clots.

Exaggerated respiratory variation ('respiratory swing')

This suggests that the patient is hypovolemic, in respiratory distress, or cardiac tamponade (📖 p144).

Problems related to other sites

Brachial artery thrombosis has particularly severe consequences. Femoral artery cannulas are more prone to *infection.* Thrombosis has potentially very severe sequelae. A longer catheter is required which may result in *damping*. It is *less accessible* intraoperatively. Dorsails pedis artery cannula pressure is 10–20mmHg higher than in the central circulation.

Intra-arterial injection

This is a serious mishap. The awake patient will complain of pain in the affected hand. Early signs include pallor, mottling, and cyanosis due to arterial spasm. Late signs included trophic skin changes, ulceration, and edema. Gangrene or contractures are a late feature of severe cases.

The management depends on the injectate:

- Leave the cannula in place.
- Inject 1000U heparin down cannula to reduce risk of thrombosis.
- Inject warm normal saline down cannula to dilute substance.
- Dexamethasone 8mg iv may reduce arterial edema.
- Maintenance anticoagulation for 7–14 days.
- Discuss with plastic surgeons early.
- Antispasmodics such as papaverine may have a role: theatres will have papaverine as it is used topically on harvested arterial conduit intraoperatively to prevent vasospasm. Topical NTG patches may help.

Insertion

Techniques of arterial cannulation are described on 📖 p244.

Removal

Arterial cannulas are routinely removed 24–48h after surgery if the patient is hemodynamically stable, off inotropic support, and has satisfactory ABGs. Firm pressure should be applied to the insertion site for 5min after removal to prevent hematoma formation.

Central venous catheters

Key facts

- All cardiac surgical patients should have central venous pressure monitoring intraoperatively and for the first 12–24h postoperatively.
- Anesthesiologists insert a central line into the right internal jugular vein immediately after intubation.
- Central venous pressure cannulas not only allow continual RAP measurements which are used to guide filling status and assess RV function but also act as an:
  - Infusion port for some drugs which cannot be given peripherally.
  - Insertion port for pulmonary artery catheter.
  - Insertion port for transvenous pacing wires.
  - Infusion port for TPN.
  - Large-bore side arm for rapid infusion volume if required.

Cannulas can be single or multilumen, sheaths (for insertion of PA catheters and pacing wires), tunnelled, or long lines.

Contraindications

- *Absolute*: SVC syndrome, infection at the site of insertion.
- *Relative*: coagulopathy. Undrained contralateral pneumothorax. Uncooperative patient. DVT of the head and neck vessels may make insertion difficult. Patients with ASDs and VSDs are at risk of CVA from air emboli caused by poor technique.

The central venous waveform

(See Fig. 5.4.)

- The waveform is composed of three upstrokes (the 'a', 'c', and 'v' waves) and two descents (the 'x' and 'y' descent).
- The 'a' wave is due to atrial systole.
- The 'v' wave is due to venous return filling the RA before the tricuspid valve opens.
- The 'c' wave is due to bulging of the closed tricuspid valve into the RA at the start of isovolumetric ventricular contraction.
- The 'x' descent occurs in atrial diastole.
- The 'y' descent occurs in ventricular diastole as the tricuspid valve opens.

See 🕮 p175 for conditions that cause changes to the normal waveform.



Fig. 5.4 The normal pressure waveform.

Complications

Immediate

Damage to nearby structures (carotid artery puncture, pneumothorax, hemothorax, chylothorax, brachial plexus injury, arrhythmias) air embolism, loss of guidewire into right side of heart.

Late

Sepsis, thromboembolism, AV fistula formation.

Common problems

Poor trace

This may be caused by arrhythmias. Check the EKG. A damped waveform can be caused by excessive lengths of compliant connection tubing, three-way taps, air bubbles, and blood clots. Check that all the three-way taps between patient and transducer are open, that the tubing is not *kinked*, and that no infusions are running into the central line. Check that the central line is still secured in its original place and that you can freely aspirate and flush all ports. Try the pressure transducer on a different port. If there is any suspicion that the central line is *misplaced* (central lines can migrate rostrally, and into the axillary vein) a CXR will confirm placement.

CVP too low

Hypovolemia, peripheral vasodilation, transducer too high.

CVP too high

LV and/or RV failure, cardiac tamponade, fluid overload, TR, AV dyssynchrony, tension pneumothorax, rarely pulmonary embolus. Commonly patient Valsalva-ing or fighting the ventilator, transducer too low, or fluids running through the central line.

Bleeding Apply direct pressure. Correct coagulopathy.

Sepsis Avoid by using good insertion technique. Remove catheter as described.

Insertion

Technique of insertion of internal jugular catheters is described on 📖 p244.

Removal

Central venous catheters are routinely removed 24–48h following surgery if the patient is not requiring inotropic support, and has no rhythm disturbances requiring central infusions or transvenous pacing wires. The patient is placed flat on the bed, and the catheter removed with pressure applied locally for 1min to prevent hematoma formation. Routine change of central venous catheters for patients requiring them longer avoids infection. Most units do this at 7 days unless indicated before. Change by railroading a new catheter over a guidewire placed down the old catheter may be sufficient in the patient with a neutrophilia or pyrexia, with a clean site.

Pulmonary artery catheter

Key facts

- In many units most cardiac surgery patients have PA catheters.
- A right heart catheterization is a one-off set of measurements from a PA catheter.
- This is indicated in patients with hypoperfusion states refractory to 1st- and 2nd-line inotropic support, and should be placed after induction in patients likely to present problems in terms of inotrope and fluid management: patients with pulmonary hypertension, poor ventricular function, transplant and VAD patients, multivalve surgery, and in off-pump CABG to guide intraoperative management.
- Uses include:
  - Pressure monitoring: RAP, RVP, PAP, PAWP.
  - Flow monitoring: cardiac output.
  - Mixed venous oxygen saturations.
  - Derived parameters: SVR, SVRI, PVR, PVRI, LVSV, VO_2, DO_2.
  - RV ejection fraction and RVEDP.
  - Temporary atrial and ventricular pacing.
  - LVSWI and RVSWI less commonly used to guide management.

Contraindications

As for central venous catheters (📖 p174). Those specific to PA include:

- *Absolute*: tricuspid or pulmonic valve stenosis/mechanical prosthesis, RA and RV masses which may embolize, tetralogy of Fallot (the RVOT is hypersensitive: hypercyanotic episode may be induced by the catheter).
- *Relative*: severe arrhythmias, coagulopathy, new intracardiac pacemaker wires which may become dislodged before they have become embedded.

Waveform

(See Fig. 5.5.)

- There is a sudden increase in systolic pressure as the catheter enters the RV.
- As the catheter enters the PA the diastolic pressure increases.
- There is a decrease in mean pressure as the catheter enters the wedge position.
- The PA waveform should reappear if the balloon is deflated at this point.

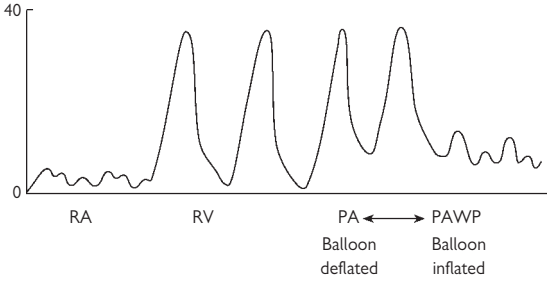


Fig. 5.5 Waveform progression during correct insertion of the PA catheter.

Catheter

Measurements may be continuous or based on intermittent bolus methods, depending on the calibration system. The catheter has four lumens:

- A proximal lumen 25cm from the tip which sits in the RA.
- A distal lumen connected to a pressure transducer sitting in the PA.
- A balloon lumen allowing balloon inflation.
- A thermistor lumen connecting to the gauge 4cm from the tip.

Complications

Complications are those of central venous cannulation (📖 p175) and those specific to PA catheters including: arrhythmias (complete heart block and VF), pulmonary artery rupture, and endobronchial hemorrhage, pulmonary infarction, knotting of catheter, valve damage, infection, thrombocytopenia, thrombus formation, balloon rupture, malplacement.

Common problems

Unable to get good trace

Check that the connection tubing to pressure transducer is free of air, kinks, clots, and closed taps. Flush and aspirate. Deflate balloon, withdraw the catheter and start again: no more than 15–25cm should pass before the waveform changes. If not, deflate balloon, withdraw and try again. Note the natural curve of the catheter, so that you know where the tip is pointing once it is inside the heart. If the catheter is failing to advance to the next chamber: (1) rotate the catheter slightly within the sheath, (2) retry with the patient in the left lateral position, (3) advance catheter slowly with balloon deflated.

Unable to wedge

The catheter is in an incorrect position. Deflate, withdraw, and try again.

Coiling and knotting

This can be avoided by not inserting excessive length: if the trace has not changed after 20cm of catheter deflate, withdraw and try again. If a loose knot is detected on CXR, insertion of a guidewire under fluoroscopy may unravel it. Alternatively, tighten the knot and withdraw the catheter with the balloon deflated, along with the introducer sheath.

Arrhythmias

Placing the patient slightly head up with right lateral tilt may help. Withdraw the catheter 1–2cm when arrhythmias occur, to terminate them.

Difficulty calibrating continual measurement systems

Common pitfalls include failure to enter the correct height and weight data, failure to provide a SvO_2 sample to calibrate, and LED not working. If basic measures don't help, change the box, leads, and catheter.

Insertion

Technique of insertion is described on 📖 p174.

Removal

PA catheters are removed as soon as practicable. The sheath may be left. The balloon ***must*** be deflated to avoid trauma to the tricuspid valve.

Mixed venous oxygen saturation

Key facts

- The *percentage saturation of mixed venous hemoglobin with O_2* (SvO_2) or 'mixed venous', is sampled from blood in the pulmonary artery and reflects the balance between O_2 consumption and delivery.
- SvO_2 is dependent on cardiac output, hematocrit, oxygenation, and oxygen consumption.
- In normal physiologic states cardiac output can increase up to 6–10× to compensate for changes in the other three parameters, maintaining SvO_2 within a normal range: this is not true following cardiac surgery.

Measurement of SvO_2

- It can be measured continuously using an oximetric pulmonary artery catheter fitted with an instrument that operates on similar principles to a pulse oximeter (📖 p156), or intermittently by withdrawing blood from the distal PA port and passing it through a standard blood gas analyzer.
- It is not adequate to sample SVC or RA blood, because the maximally deoxygenated blood from the coronary sinus will not be fully admixed, and SvO_2 will normally be overestimated. Peripheral venous blood sampling reflects the perfusion and oxygenation of peripheral tissue, and is not a good indicator of cardiac output.

Interpretation of SvO_2

From the equation (see Box 5.7) it is clear how an increased cardiac output can compensate for a low SaO_2, and how increased oxygen extraction can compensate for a low cardiac output.

- A low SvO_2 (<60%) indicates inadequate oxygen delivery. The lower the SvO_2, the more severe the deficit: SvO_2 <40% suggests a pre-terminal state.
- Cardiac output can increase by up to 6 times to compensate for low blood oxygen content, and oxygen extraction can increase by a similar amount to compensate for poor cardiac output: a low SvO_2 suggests that the limits of cardiac compensation have been reached.
- A normal SvO_2 (65–80%) does not always equate with adequate oxygen delivery: an overwedged PA catheter, regional hypoperfusion, left-to-right cardiac shunts, and carbon monoxide poisoning may all give falsely reassuring readings.
- A high SvO_2 (>90%) is most commonly caused by readings being taken from an overwedged PA catheter, but could also be caused by left-to-right shunt
- SvO_2 is a very dynamic measure of cardiac output, tissue perfusion and oxygenation: Fig. 5.6 shows the effects of simple clinical maneuvers on SvO_2.
- Oxygen consumption can easily increase 10-fold in shivering, septic or agitated patients, or in patients with seizures. In normal patients cardiac output can increase up to 10-fold to compensate. This is neither possible nor desirable in post-cardiac surgery patients: so oxygen extraction increases and SvO_2 falls.

- Always look at the SvO_2 in context—a low SvO_2 may indicate poor contractility, or respiratory compromise, hypovolemia, profound anemia, status epilepticus, or sepsis, etc. Similarly a normal SvO_2 does not always equate with a normal cardiac output.

High SvO_2

- Artifact:
  - Overwedged PA catheter.
  - Sampling from SVC, left-to-right shunt.
- Increased systemic O_2 delivery:
  - High FiO_2.
- Decreased systemic O_2 extraction:
  - Hypothermia.
  - Sepsis.
  - Pharmacological paralysis.

Low SvO_2

- Decreased systemic O_2 delivery:
  - Hypoxia, e.g. after suctioning.
  - Anemia.
  - Low cardiac output (hypovolemia, arrhythmias, poor contractility).
- Increased systemic O_2 extraction:
  - Hyperthermia.
  - Sepsis.
  - Shivering, seizures, agitation.

Examples	VO_2 (mL/min)	Hb (g/dL)	CO (L/min)	SaO_2 (%)	SvO_2 (%)
Shivering	1000	13	10	95	45
Anemia	250	6	5	95	36
Low CO	250	8	3	95	26
Hypoxia	250	7.5	5	90	43

Box 5.7 Calculation of SvO_2

$$SvO_2 = SaO_2 - [VO_2/(CO \times Hb \times 13.9)]$$

SvO_2 = mixed venous saturation	65–80%
SaO_2 = arterial saturation	95–100%
VO_2 = oxygen consumption	200–250mL/min
Hb = hemoglobin concentration	10–13g/dL
CO = cardiac output	4–6L/min

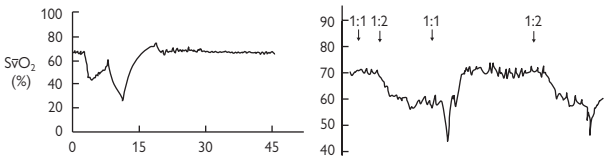


Fig. 5.6 Changes in SvO2 with (a) ET tube suctioning, (b) weaning an IABP.

Other forms of monitoring

Left atrial pressure

Key facts

- LA catheters are indicated where reduced LV compliances or pulmonary vascular disease, congestion, and edema mean that PAWP measurements are no longer a good approximation of LA pressures and LV preload.
- They are also indicated in corrective surgery for congenital lesions where PA placement is impossible.

Contraindications

Sepsis and coagulopathy are relative contraindications.

Insertion

Insertion is done intraoperatively via the right superior pulmonary vein, the roof of the LA posterior to the SVC and aorta, or in infants via the LA appendage. The fine-bore catheter is usually passed anterior to the heart and out to the skin at the base of the median sternotomy incision in the same way as a temporary pacing wire.

Waveform

(See Fig. 5.7.)

- Normal LAP is 10–14mmHg.
- PAWP normally exceeds LAP by 2–3mmHg, and the difference increases in the first 12h post bypass, possibly due to pulmonary edema and the effects of ventilation.
- LA pressure does not reflect LVEDP in the presence of mitral disease.
- MR can be detected on the trace as increased 'v' waves, and an increase in the mean LAP. This may be a sign of ischemia, arrhythmias, or overfilling.

Problems

- These are similar to those described for central lines: transducer setting, damping. Particular care must be taken to keep air and clots excluded from the system, as even small amounts of systemic emboli could cause myocardial ischemia and stroke.

Removal

Normally the LA catheter should be removed within 48h, preferably before the patient is extubated and the mediastinal drains have been removed, in case of bleeding from the insertion site. Any coagulopathy should be corrected. The LA line should be switched off for 2–3h and then removed.

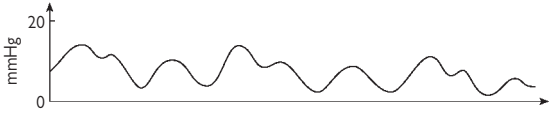


Fig. 5.7 Left atrial pressure waveform.

Doppler hemodynamic monitoring

Key facts

- Blood velocity can be measured by using the change in frequency of an ultrasonic beam as it reflects off a moving object, i.e., erythrocytes. This can be done using a small transesophageal US probe.
- To achieve accurate measurements of cardiac output, three conditions must be met:
  - The cross-sectional area of the aorta must be known.
  - The ultrasound beam must be parallel to the flow of blood.
  - The beam should be stable between measurements.
- Newer probes contain two transducers, one of which is an M-mode transducer which continuously measures the aortic diameter, while a second pulsed Doppler measures beat-to-beat flow. A continuous measure of cardiac output and derived parameters are displayed. The instruments indicate if positioning is poor.

Indications

Where a PA catheter cannot be floated, and cardiac output measurements and derived parameters are needed to guide inotropic therapy.

Contraindications

Esophageal rupture, esophageal varices, recent esophageal surgery.

Insertion

- Sedate the patient.
- Measure approximate insertion depth on the patient (the 3rd intercostals space anteriorly corresponds to T5/6).
- Insert probe facing the patient so that the probe is just beyond T6.
- Withdraw to obtain best signals.
- Rotate to gain Doppler signal of flow in the aorta, and image proximal and distal aortic walls.
- Validate the combined M-mode and Doppler signals.

Interpretation

Aortic diameter (which may be measured or calculated from patient nomograms) continual measurement of cardiac output, stroke volume, and systemic vascular resistance can be displayed. Aortic stroke volume is about 70% of total stroke volume. Maximal acceleration, normally 8–12m/s^2, is the initial acceleration in blood flow before the effect of systemic vascular resistance comes into play. It is a sensitive indicator of left ventricular contractility. Maximal acceleration <5m/s^2 indicates very poor contractility.

Problems

The main problem with transesophageal Doppler is that measurements are positional, and in inexperienced hands they can be highly user dependent.

Cardiac function after surgery

Key facts

Pathophysiological changes postoperatively were summarized on 📖 p186. Changes in the cardiovascular system are reiterated here.

- ***Myocardial compliance falls*** because of myocardial edema: higher filling pressures are required to generate the same stroke volume.
- ***Contractility is impaired*** as a result of myocardial edema, metabolic derangements, myocardial stunning, and ischemia.
- ***Myocardial function continues to decline for 6–8h*** postoperatively as a result of ischemia-reperfusion injury, before returning to baseline.
- ***Hypothermia*** is common, predisposing to ventricular arrhythmias (VF is common at temperatures below 32°C), coagulopathy and metabolic acidosis, and also raising afterload increasing myocardial O_2 demand. Hypothermia does, however, provide neuroprotection.
- The high SVR helps, however, to maintain MAPs until the patient is rewarmed: it is vital to ensure that adequate volume replacement and if necessary inotropic support are given, to avoid ***hypotension during rewarming***.
- Patients may ***overshoot*** to temperatures above 37.5°C because of abnormalities in central thermoregulation, resulting in pronounced peripheral vasodilatation and hypotension, and associated with a poorer neurological outcome.
- Vasodilatation and capillary leak mean that there is a ***progressive requirement for volume*** resuscitation to achieve satisfactory filling pressures which must be balanced against volume overloading the patient which has a particularly deleterious effect on respiratory function.
- Electrolyte and acid–base disturbances and myocardial irritability mean that ***arrhythmias*** are common: a third of patients will have AF postoperatively.

Low cardiac output

The aim of cardiovascular management postoperatively is to achieve a satisfactory cardiac output. If cardiac output measurements are available (PA catheter or transesophageal Doppler) a cardiac index of >2.5L/min/m^2 is satisfactory. In the absence of cardiac output measurements a MAP of 70–80mmHg, with a urine output of 1mL/kg/h, base deficit of <2, skin temperature of 36.5–37.5°C, and palpable pedal pulses (if these were present preoperatively) mean that cardiac index is most likely > 2.5L/min/m^2.

Causes of low cardiac output

The causes of low cardiac output and hypoperfusion states are summarized in Box 5.8. The etiology can be divided into:

- Preoperative.
- Operative.
- Postoperative causes, which can be further divided into pump failure (contractility, arrhythmias, mechanical), decreased preload, and increased afterload.

Box 5.8 Causes of low cardiac output

Preoperative

- *Previous MI* resulting in localized or global hypokinesia.
- Recent myocardial infarction resulting in global hypokinesia.
- Right or left ventricular systolic or diastolic dysfunction secondary to valvular heart disease or cardiomyopathy (📖 pp76–77).

Intraoperative

(Ask the surgeon and the anesthesiologist.)

- *Poor myocardial protection.*
- Incomplete revascularization.
- Kinking or occlusion of coronary artery (e.g. Cx during mitral).
- Air embolus down coronaries (usually resolves before transfer).
- Incomplete repair of valvular lesions.
- Hypotension before or after bypass.

Postoperative

(1) Pump failure

Decreased contractility:

- *Myocardial stunning*, poor myocardial protection.
- *Myocardial ischemia* due to coronary artery or conduit spasm.
- Hypoxia, hypercarbia, *acidosis*, hyperkalaemia.
- Drugs including propofol, amiodarone, β-blockers.

Arrhythmias:

- *Tachycardias* reducing filling time (sinus tachycardias, AF, VT).
- *Bradycardias.*
- *Loss of atrial 'kick'* (AF, flutter, junctional rhythms, CHB, VVI pacing).
- Ventricular arrhythmias (VT, VF).

Mechanical:

- Incomplete repair of valvular lesions, *SAM* (📖 p190).
- Ventricular rupture, atrioventricular dehiscence, aortic dissection.

(2) Decreased LV preload

- *Hypovolemia (hemorrhage, polyuria, inadequate fluid replacement)*
- *Vasodilatation (warming, sepsis, vasodilating drugs, e.g. NTG, milrinone, SNP, sedatives, narcotics, protamine, blood products).*
- *RV dysfunction, pulmonary hypertension.*
- *Cardiac tamponade*, tension pneumothorax.
- Sepsis, anaphylaxis, adrenal insufficiency, protamine reaction.

(3) Increased afterload

- Excessive vasoconstriction (hypothermia, vasoconstrictors).
- Fluid overload.
- Poor IABP timing.
- Valve obstruction, e.g., mechanical leaflet blockage.

More frequently seen problems in *italics.*

Managing low cardiac output

Management of common postoperative scenarios is described on 📖 pp128–129. Causes of low cardiac output are described on 📖 p183. The general management of low cardiac output is described here.

Key concept

Don't treat one number: look at the whole patient, and look at trends.

Signs of low cardiac output

Low cardiac output is a cardiac index <2.0L/min/m^2, but any of the following problems may be a sign of a low cardiac output state:

- Hypotension (MAPs <60mmHg).
- Tachycardia (heart rate >90bpm).
- Oliguria (urine output <1mL/kg/h).
- Metabolic acidosis (pH <7.4, lactate >3.0mL/L, base deficit >−2).
- Low mixed venous O_2 saturations (<65%).
- Pale or mottled, cold extremities.
- Pulmonary congestion and hypoxia.

Assessment of low cardiac output

Make sure you know the perioperative history: look for poor or hypertrophied LV, operation, difficulty weaning from bypass, coagulopathy risks

- Examine the patient: chest tube drainage, SaO_2, breath sounds, murmurs, peripheral perfusion, skin temperature, occult bleeding. agitation.
- Reassess hemodynamics: MAPs, CVP, PAWP, cardiac index, SVRI.
- Look at concentration of inotropes, pressors, sedation.
- Look at fluid balance: amount in vs. urine output, blood loss, etc.
- Look at EKG: heart rate and rhythm, ST segment changes.
- Look at blood gases: pH, $PaCO_2$, PaO_2, SVO_2, lactate, base deficit, blood glucose, hematocrit, potassium.
- CXR: pneumothorax, hemithorax.
- Echo: tamponade, LV and RV dysfunction, SAM (📖 p141).

Management of low cardiac output

Low cardiac output with low CVP (<14–16mmHg)

- Cardiac contractility is usually *not* the problem.
- Look for and treat the commonest causes which are *hypovolemia* (bleeding or polyuria) and *vasodilatation*.
- (1) Stop or reduce vasodilating infusions, e.g. NTG, milrinone, diltiazem.
- (2) Give 500mL of NS or colloid—one of three things will happen:
  - CVP and MAPs rise and cardiac index normalizes.
  - CVP rises dramatically with little improvement in MAPs or cardiac index: stop giving fluid and look for other problems (see 📖 p143).
  - CVP and MAPs rise transiently: watch CVP and repeat fluid bolus until hemodynamics are better, or you have given >2L fluid with low CVP in which case start a low dose of vasoconstrictor (e.g. norepinephrine 0.03 micrograms/kg/min), or you have high CVP (see 📖 p142).

Low cardiac output with high CVP (>18mmHg)

- This is more worrying as cardiac contractility is usually the problem.
- Look for and treat the commonest causes:
  - LV and/or RV dysfunction.
  - Arrhythmias.
  - Tamponade.
- (1) Correct respiratory or metabolic acidosis, electrolyte abnormalities, anemia and hypoxia (📖 p189). Inotropes are relatively ineffective at low pH, and the myocardium is much more prone to arrhythmias.
- (2) Check EKG for ischemia and treat aggressively (see 📖 p185).
- (3) Treat arrhythmias: if bradycardic pace at 80–90bpm: beware loss of atrial kick in hypertrophied LV (📖 p140) which may result in lower cardiac output, if AF aim to convert to sinus (📖 p218).
- (4) Treat poor contractility with inotrope (see 📖 p198 for choice), e.g., 0.03–0.05 micrograms/kg/min epinephrine, or 3–5 micrograms/kg/min dobutamine, or if high SVR consider milrinone 0.3–0.5 micrograms/kg/min.
- (5) *Escalating inotrope requirements mandate an echo to look for:*
  - Tamponade requiring re-exploration (📖 p144).
  - Severe LV or RV dysfunction requiring IABP (📖 p194) or NO (📖 p191).
  - Ongoing ischemia requiring intervention (📖 p185).
  - SAM requiring more volume and contraindicating inotropes (📖 p190).

Postoperative myocardial ischemia

Etiology

- Kinking of conduits that are too long or follow a tortuous path.
- Vasopasm of coronary arteries, or arterial conduits: increased α-adrenergic tone, adrenergic agents such as norepinephrine, air emboli.
- Thrombosis of grafts: hypercoagulable and low-flow states (such as peri-arrest) predispose to graft thrombosis.
- Endarterectomized vessels are prone to thrombosis.

Diagnosis

ST segment elevation, ventricular arrhythmias, heart block, and deteriorating cardiac output. The differential diagnosis includes the causes of low cardiac output states described on 📖 p183. TEE may demonstrate regional wall motion abnormalities.

Prevention

Good operative technique (everted anastomoses, exclusion of air from the coronary circulation, minimal handling of the conduits and coronary endothelium). Some units give all CABG patients NTG infusion (📖 p138).

Management

- Optimize oxygenation (📖 p189), preload, contractility and afterload (📖 p186).
- Start NTG infusion (200mg in 250mL 5% dextrose at 5mcg/min).
- Persistent ischemia with hemodynamic compromise is an indication for an IABP (📖 p194) if there is no surgical option (e.g., very poor or inaccessible targets) and the patient has no contraindications (📖 p194).
- Otherwise ischemia with hemodynamic instability mandates re-exploration in the first 12h if a surgical cause is suspected or cardiac catheterization if cause is unclear (>12h irreversible infarct).

Optimizing preload and afterload

Key facts

- The aim is to augment cardiac output while minimizing the work of the heart and hence myocardial oxygen demand.
- The most important determinant of myocardial oxygen demand is LV wall tension. The more pressure the LV is required to generate, the greater the myocardial oxygen demand. Wall tension is determined by preload and afterload. Optimizing these parameters, so that LV contraction occurs at smaller LV end-diastolic volume and against less resistance, reduces myocardial oxygen demand most effectively.

Preload

Optimizing preload so that the heart functions in the upper part of the Frank–Starling curve, before the plateau, increases contractility.

Definition of preload

- Preload is a measure of left ventricular end-diastolic fiber length, which is proportional to left ventricular end-diastolic volume (LVEDV) and pressure (LVEDP). It is determined by filling pressure, filling time, and ventricular compliance.
- In the clinical setting this is assessed indirectly by measuring *LAP, PAWP,* and *RAP* (LAP is closest and RAP least close to LVEDP).

- Optimize preload by giving *colloid* to increase it or *loop diuretics* to decrease it, aiming for an RAP of 12–16mmHg, a PAWP of 10–14mmHg.
- Patients with chronic MR, or LV hypertrophy (AS, MS, HOCM), require higher filling pressures, sometimes over 20mmHg.
- Trends are more important than absolute values.
- If filling pressures do not noticeably increase with filling, or increase and quickly fall back, then it is likely that the patient is still relatively hypovolemic, even if the absolute pressures are quite high.
- If filling pressures rapidly increase with volume and remain raised, with systemic pressures remaining the same or even decreasing, then the patient is overfilled, even if absolute filling pressures are relatively low.
- In patients with *poor ventricles* there may be very little margin between euvolemia and overfilling. Use small fluid challenges (<250mL colloid) and monitor filling and arterial BP closely.
- *Excessive preload* is deleterious for a number of reasons: it increases ventricular wall tension which exacerbates ischemia as MVO_2 is increased and myocardial perfusion is decreased, and it leads to pulmonary edema, splanchnic congestion, and potentially cerebral edema.
- Failure of filling pressures to rise with adequate volume may be due to *vasodilatation or capillary leak*: it is sometimes necessary to vasoconstrict a patient with a small amount of norepinephrine (4mg in 250mL 5% dextrose at a rate of 0.03–0.06 micrograms/kg/min) to achieve satisfactory preload (📖 pp200–201).

Afterload

The aim here is to optimize afterload. Reducing afterload reduces the work of the heart: the pressure work of the LV (wall tension) is the major determinant of myocardial oxygen demand.

- Care is required in *low cardiac output* states when using vasodilators to reduce afterload: a high SVR is sometimes the only mechanism by which BP, and hence coronary perfusion, is maintained when cardiac output is marginal.

Definition of afterload

- Afterload is a measure of LV wall tension during systole: it is determined by preload (see 📖 p130) and systemic vascular resistance (SVR).
- In the clinical setting SVR can be calculated from measurements of *cardiac output* and *BP* (📖 p131), or estimated from palpation of peripheral pulses and skin temperature.

- *Coronary perfusion pressure* is the difference between aortic diastolic pressure and coronary sinus or right atrial pressure: excessive reduction in afterload will contribute to myocardial ischemia by reducing aortic BP and hence coronary perfusion pressure.
- Afterload is raised in the *hypovolemic* patient because of reflex vasoconstriction. It may be possible to correct this with careful volume infusion often combined with inodilators, e.g., milrinone.
- *Hypothermia* is a common cause of raised SVR postoperatively. Warm the patient using warming blankets, and consider infusion warmers, particularly if large-volume transfusions are needed.
- A variety of *vasodilators can be used to lower SVR* (📖 p202): NTG (200mg in 250mL at a rate of 5mcg/min) is commonly used in most units as it has a beneficial effect on myocardial perfusion, in addition to its effect on the peripheral vasculature; SNP is a particularly potent vasodilator that has a less beneficial effect on myocardial perfusion than NTG and as such is reserved as a second-line adjunct in the control of hypertension (📖 p202).
- There are a number of *inotropes with vasodilatory properties* that reduce afterload: these are milrinone, dopexamine, and dobutamine in decreasing order of ability to reduce afterload and they should be considered in the context of low cardiac output.
- An increase in SVR (and hence afterload) may be the aim: patients with good cardiac outputs (CI >2.5) and poor systemic pressures as a result of *peripheral vasodilatation* are treated with vasoconstrictors such as norepinephrine (4mg in 50mL at a rate of 0.03–0.3 micrograms/kg/min).
- Beware of reducing afterload in patients with a *residual LVOT obstruction*, such as those with septal hypertrophy, or small aortic valve prostheses, as reducing SVR will lead to reduced diastolic pressures resulting in reduced coronary blood flow and myocardial ischemia.

Contractility

Key facts

- The ability to change contractility is unique to cardiac muscle: skeletal muscle cannot change intrinsic contractility.
- Several factors increase myocardial contractility (positive inotropes, e.g., endogenous catecholamines, β_1-adrenergic agents, Ca^{2+}) or decrease it (negative inotropes, e.g., β-blockers, acidosis, hypoxia).
- Although the official definition implies 'fixed preload and afterload', in the clinical setting contractility equates with force of contraction, and optimizing preload and afterload will help increase this.
- Force of contraction is maximal at the midpoint on the Frank–Starling curve which corresponds to a small but not empty LVEDV.

Definition of contractility

- Contractility is formally defined as a measure of the strength of myocardial contraction, at a given preload and afterload.
- It is assessed in the clinical setting by measuring ejection fraction, cardiac index, and LVSWI.
- MAPs and SVO_2 are very indirect measures of contractility.

Positive inotropic effect

Contractility can be increased through two mechanisms. There is a fast acting mechanism important in clinical practice, which involves an increase in Ca^{2+} uptake by myocytes. This can occur through increased stimulation of α_1- and β_1-adrenergic receptors, or via direct effects on electrolyte concentrations. The second process, less important in immediate postoperative care as it takes place over several weeks, results in upregulation of ATPase receptors, which has a similar effect on myocyte electrolyte concentration gradients. Positive inotropes include:

- Sympathetic nerve stimulation (both mechanisms).
- Epinephrine (α_1- and β_1-adrenergic agonist).
- Norepinephrine (very strong α_1 and weaker β_1-adrenergic agonist).
- Dopamine (weaker α_1- and β_1-adrenergic agonist, DA agonist).
- Dobutamine (very weak α_1- and stronger β_1-adrenergic agonist).
- Ephedrine (weak α_1- and β_1-adrenergic agonist).
- Metaraminol (strong α_1- and weak β_1-adrenergic agonist).
- Milrinone (increases Ca^{2+} influx by inhibiting inactivation cAMP).
- Calcium (increases Ca^{2+} influx).
- Digoxin (inhibits Na^+/K^+ ATPase to raise intracellular Ca^{2+}.

Negative inotropic effect

These work through mechanisms, including α_1- and β_1-adrenergic receptor blockade, reduction in uptake of Ca^{2+}, reduction in the speed of spread of the myocardial action potential, and reduction in energy dependent processes governing contraction. Negative inotropes include:

- Parasympathetic nerve stimulation.
- β-blockers and calcium channel blockers.

- Amiodarone.
- Acidosis.
- Hypoxia.
- Inflammatory cytokines.
- Electrolyte abnormalities including ⇈K^+, ⇊K^+, ↓Mg^+, ↓Ca^{2+}.

Optimizing contractility

- Treat hypoxia and hypercapnia (📖 pp160–161).
- Optimize myocardial perfusion (📖 p185).
- Tread acidosis and metabolic abnormalities (📖 pp170–171).
- Treat sepsis (📖 p284).
- Optimize preload and afterload (📖 pp186–187).
- Select inotropic support (📖 pp198–199).
- Mechanical support (📖 pp194, 496, 500) may be required.

Optimizing myocardial oxygen supply: demand

Optimizing preload, contractility and afterload (📖 p186, 📖 p188)

The more pressure the LV has to generate, the greater the MVO_2. Optimizing preload and afterload, so that LV contraction occurs at smaller LVEDV and against less resistance, reduces MVO_2 most effectively.

Peripheral vascular resistance

- Intense peripheral vasoconstriction due to hypothermia, hypovolemia, or α_1- agonists can maintain adequate MAPs despite low cardiac output. End-organ perfusion is compromised leading to mesenteric ischemia, ATN, and hepatic ischemia, which has a poor prognosis. Excessive afterload increases work for the heart, increasing MVO_2 and risk of ischemia.

Arrhythmias

- Poor myocardial function may be disguised by reflex tachycardia that compensates for small stroke volumes to maintain cardiac output. Tachycardia may be a sign of failing LV and exacerbates myocardial ischemia. Management of arrhythmias that cause low CO is described on 📖 pp216–221.

Intra-aortic balloon pump (📖 pp192–195)

An IABP (📖 p192) improves the ratio between myocardial oxygen supply and demand by reducing afterload and improving coronary perfusion.

Improving myocardial perfusion

Coronary perfusion pressure is equal to aortic diastolic pressure minus coronary sinus or right atrial pressure. Myocardial perfusion is also dependent on diastolic filling time.

- If diastolic pressures are low because the patient is profoundly vasodilated, rather than because of a low cardiac output state, start a norepinephrine infusion to raise diastolic pressure >55mmHg.
- If diastolic pressures are low because of low CO state, despite optimizing preload, afterload, and contractility, an IABP may eventually be necessary to treat myocardial ischemia.
- In both cases treat tachycardia aggressively to optimize diastolic filling time: the optimum heart rate postoperatively is 80–100bpm (📖 p222).

Vasoplegia, SAM, and RV failure

Vasoplegia

- Vasoplegia is profound vasodilatation causing hypotension refractory to high dose vasoconstrictors despite normal or high cardiac output.
- It complicates 10% of patients post cardiopulmonary bypass: risk factors include preoperative ACE inhibitors, β-blockers, critical preoperative states, sepsis, low EF, and inodilators such as milrinone.
- High-dose norepinephrine and Pitressin® may not adequately restore arterial pressure and can cause splanchnic, mesenteric, and peripheral malperfusion and ischemia, despite low overall SVR.
- *Methylene blue* (bolus 2mg/kg, infusion 0.6mg/kg/h) blocks cGMP-mediated vasodilatation leading to rapid restoration of vascular tone.

Systolic anterior motion

- SAM is abnormal motion of the anterior leaflet of the mitral valve into the LVOT during systole (Fig. 5.8) resulting in:
  - Anterior displacement of leaflet coaptation.
  - LVOT obstruction (gradients can reach 80mmHg).
  - Mitral regurgitation (posteriorly directed jet).
- It is usually due to excess mitral leaflet tissue compared to the size of the mitral annulus, and the LVOT—*inotropes, tachycardia, and hypovolemia make this problem worse!*

Risk factors

- SAM occurs in hypertrophic obstructive cardiomyopathy.
- It is encountered on ICU as a complication of about 5–10% of mitral valve repairs where it is usually confined to patients with degenerative disease and excessive redundancy of the anterior and/or posterior leaflets, usually with an undersized rigid ring. It is not a feature of:
  - Ischemic mitral repair (where leaflet tissue is deficient and tethered).
  - Mitral valve replacement (where the anterior leaflet is resected).
- SAM may occur more commonly with a rigid annuloplasty ring, possibly as a result of reduction of the anteroposterior annulus diameter, but it does also occur with flexible rings, and in repairs without rings.

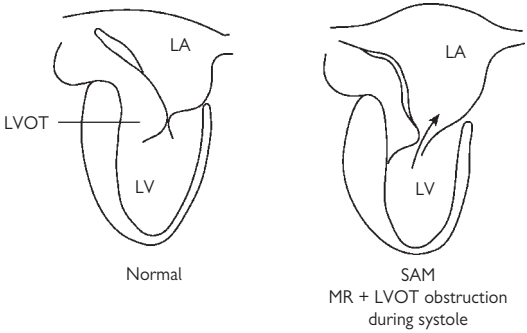


Fig. 5.8 Longitudinal section across the mitral and aortic valve and left ventricle, showing abnormal systolic anterior motion of the mitral valve.

- Small (underfilled), hyperdynamic (inotropes), hypertrophic, tachycardic left ventricle exacerbate the problem.

Assessment and management

- Hemodynamic compromise (high CVP, high PAWP, low cardiac output, poor MAPs) and a pansystolic murmur after degenerative mitral repair should raise the suspicion of SAM, particularly if a tendency towards SAM was noted on the post-CPB TEE.
- Assess the LVOT gradient, and degree of MR by TEE.
- Treatment consists of careful volume loading, reducing inotropic agents, and increasing afterload, under TEE guidance
- Mild to moderate SAM is treated conservatively: it normally resolves.
- In severe LVOT obstruction, removal of the annuloplasty ring and/or placing neochordae to move coaptation posteriorly improves SAM.
- Aggressive quadrangular resection and sliding leaflet plasty removes redundant posterior leaflet tissue so that the line of coaptation is well away from the LVOT: these are used in conjunction with accurately sized rings (📖 p411) to avoid SAM in high-risk cases.

Right ventricular failure

Risk factors

- Preoperative severe RV dysfunction (look for preoperative markers such as RV dilation, severe TR, congestive liver dysfunction).
- Poor myocardial protection, very long cross-clamp time, injury to right coronary button during root surgery even in non-dominant RCA, air.

Assessment and management

- Hemodynamic compromise (high CVP, RVEDP equalizing with CVP suggesting no RV pressure augment) with low to normal PAWPs and low cardiac output because of inadequate LV filling, often arrhythmias.
- *This picture is similar to tamponade: get an echo to make diagnosis.*
- This will usually respond to a similar approach to poor LV function but with some important differences, that reflect the fact that systemic perfusion pressure has a greater role in RV perfusion, the RV is much more sensitive to over-filling, and pulmonary vascular resistance:
  1. Correct respiratory or metabolic acidosis, electrolyte abnormalities, anemia and hypoxia (📖 p189). Inotropes are ineffective at low pH, and the myocardium is much more prone to arrhythmias.
  2. Check EKG for ischemia and treat aggressively (see 📖 p185).
  3. Treat arrhythmias: if bradycardic pace at 80–90bpm: if AF aim to convert to sinus (📖 p218).
  4. Start inotrope, e.g. 0.05–0.1mcg/kg/min epinephrine, ± norepinephrine or Pitressin® aiming for MAPs >80–90mmHg.
  5. Do not fluid overload: if CVP >25mmHg consider diuresing patient, or placing bed in reverse Trendelenburg position.
  6. Minimize PVR: inhaled NO or epoprostenol for intubated patients run less risk of dropping MAPs than inodilators, e.g., milrinone
  7. IABP will improve coronary perfusion and help failing RV, although in extreme circumstances a temporary RVAD may be needed (📖 p496).
- The occasional patient with SAM and RV dysfunction is a challenge as the treatments are completely opposite: a definitive surgical solution, and inhaled NO (📖 p203) are the mainstays of treatment.

IABP insertion

Key facts

- The IABP is a long polyethylene balloon filled with helium, 2–50mL in size, that sits in the descending aorta just distal to the left subclavian artery (see Fig. 5.9a, p194).
- The balloon inflates in early diastole improving coronary perfusion, and deflates just before systole reducing afterload.
- The MVO_2 supply:demand ratio, and cardiac output are improved.

Indications

- Preoperatively: cardiogenic shock, myocardial ischemia or for high-risk coronary angiography, e.g., pinhole left main stem with angina.
- Intraoperatively to assist weaning from bypass if very poor LV or RV.
- Postoperative poor cardiac output refractory to inotropic support, or myocardial ischemia without surgical or interventional solution.

Contraindications

- ≥2+ AI (balloon inflation during diastole will worsen AI).
- Aortic dissection (may worsen dissection or cause free rupture).
- Severe peripheral vascular and aortic atherosclerosis.

Insertion

The IABP is usually inserted percutaneously with a Seldinger technique. In addition to the IABP pack and the balloon pump console, you will require four large sterile drapes, prep, sterile gown and gloves, hand-held suture, 11 blade, and 10mL of 1% lidocaine if the patient is awake.

- Locate the femoral artery pulse (mid-inguinal ligament).
- Prep and drape. Check that the wire runs smoothly through the wire holder. Draw back firmly on the 60mL syringe connected to the IABP to ensure that the balloon is collapsed, remove the IABP firmly from its packaging, and withdraw the wire from the sidearm.
- Advance the large pink needle from the IABP set on a 10mL syringe, at 45° into the femoral artery until you can easily aspirate arterial blood.
- Remove the syringe, insert and advance the guidewire which should advance without resistance. Withdraw 2cm if ectopics appear.
- Nick the skin over the wire with the 11 blade and advance the smallest of the three dilators provided with the set over the wire and into the femoral artery: you only need the largest dilator if using IABP sheath.
- Leave the dilator *in situ*, and measure IABP so that the tip is at the 3rd rib space, advancing the proximal end of the plastic guard to the groin.
- There is usually a wire inside the IABP that needs to be completely removed first before the IABP will pass over the femoral guidewire.
- Withdraw the dilator, press firmly over the guidewire to prevent bleeding and insert the IABP over the wire and into femoral artery, up to the proximal guard, keeping tension on the end of the guidewire.
- Remove the guidewire.
- Now quickly take the manometry line, flush it and connect with a three-way tap to the IABP, aspirate blood and when the system is free of air, open it to the IABP console: an arterial trace should be visible.

  - If this is not done rapidly the manometry lumen may clot off (the manometry lumen is very small to minimize impact on the flow of helium) and a new IABP will need to be inserted.
  - Newer models transduce pressures without a manometry line.
- Next remove the yellow internal marker and attach the port in the pack to the side arm and hand out to the balloon console.
- In an emergency the IABP can be operated at once, using the augmentation trace to decide if positioning is adequate; a TEE is helpful to check positioning; otherwise a CXR should show the proximal tip of the IABP just distal to the left subclavian.

Alternative methods of insertion

- If the femoral artery is difficult to palpate, a cut-down is necessary (📖 p120). Using 5/0 Prolene place a small purse-string in the femoral artery, and insert needle through skin into artery under direct vision.
- Rarely peripheral vascular disease is so severe that femoral insertion is impossible. The IABP can be inserted retrogradely via the ascending aorta. Either a purse-string is created as for aortic cannulation, with short snares left *in situ*, for later removal, or the IABP inserted through a Dacron graft is anastomosed to the ascending aorta. If the graft is brought to the skin, the IABP can be removed without resternotomy.

Complications

Vascular: distal limb ischemia is the commonest complication, causing IABP removal in >10% of patients. Risk is reduced by not using a sheath for insertion as this occupies a large proportion of the femoral artery lumen. Vascular complications also result from aortic dissection, iliac artery rupture, hematoma, and embolization of atherosclerotic plaques, thrombus, and cholesterol microemboli. Renal ischemia may result if the balloon is too low. Regular assessment of distal pulses by palpation or Doppler is mandatory, as is heparinization. If the balloon is left on a 1:3 ratio for over an hour, or deflated, thrombus is likely to form on the balloon leading to embolic complications. If distal limb ischemia develops either remove or place the IABP in the contralateral artery without a sheath (or very rarely directly into the aorta).

Thrombocytopenia: the mechanical action of the IABP destroys circulating platelets, which may be further depleted by HIT. Check platelet counts daily and consider alternative anticoagulation and/or IABP removal if platelet counts fall below 60.

Sepsis: sepsis is rare but associated with a high mortality. Positive blood cultures require removal of the IABP.

Balloon rupture: if blood appears in the balloon tubing it is likely that the balloon has ruptured. In view of the risk of releasing helium into the systemic circulation, inflation should be discontinued immediately and the balloon removed. Place the patient in Trendelenburg to minimize cerebral air emboli. Helium is an inert gas that is quickly absorbed.

- *Other complications*: AV fistula, false aneurysm, and hematoma formation following removal may require surgical exploration and repair. If the patient flexes hip or sits up the IABP tip can migrate and perforate the aortic arch or occlude the left subclavian or left common carotid.

IABP management

IABP basics

- The IABP end-diastolic pressure (A) is lower than the patient's aortic end-diastolic pressure without the balloon pump (a) showing that balloon deflation is well timed: deflation offloads the LV (Fig. 5.9b).
- IABP diastolic augmentation (C) occurs at the dicrotic notch (c): it should be higher than the patient's diastolic pressure (c), is ideally higher than systolic pressure (b): it is the coronary perfusion pressure.
- Assisted systolic pressure (B) is lower than the patient's unassisted systolic pressure (b) because augmented end-diastolic pressure and afterload are reduced by the IABP (Fig. 5.9b, bottom panel).
- The improvement in cardiac output and myocardial oxygen supply: demand ratio is the key: systolic pressures are often higher with the IABP switched off but cardiac function may be worse.

Which pressure to go by? Use systolic not mean pressures. The patient monitor is usually used, but it does not distinguish between diastolic augmentation and systole (denoting the higher as systolic pressure). The IABP console can ('Aug' on the IABP console is the diastolic augmented pressure (C), console 'systolic' is unassisted systole (b)): but *always* check that the IABP transducer is correctly levelled if you are using that.

IABP timing

- IABP should inflate at beginning of diastole and deflate at the end of diastole: there are several triggering modes that can be used.
- *EKG*: EKG leads attached either to the patient or the bedside monitor are plugged into the IABP console. Inflation occurs at the peak of the T wave and deflation on or just before the R wave. Use bipolar cautery.
- *Pressure*: inflation occurs at the dicrotic notch and deflation just before the beginning of the aortic upstroke. Use this mode if interference from monopolar electrocautery, or a poor EKG, makes EKG timing ineffective.

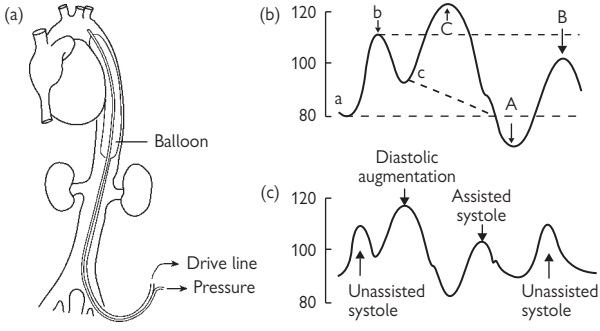


Fig. 5.9 (a) Correct IABP position just distal to left subclavian, and above renal arteries. (b) Top: 1:2 IABP augmentation trace, bottom: 1:3 IABP trace.

- *Pacing*: inflation occurs at the peak of the T wave, deflation occurs on the R wave as for EKG timing, but pacing spikes are filtered out so that that they are not misinterpreted as QRS complexes.

Common problems

Ineffective augmentation: (Fig. 5.10). Too short an augmentation time reduces coronary perfusion pressure. Too early inflation increases LV afterload, too late and coronary perfusion is suboptimal. Arrhythmias, rapid heart rates, and pacing spikes may cause ineffective augmentation or prevent balloon functioning altogether. Move the sliders on the console to obtain the optimal augmentation trace (Fig. 5.10). In older models rapid heart rates require augmentation to be dropped to 1:2.

Volume loss: this will be indicated by the console alarms and is caused by a gas leak in the system, or by incomplete unwrapping of the balloon. Check the connectors are secure. If the alarm persists there may be a gas leak from the balloon, in which case it should be replaced.

Weaning

Wean when pharmacological support has been reduced to moderate levels, and patient can tolerate lying flat. Inflation is reduced to 1:2 for 6h. ***If no angina, ST segment changes, fall in urine output, increase in heart rate or PVCs, cardiac index maintained >2.0, SVO_2 >65, minimal rise in PAWP,*** the ratio is decreased to 1:3. Heparin is stopped and coagulation checked. If platelets >90 and the APTT <50 the IABP may be removed. 1:3 mode should not last >1h because of the risk of thrombus.

Removal

Balloons inserted surgically should be removed surgically. Balloons inserted percutaneously may be removed with aseptic technique, by cutting the sutures fixing them to the skin. ***Ensure IABP is deflated.***

- Press the assist-standby button on the console, and ***check the trace again to ensure the balloon is not inflating.***
- With plenty of sterile gauze swabs available swiftly pull the IABP out in the direction of insertion and press firmly over the arteriotomy which is 1–2cm ***proximal*** to the skin insertion for 15–30min.
- Monitor the patient for the next 24h for hematoma formation and distal limb ischemia.

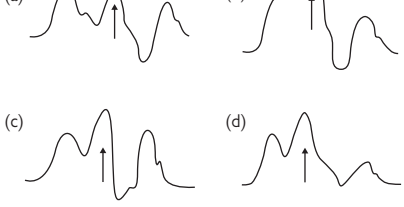


Fig. 5.10 Poor arterial IABP 1:2 augment traces: (a) Late inflation, (b) early inflation, (c) early deflation, (d) late deflation.

Pharmacological support

Calculating drug doses

Conversion to milligrams (mg) or micrograms (μg or mcg) per millilitre

- A drug labelled x% contains xg/dL. Multiply **x** by 10 to get the number of grams per litre, or the number of milligrams per millilitre.
- Concentrations stated as a ratio are converted to milligrams or micrograms per millilitre: 1:1000 = 1g/thousand mL = 1 milligram/mL; 1:1,000,000 = 1g/million mL = 1 microgram /mL.

Examples

- Lidocaine 2% contains 2g/dL, or 20g/L, or 20mg/mL.
- Epinephrine 1:10,000 = 1g/10,000mL = 0.1 mg/mL, or 100mcg/mL.

Calculating infusion rates

- First calculate the *desired dose* to be infused per minute by multiplying the patient's weight in kilograms by the dose in micrograms per kilogram per minute.
- Next calculate the *concentration* (micrograms/mL) of the drug in the syringe driver or infusion bag by multiplying the amount of the drug (micrograms) in the infusion unit by 1000 divided by the amount of diluent (mL).
- Finally calculate the *infusion rate* (mL/min) by dividing the desired dose rate (micrograms/min) by the concentration (micrograms/mL). Multiply this by 60 to get the infusion rate per hour (mL/h) (see Box 5.9 for an example). Common infusions are listed in Tables 5.1 and 5.2.

Box 5.9 Example infusion calculation

- A 70kg patient requires a norepinephrine infusion at 0.03 micrograms/kg/min. The syringe driver is prescribed at 4mg in 50 mL 5% Dextrose.

1) The *desired dose* is 70 × 0.03 = 2.1 micrograms/min
2) The *concentration* is 4 × (1000/50) = 4 **x** 20 = 80 micrograms/mL
3) The infusion rate = 2.1/80 = 0.026mL/min

or 0.026 **x** 60 = 1.5mL/h

$$\text{Infusion rate} = \frac{\text{Wt (kg)} \times \text{prescribed dose (micrograms/kg/min)} \times 60\ \text{(mL/h)}}{\text{Drug in pump(mg)} \times (1000/\text{diluent in pump (mL)})}$$

Table 5.1 Concentrations, rates, and starting doses for vasoactive drug infusions

Drug	Mg in 250 mL	Dilution (micrograms/mL)	Range (micrograms/kg/min)	Patient wt (kg)		
				50	70	100
Epinephrine	4	16	0.03–0.5	9.5	13	19
Dopamine	250	1000	3.0–10.0	9.0	12.6	18
Dobutamine	250	1000	3.0–10.0	9.0	12.6	18
Dopexamine	250	1000	3.0–10.0	9.0	12.6	18
Norepinephrine	4	16	0.03–0.3	9.5	13	19
Milrinone	40	160	0.3–0.9	9.0	13	18
Isoprenaline	25	100	0.01–0.3	0.5	0.7	1.0
Nitroglycerine	25	100	5–200 micrograms/min	1.5	1.5	1.5
SNP	50	200	0.1–10.0	8	11	16
Diltiazem	250	1000	5–10	5	5	5
Labetalol	250	1000	1–4mg/min	2	2	2
Pitressin® (vasopressin)	250 units	1 unit/mL	2.4–8 units/h	2.4	2.4	2.4

The numbers in Table 5.1 under patient weight heading show the drug infusion rate in mL/h required to achieve the stated starting dose. Not all drips are adjusted by patient weight.

Table 5.2 As regimes used at various institutions vary, this table is blank so that alternative infusion regimes can be entered

Drug	Concentration	Range

Inotropes

Inotropes increase inotropy (contractility)

- α-adrenergic receptor stimulation leads to ↑SVR and PVR.
- β_1-stimulation leads to increased contractility, heart rate, and conduction.
- β_2-stimulation causes peripheral dilatation and bronchodilation.
- *DA* stimulation causes coronary, renal, and mesenteric vasodilatation.

Epinephrine (adrenaline) 0.03–0.2 mcg/kg/min

Catecholamine produced by the adrenal medulla.

Action: Direct agonist at α-, β_1-, and β_2- receptors, increased glycogenolysis, lipolysis, gluconeogenesis—leads to increased blood glucose and lactate.

Pharmacodynamics: instant onset, half-life 2min. Metabolized by MAO.

Indications: (1) cardiac arrest (asystole, VF, EMD), (2) anaphylaxis, (3) low cardiac output states, (4) bronchospasm.

Clinical u se: (1) *cardiac arrest*: high dose. 0.5–1mg bolus iv (5–10mL of 1:10 000); (2) *anaphylaxis*: low to moderate dose. 0.5–2mL of 1:10 000 bolus iv; (3) *low cardiac output* states: bolus 0.5–2mL of 1:10 000 iv (1mL of 1:10 000) then infusion 0.03–0.2 micrograms/kg/min); (4) bronchospasm: 5mL of 1:1000 epinephrine nebulized.

- Administration by central line: extravasation causes skin necrosis.
- Infusions require continual arterial line blood pressure monitoring.
- With high-dose epinephrine extreme hypertension, CVA or MI may result. ***Reserve bolus administration for the peri-arrest patient***.
- Monitor SVR: a vasodilator may be required.
- Myocardial oxygen demand is increased.
- Ischemia due to vasoconstriction at high doses may occur: lactic acidosis is common at moderate doses, but this usually reflects enhanced glycogenolysis and inhibition of pyruvate dehydrogenase (📖 p65).
- Serum lactate and glucose rises, particularly in diabetic patients: low glucose suggests hepatic ischemia which is a poor prognostic sign.
- Epinephrine is an effective bronchodilator and stabilizes mast cells: it is useful in the treatment of anaphylaxis and bronchospasm.

Milrinone 0.3–0.9 micrograms/kg/min

Bipyridine derivative that inhibits phosphodiesterase.

Action: potent inotropic and vasodilator effects.

Pharmacodynamics: half-life is 4h. Metabolism is hepatic, with 30% excreted unchanged in the urine.

Indication: low cardiac output state in the setting of raised SVR and/or pulmonary hypertension. Avoid in patients at risk of vasoplegia (📖 p190).

Clinical use: optimize preload. Milrinone is commonly started together with a low dose of norepinephrine to partially offset its vasodilator effect.

- Administration by central line, monitor with arterial line.
- May cause thrombocytopenia.
- Milrinone increases cardiac output by increasing contractility and decreasing afterload, reducing myocardial oxygen demand.
- Less risk of arrhythmias than other inotropes.

Dobutamine 3–10 micrograms/kg/min

Synthetic catecholamine.

Action: direct β_1-agonist (inotropy), small β_2 effect (↓SVR) with minimal α- or DA receptor action. Metabolite of dobutamine is an α-antagonist.

Pharmacodynamics: half-life is 2min. Hepatic metabolism.

Indication: low cardiac output secondary to increased afterload.

Clinical use: as for dopamine. Reduce vasodilator infusions, as dobutamine acts like a combination of an inotrope and a vasodilator.

- Administration by central line, monitor with arterial line.
- Improvement in urine output and splanchnic perfusion secondary to increased cardiac output and splanchnic vasodilatation.
- Reduction in afterload reduces myocardial oxygen demand.
- Dose dependent and reflex tachycardia.

Dopamine 3–10 micrograms/kg/min

A catecholamine precursor to norepinephrine and epinephrine.

Action: α, β_1, β_2 and DA_1 agonist, and release of stored neuronal NA. At the low to middle of the dose range β and DA effects (↑HR, contractility, ↓SVR at low doses) predominate. Above 10 micrograms/kg/min α effects predominate (increased SVR, PVR, arrhythmias).

Pharmacodynamics: fast onset, slow offset. Metabolized by MAO.

Indication: (1) low cardiac output states; (2) renal insufficiency.

Clinical use: Optimize preload, commence dopamine at 3 micrograms/kg/min and titrate upwards depending on BP/cardiac output. If no improvement at high doses reassess with a view to a second inotrope.

- Mixed α, β_1, β_2, and DA_1, seen by some as a disadvantage.
- Administration by central line: extravasation causes skin necrosis.
- Improvement in urine output is secondary to increased cardiac output rather than any direct effect on renal vasculature.
- Tachycardia and arrhythmias may occur.
- Systemic vasodilatation occurs at low doses.
- Pulmonary vasoconstriction may occur.
- Dopamine can reduce respiratory drive.

Dopexamine 3–10 micrograms/kg/min

A synthetic analogue of dobutamine with primary vasodilator action.

Action: potent DA_1, β_2 agonist (↑CO, HR, and vasodilation). Little effect on α- and β_1-receptors. Inhibits norepinephrine uptake, potentiating norepinephrine effects.

Pharmacodynamics: half-life 6–10min. Hepatic metabolism.

Indication: low cardiac output secondary to increased afterload.

Clinical use: as for dopamine. Reduce vasodilator infusions, as dopexamine acts like a combination of an inotrope and a vasodilator.

- Administration by central line, monitor with arterial line.
- Improvement in urine output and splanchnic perfusion secondary to increased cardiac output and splanchnic vasodilatation.
- Reduction in afterload reduces myocardial oxygen demand.
- Dose dependent and reflex tachycardia, but less than dobutamine.

Pressors

Pressors cause vasoconstriction

- α_2- adrenergic stimulation increases SVR and PVR
- α_1- adrenergic stimulation ↑contractility without ↑HR.

Norepinephrine 0.03–0.3 micrograms/kg/min

Catecholamine produced by the adrenal medulla.

Action: direct α- agonist: potent vasoconstriction + β_1 effect = ↑BP.

Pharmacodynamics: immediate onset, half-life 2min. Metabolized by MAO.

Indication: hypotension because of low SVR in good to high CO states.

Clinical use: low-dose norepinephrine (0.03–0.06 micrograms/kg/min) is often started empirically in the well-filled patient with warm peripheries, where low BP and filling pressures are thought to be a result of profound vasodilation. Norepinephrine requirements may be very high (>1.0 microgram/kg/min) in severe sepsis.

- Administration by central line: extravasation causes skin necrosis.
- In hearts with borderline CO norepinephrine should be avoided or used in conjunction with another inotrope, for the reasons described.
- Norepinephrine increases afterload and contractility, resulting in increased myocardial oxygen demand.
- There is increased risk of vasopasm of arterial conduit resulting in ischemia, and rarely infarction.
- End-organ ischemia may result from the intense vasoconstriction.
- Norepinephrine is contraindicated in pregnancy.
- Great care must be taken in changing infusions in norepinephrine dependent patients as catastrophic falls in systemic pressures may in occur in the short time it takes to disconnect one pump and run a second infusion through.

Metaraminol (Aramine®) bolus 0.5–1.0mg iv

Potent synthetic sympathomimetic amine.

Action: direct α-agonist: potent vasoconstrictor + β_1 effect = ↑BP

Pharmacodynamics: metaraminol has a longer duration of action than norepinephrine. Immediate onset, half-life 4–5 min. Metabolized by MAO.

Indication: temporizing measure to treat hypotension.

Clinical use: boluses of metaraminol are sometimes given to improve blood pressure prior to institution of bypass or during patient transfer. If repeated boluses of metaraminol in the ICU achieve a good but not sustained response, an infusion of norepinephrine is indicated.

- Metaraminol increases afterload and contractility, resulting in increased myocardial oxygen demand.

Phenylephrine (40–180 micrograms/min)

Potent synthetic sympathomimetic amine.

Action: pure α-agonist: potent vasoconstrictor. Causes reflex decrease in heart rate. Tachyphylaxis (a rapid decrease in the response to the drug) may be seen after a few hours.

Pharmacodynamics: immediate onset, half-life 30–40min. Metabolized by MAO.

Indication: hypotension because of low SVR in good to high CO states.

Clinical use: low-dose phenylephrine may be started empirically in the well-filled patient with warm peripheries, where low BP and filling pressures are thought to be a result of profound vasodilation.

- Administration by central line: extravasation causes skin necrosis.
- In hearts with borderline CO phenylephrine should be avoided or used in conjunction with another inotrope, for the reasons described.
- Phenylephrine increases afterload and contractility, resulting in increased myocardial oxygen demand.
- There is increased risk of vasospasm of arterial conduit resulting in ischemia, and rarely infarction.

Vasopressin (Pitressin®) 2.4–8 units/h

Synthetic analogue of ADH.

Action: acts at V1 (arterial smooth muscle) receptors, not α-receptors. Also acts on V2 renal tubule receptors.

Pharmacodynamics: very rapid onset, half-life 10–20min. Metabolized in liver and kidneys.

Indication: hypotension because of low SVR in good to high CO states.

Clinical use: Vasopressin may be started empirically in the well-filled patient with warm peripheries, where low BP and filling pressures are thought to be a result of profound vasodilation. In patients with vasoplegia, addition of Vasopressin to norepinephrine or phenylephrine reduces the dose of catecholamines required to restore vascular tone, and may therefore avoid the adverse effects on end-organ perfusion associated with high dose catecholamine therapy.

- Administration by central line: extravasation causes skin necrosis.
- Vasopressin has no effect on adrenergic receptors: for the same improvement in systemic arterial pressure achieved by norepinephrine, there may be less adverse effects on splanchnic and renal perfusion.

Methylene Blue

Action: Blocks cGMP-mediated vasodilatation leading to rapid restoration of vascular tone in vasoplegic patients.

Pharmacodynamics: onset 5–10 minutes, half-life several hours. Renal excretion.

Clinical use: Bolus 2mg/kg, infusion 0.6mg/kg for 6–12 hours in patients with vasoplegia refractory to maximal pressor support.

Vasodilators

Nitroglycerin (NTG) 5–200 micrograms/min

Also known as glyceryl trinitrate (NTG), or trinitroglycerine.
Venodilator. Increases cGMP production.

Action: at low doses dilates veins reducing venous return to the heart, decreasing heart size and preload, stroke volume and cardiac output. This reduces myocardial oxygen demand. Dilates coronary vasculature reducing vasospasm, redistributing more flow to ischemic myocardium. This has an antiarrhythmic effect.

Pharmacodynamics: metabolized by liver and smooth muscle to nitrites. Half-life 1–3min.

Indications: (1) in many units NTG is started routinely as prophylaxis against myocardial ischemia and coronary spasm; (2) myocardial ischemia; (3) coronary and conduit vasospasm; (4) hypertension; (5) pulmonary hypertension.

Clinical use: NTG is routinely started at the lowest end of the range above and titrated against BP to maintain MAPs 70–75mmHg.

- Administration may be by peripheral line, sublingually, or topical NTG patches 5–15mg over 24h.
- NTG selectively reduces preload, except at high doses.
- NTG lowers PVR and is a useful adjunct in the treatment of pulmonary hypertension and right heart failure pre- and postoperatively.
- Nitrites, the metabolites of NTG, oxidize hemoglobin to methemoglobin: impaired oxygen transport characterized by elevated PaO_2 but poor SaO_2 may result if patients receive large doses of NTG over several days, or if renal and hepatic excretion is impaired.
- Tolerance can be avoided or treated by stopping NTG infusion for several hours.

Labetalol (1–4mg/min)

β-blocker with α-blocking properties and a vasodilatory effect.

Actions: like all β-blockers labetalol reduces BP by negative inotropic ad chronotropic effects. Contractility, stroke volume, cardiac output and heart rate are all reduced. SVR and diastolic pressure are largely unchanged. α-blocking effects prevent reflex vasoconstriction.

Pharmacodynamics: onset 3–5min, half-life 3h.

Indications: labetalol, like other β-blockers, should not be used in borderline or low cardiac output states to reduce heart rate. It is indicated for control of hypertension with satisfactory cardiac output, particularly hyperdynamic, hypertrophic, tachycardic hearts.

Clinical use: labetalol can be given as a bolus (0.25mg/kg/min iv over 2min with subsequent doses at 15min intervals) or as an infusion.

- AV conduction is slowed as β-blockers depress the AV node: complete heart block may be precipitated.

SNP (sodium nitroprusside) 0.3–10 micrograms/kg/min

Vasodilator. Converted to NO in vascular smooth muscle which increases cGMP production.

Action: venodilation reduces venous return to the heart, decreases heart size and preload, stroke volume, and cardiac output. Arterial vasodilation reduces afterload. Myocardial oxygen demands falls. At low doses the proportional decrease in preload is greater than the decrease in afterload. Reduces PVR.

Pharmacodynamics: half-life 1–2min. SNP reacts with hemoglobin to release highly toxic free cyanide ions (CN^-). Normally a hemoglobin molecule binds four cyanide ions to give cyanomethemoglobin which is non-toxic. Patients on high doses of SNP, or severe renal or hepatic dysfunction are at risk of cyanide and thiocyanate toxicity, where CN^- ions are no longer inactivated by binding with hemoglobin. CN^- inhibits cytochrome oxidase preventing mitochondrial oxidative phosphorylation, resulting in tissue hypoxia despite adequate PaO_2.

Indications: treatment of hypertension refractory to high-dose NTG.

Clinical use: infusions are started at the lower end of the dose range, with high dose NTG still running, and titrated against BP.

- Administration via a central line. Monitor BP with arterial catheter.
- Infusions should be tapered slowly to avoid rebound ↑SVR and PVR.
- Solution must be protected from light.
- *SNP toxicity*: tachyphylaxis, elevated mixed venous PO_2 (secondary to decreased cellular oxygen extraction), metabolic acidosis, fatigue, nausea, miosis, psychosis, hyper-reflexia, and seizures are signs of SNP toxicity. Discontinue SNP immediately. Ventilate with 100% oxygen. Treat with either sodium thiosulphate 12.5g iv slow bolus (mild or chronic SNP toxicity), or sodium nitrite (300mg iv slow push) and amyl nitrite (severe SNP toxicity).

Nitric oxide 0.05–50 parts per million

Naturally produced from L-arginine in endothelial cells: ↑cGMP in cells.

Actions: venodilation. Administration by inhalation results in selective pulmonary vasodilation and improvement in VQ mismatch, as it vasodilates areas of lung that are well ventilated. It is rapidly inactivated once it crosses the alveolar membrane, systemic effects are minimal.

Pharmacodynamics: half-life in blood 6s.

Indications: severe pulmonary hypertension, RV failure.

Clinical use: dilute NO is blended into the ventilator inlet gas. The lowest effective concentration should be utilized and responses monitored with a PA catheter 40ppm is maximal dose. Wean slowly.

- NO is not injected between the ventilator and the patient to avoid formation of toxic NO_2.

Epoprostenol (Flolan®) 3–50 ng/kg/min

- Synthetic analogue of prostaglandin (vasodilator, antithrombotic).
- ***Actions:*** administration by inhalation results in selective pulmonary vasodilation and improvement in VQ mismatch, as it vasodilates areas of lung that are well ventilated. It is rapidly inactivated once it crosses the alveolar membrane, systemic effects are minimal.
- ***Pharmacodynamics:*** half-life 20min: rebound hypertension.
- ***Indications:*** severe pulmonary hypertension, RV failure.
- ***Clinical use:*** nebulized into the ventilator inlet gas.

Antiarrhythmics

Potassium chloride (10–40mmol)

The myocardial cell membrane is highly permeable to potassium. The high concentration of potassium ions inside myocardial cells (140mmol/L) compared to low concentrations in extracellular fluid (4mmol/L) makes up the bulk of the resting cell membrane potential (📖 pp56–57).

Action: *hyperkalemia* (📖 p271) decreases the resting membrane potential and the intensity of the action potential. At high concentrations (>6.0mmol/L) this results in AV block, and reduction in the force of contraction. At concentrations >10mmol/L the heart arrests in diastole. *Hypokalemia* (📖 p270) increases the resting membrane potential resulting in increased excitability and arrhythmias such as atrial fibrillation.

Pharmacodynamics: 800mmol of K^+ enters glomerular filtrate per day: most is reabsorbed in the proximal tubules and loops of Henle (📖 p69).

Indication: maintenance of serum K^+ 4.5–5.0mmol/L postoperatively.

Clinical use: 10–20mmol in 50mL 5% dextrose can be administered by central line over 20min, or 40mmol in 1L 5% dextrose can be administered peripherally over 8h. ***Never bolus:*** can cause asystole

- Cardiac arrhythmias including asystole can result if infusion is too fast
- Respiratory and metabolic alkalosis exacerbate hypokalemia.
- Potassium should be administered cautiously in patients with renal impairment as hyperkalemia can quickly result.

Magnesium sulfate (1–2g)

The normal serum Mg^{2+} is 1.0mmol/L. Magnesium promotes cardiac electrical stability by affecting ion transport across membranes (📖 pp56–57).

Action: arrhythmias associated with magnesium depletion can be minimized with magnesium replacement.

Pharmacodynamics: renal excretion

Indication: Mg^{2+} may be given empirically to treat arrhythmias in cardiac surgical patients who are normally Mg^{2+} depleted.

Clinical Use: 1–2g magnesium sulfate in 50mL 5% dextrose may be given via central line over 20min, 1–2g MgCl may be given orally.

- <1% of total body magnesium is in the serum: intracellular levels do not correlate well with serum magnesium.

Calcium chloride (250–1000mg iv)

Calcium is an inorganic substance physiologically active only as Ca^{2+}

Action: calcium plays an important role in excitation–contraction coupling. It is a positive inotrope and reverses myocardial depression.

Pharmacodynamics: calcium is incorporated into bone and binds to protein, heparin, and citrate.

Indication: hypocalcemia, hyperkalemia, massive blood transfusion (📖 p149), decreased myocardial contractility from calcium channel blockers, protamine, and peri-arrest.

Clinical use: calcium chloride should be given through a central line: peripheral administration can cause sclerosis of veins.

- Can cause severe bradycardia or complete heart block.

Amiodarone (150–300mg bolus iv)

Amiodarone is an iodinated benzofuran.

Action: increases the refractory period of all cardiac tissues: atria, ventricles, and conducting system. It reduces sinus node automaticity. It is a non-competitive α- and β-antagonist, with a vasodilatory action.

Pharmacodynamics: half-life is 20–100 days. Elimination is by hepatic metabolism. High lipid solubility results in tissue accumulation.

Indication: AF, ventricular and supraventricular tachycardia.

Clinical use: loading iv: 300mg in 50mL 5% dextrose over 1h followed by 900mg in 250mL 5% dextrose over 23h, then reducing dose of 600mg per 24h for 1 week, then 400mg per 24h for 1 week. Loading orally: 400mg tid for 24h, then 200mg tid for 1 week, then 200mg bid for 1 week. Maintenance dose is 200mg per 24h. Amiodarone should be discontinued after 4–6 weeks if the patient reverts to sinus rhythm to reduce the risk of toxicity.

- Arrhythmias including torsades, and hypotension can occur.
- Amiodarone is a negative inotrope: avoid in patients with poor LV.
- Amiodarone can cause hepatitis, pulmonary fibrosis, hypothyroidism or hyperthyroidism, neurologic and gastrointestinal problems. Liver, respiratory, and thyroid function should be monitored regularly.
- Amiodarone potentiates warfarin, digoxin, and diltiazem.

Digoxin (maintenance 62.5–250 micrograms od)

A glycoside derived from the foxglove plant.

Action: digoxin increases intracellular calcium by inhibiting Na^+, K^+- ATPase. Increased Ca^{2+} release from the sarcoplasmic reticulum resulting in increased force of contraction. Ventricular automaticity is increased, but atrioventricular conduction is decreased.

Pharmacodynamics: onset 30 min. Half-life 1.7 days. Renal excretion.

Indication: fast AF and supraventricular tachycardias.

Clinical use: adult loading dose is up to 1250 micrograms over 24h in iv increments of 125–250 micrograms. Maintenance dose is titrated by serum levels: 1–2.0nmol/L 6–8h post dose.

- Digoxin will slow the rate of these arrhythmias, not achieve cardioversion: its inotropic effect is particularly beneficial in poor LV.
- Digoxin toxicity results in arrhythmias including fast AF, VT, and VF, AV block, and GI and CNS symptoms.
- *Digoxin toxicity* is exacerbated by $\downarrow Mg^{2+}$, $\uparrow Ca^{2+}$, and $\downarrow K^+$, alkalosis and acidosis, renal failure and hypothyroidism, all of which can result in toxicity despite low serum digoxin levels. Treat by correcting electrolytes. Digibind® 1 vial (38mg) in 4mL water iv is an antidote.

Lidocaine (1mg/kg iv bolus)

Tertiary amine local anesthetic that blocks Na^+ channels.

Action: activity localized to ventricles decreasing automaticity. Lidocaine can convert ventricular escape rhythm to asystole.

Pharmacodynamics: duration 15–30min. Hepatic metabolism.

Indication: ventricular arrhythmias. Lidocaine has *no* atrial action.

Clinical use: overdose results in confusion and seizures, respiratory depression, and sedation, but very safe drug within therapeutic range.

Chronotropes

Chronotropes increase heart rate

- Pacing is a quicker, more easily reversible option than drug boluses and infusions for raising heart rate in most patients.
- β_1 stimulation leads to ↑contractility, heart rate, and conduction.
- *Muscarinic cholinergic* activity is predominantly parasympathetic.

Atropine (0.3–1mg iv bolus)

Atropine is a belladonna alkaloid.

Action: a competitive antagonist at muscarinic cholinergic receptors, reducing parasympathetic tone to 'reveal' underlying sympathetic tone.

Pharmacodynamics: almost instant onset. When given iv offset is 15–30min. When given im, sc, or po offset 4h. Renal elimination.

Indication: bradyarrhythmias, reduction of oral secretions.

Clinical use: normally given as in iv bolus, starting with 0.3 micrograms iv, repeated up to 1.0mg as necessary.

- Atropine can also be given im, sc, po, and via the ET tube.
- Low doses (<0.2 micrograms) can exacerbate bradycardia.
- Antimuscarinic cholinergic side effects include urinary retention, sedation and confusion.

Isoprenaline (0.01–0.3 micrograms/kg/min)

Synthetic catecholamine

Action: direct β_1 (↑contractility, conductivity, and heart rate) and β_2↑ (vasodilation and bronchodilation) effect. No α activity.

Pharmacodynamics: plasma half-life 2min. Hepatic metabolism by MAO, 40% conjugated, 60% excreted unchanged.

Indication: bradycardia unresponsive to atropine resulting in low cardiac output states (📖 p221), AV block and β-blocker overdose. Indications pretty limited as usually AV pacing or low-dose epinephrine can achieve similar results without associated vasodilatation.

Clinical use: give 2mg isoprenaline in 250mL 5% dextrose, at a rate of 0.02–0.3 micrograms/kg/h).

- Isoprenaline can be safely given through a peripheral line.
- Cardiac output is increased as a result of increased heart rate, contractility, and reduced afterload.
- An infusion gives a sustained improvement in heart rate that may not be achieved with atropine boluses.
- Peripheral vasodilation may result in hypotension: the combination of tachycardia, decreased diastolic time, increased contractility, and hypotension can precipitate myocardial ischemia because of the mismatch in myocardial oxygen supply and demand.

- Arrhythmias may be precipitated by the increased conductivity.
- Profound coronary vasodilation can theoretically result in diversion of myocardial blood flow from ischemic myocardium.

Inotropes with a chronotropic effect

The following inotropes also tend to result in an increase in heart rate, either as a result of β_1 effects, or as a result of reflex tachycardia associated with vasodilation:

- Epinephrine.
- Dopamine.
- Dopexamine.
- Dobutamine.

Diuretics

Key facts

- Diuretics cause net loss of sodium and water from the body by action on the kidney: their primary effect is to decrease Na^+ and Cl^- reabsorption from the filtrate: increased water loss is secondary.
- Diuretics act by: (1) direct action on the cells of the nephron (loop, thiazide and K^+- sparing diuretics) or (2) modifying the content of the filtrate (osmotic diuretics).

Loop diuretics

Furosemide, bumetanide (both sulphonamides).

Action: loop diuretics block reabsorption of electrolytes from the loop of Henle, increasing excretion of water and Na^+, Cl^-, K^+, Ca^{2+}, and Mg^{2+}. As much as 25% of the glomerular filtrate may pass out of the nephron resulting in an intense diuresis. Bumetanide is 40–50× more potent than furosemide. Both caused a transient vasodilation and ↓CO.

Pharmacodynamics: given iv onset is 30min, orally onset is 4–6h.

Indication: volume overload, pulmonary edema, hyperkalemia.

Clinical use: boluses iv or po, (10–120mg), or iv infusion (1–25mg/h).

- Loop diuretics cause ↓K^+, metabolic alkalosis (as a result of loss of H^+ and K^+ ions), and rarely ototoxicity.
- Allergic reactions occur in patients sensitized to sulphonamides.
- Although loop diuretics are used to convert oliguric to polyuric renal failure *the temporary increase in urine output has no prognostic effect on the progression of acute tubular necrosis*. In hypovolemic states loop diuretics may exacerbate acute tubular necrosis by provoking a diuresis, worsening hypovolemia, cardiac output, and renal ischemia.

Thiazide diuretics (potassium-depleting diuretics)

Bendrofluazide, bendroflumethiazide, indapamide, ***metolazone***

Action: these have a moderate diuretic effect. They inhibit reabsorption of Na^+ and Cl^- in the distal tubule. Potassium excretion is significant.

Pharmacodynamics: onset 1–2h, duration 12–24h. Excreted by glomerular filtration and tubular secretion.

Indication: cardiac failure and hypertension.

Clinical uses: bendrofluazide po 2.5–5mg od. Indapamide 2.5–5.0mg od.

- Thiazides compete with uric acid for excretion, raising serum urate.
- Hypersensitivity reactions include pancreatitis and pulmonary edema.

Potassium-sparing diuretics

Spironolactone (aldosterone antagonist), ***amiloride.***

Action: spironolactone blocks the sodium-retaining and potassium-secreting action of aldosterone in the distal tubules. It also reduces hydrogen ion secretion. Amiloride acts directly on sodium channels, preventing sodium uptake in the distal tubule.

Pharmacodynamics: spironolactone half-life 10min, but active metabolite has a half-life of 16h. Amiloride has a half-life of 24h.

Indication: prevention of hypokalemia. Ascites, pleural and pericardial effusions caused by congestive cardiac failure. Nephrotic syndrome.

Clinical uses: spironolactone 25–50mg od. Amiloride 2.5–5mg od (sometimes in conjunction with furosemide as Frumil® 5/40 or 2.5/20).

Osmotic diuretics

Mannitol (polysaccharide).

Action: pharmacologically inert substance which is filtered in the glomerulus but not reabsorbed by the nephron. In the proximal tubule passive water reabsorption is limited by the presence of the non-reabsorbable solute within the tubule, as is Na^+ reabsorption. There is a large increase in the amount of water excreted with a proportionally much smaller increase in the amount of Na^+ excreted. Unlike the loop diuretics, mannitol retains its efficacy even in low glomerular filtration states. In acute renal failure due to hypovolemic states where the glomerular filtration rate is reduced, and absorption of water and Na^+ in the proximal tubule is almost complete, the distal nephron is at risk of irreversible damage. Mannitol protects against this by increasing the volume of filtrate passing through the distal tubules. Mannitol is also a free radical scavenger. Because proportionally much less Na^+ than water is excreted, mannitol is not useful in treating conditions associated with Na^+ retention.

Pharmacodynamics: rapid onset, short half-life.

Indications: Prevention of renal failure secondary to hypoperfusion or hemoglobinuria, and treatment of cerebral and pulmonary edema.

Clinical use: Initial dose 12.5g to a maximum of 0.5g/kg iv. 12.5g is routinely added to the pump prime in some centers to maintain a diuresis on bypass.

- Mannitol initially expands intravascular volume: this may exacerbate congestive cardiac failure.
- Mannitol causes hypotension if administered too rapidly.

When should diuretics be given for oliguria?

- When you are reasonably confident that pre-renal causes, e.g., hypovolemia, hypotension, low cardiac output and post-renal causes such as Foley catheter problems or obstruction are not the cause, *and*
- Positive fluid balance will be detrimental to the patient now or in the near future, e.g., patient with borderline lung function, severe LV or RV dysfunction, obvious fluid overload.
- There are some broad rules of thumb:
  - Almost all patients need diuresis once they leave ICU.
  - Most ICU patients will do fine whether you give fluid or diuretic for low urine output, and it isn't worth a protracted debate. *But*:
  - Oliguric patients with good hemodynamic indices on minimal pressors can safely be diuresed early if extubation is a priority.
  - Oliguric patients needing large doses of pressor to maintain MAPs most likely need fluid to avoid renal malperfusion, and careful assessment of the underlying problem (📖 pp184–185).
  - Septic (📖 p239) and vasoplegic (📖 p190) patients have high fluid requirements, and diuresis is rarely indicated in the acute phase.
  - Patients with SAM (📖 p190) or LV hypertrophy should be diuresed very, very cautiously: sometimes a little fluid overload is safer.

EKG monitoring

Indications

All cardiac surgical patients should have a 12-lead EKG preoperatively, and daily 12-lead EKGs postoperatively, as well as EKG monitoring intraoperatively and for 48h postoperatively. This facilitates:

- Diagnosis of ischemia.
- Diagnosis of arrhythmias and conduction defects.
- Diagnosis of electrolyte disturbances.

Contraindications

There are no contraindications to electrocardiography.

Techniques

The electrodes and EKGs are shown in Fig. 5.12.

3-electrode system: this system is used for continual monitoring, e.g., during surgery. It utilizes three electrodes, placed on the right arm, left arm, and left leg. The limb leads are placed on the back of the shoulders and hips to avoid interfering with the operative field. Many modifications of the system exist to aid diagnosis of arrhythmias and ischemia.

5-electrode system: five leads, four each on one extremity and one chest lead placed in the anterior axillary line in the 5th intercostals space gives additional sensitivity for diagnosis of ischemia.

12-electrode system: three extremity leads and six chest leads allow a 12-lead EKG to be produced (three augmented leads are produced by combining the extremity leads). This is the most sensitive diagnostic EKG. It is not used for monitoring.

Common problems

Poor trace: ensure good electrical contact. Skin should be hair and oil free. Change adhesive electrodes for fresh ones.

Wandering baseline: Check electrical contact, untwist cables, make sure that patient is lying still.

Interference causing fine oscillations of baseline: electrical interference is caused by poor insulation and filtering. Check cable insulation, replace cable if necessary. Skeletal muscle tremor including Parkinson's disease can also cause interference: ensure patient is warm and relaxed.

Basic EKG interpretation

This is described on 🕮 p212.

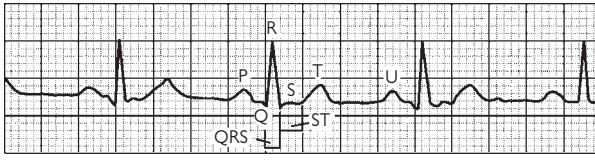


Fig. 5.11 Normal sinus rhythm.

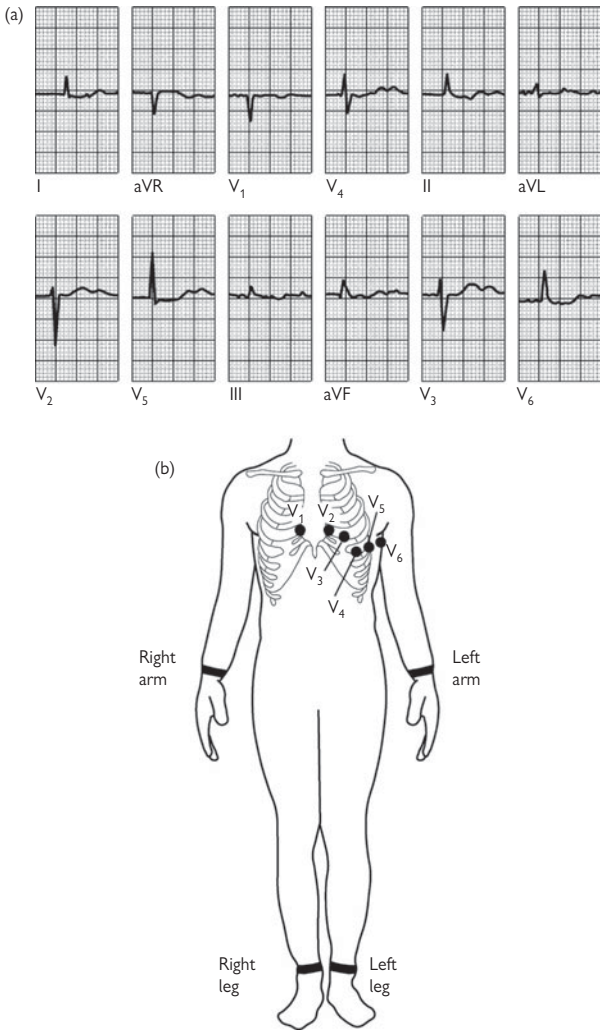


Fig. 5.12 (a) EKG readings and, (b) Electrode placement and 12 frontal plane axes for a 12-lead EKG.

Basic EKG interpretation

Normal sinus rhythm

Fig. 5.11, p210, shows normal sinus rhythm. Note the P wave, followed in 0.12–0.2s (3–5 small squares) by a QRS complex that lasts <0.12s (<3 squares), and which is followed by a T wave. The ST segment is flat and is at the isolectric point (the same level as the PQ segment).

Myocardial ischemia

Features: Fig. 5.13 shows an EKG trace typical of myocardial ischemia. Note how the ST segments are elevated. ST segment elevation in leads II, III and aVF reflects inferior ischemia (right coronary artery system), in leads V1–V6 reflects anterior ischemia (left anterior descending artery) and V4–6 suggests septal ischemia (posterior descending and circumflex arteries). New ST segment elevation as little as 1mm in these leads is significant and should be acted on (📖 p185). Note. ST segment is NOT well correlated with ischemia when conduction is abnormal as in left bundle branch block, complete heart block, or ventricular pacing.

Therapy: 📖 p185.

Acute MI or pericarditis?

- ST elevation (STE) in pericarditis is concave: in AMI it is flat or convex (Fig. 5.13a).
- STE in pericarditis is diffuse: in AMI it is in coronary distribution.
- PR depression in pericarditis, Q wave in AMI.
- T wave inversion only AFTER ST segments normalize in pericarditis, whereas in AMI it may coexist with STE.

Myocardial infarction

Features: Fig. 5.13 shows EKG traces typical of myocardial infarction. Note ST segment elevation followed by ST segment depression, Q waves (downwards deflection after the P wave rather than upwards) and later T wave inversion in the territories described.

Therapy: 📖 p185.

Pericarditis

Features: this is occasionally a feature of postoperative patients' EKGs at day 5 post surgery. Note concave upwards ST elevation in all leads.

Therapy: 📖 ***p264.***

Hyperkalemia and hypokalemia

Features: although regular blood gas sampling should mean that hyperkalemia is never diagnosed as a result of EKG changes, patients in renal failure, and longer-term patients without arterial lines are at risk of hyperkalemia. Note the flattening of the P wave, tenting of the T wave, and widening of the QRS complex. The sine waveform heralds cardiac arrest. In hypokalemia U waves are a feature.

Therapy: 📖 pp270–271.

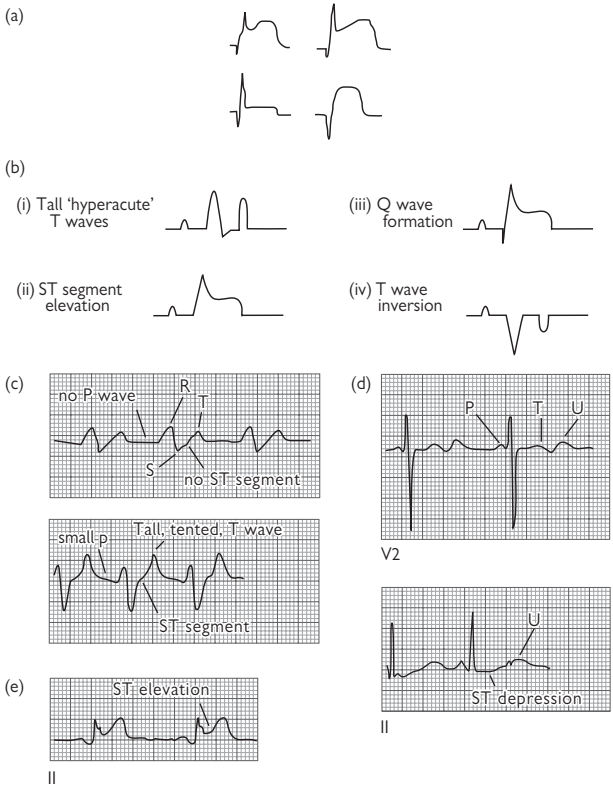


Fig. 5.13 (a) Possible morphology of ST segment elevation in acute MI. The commonest causes of ST segment elevation are not ischemia/infarction and ST segment elevation therefore has poor sensitivity for AMI but it's an important diagnosis to get right. Concave shapes are usually not AMI. (b) Temporal changes seen in acute MI. (c) EKG changes in hyperkalemia. (d) 'U' waves in hypokalemia. (e) Concave upwards ST segment elevation, characteristically seen in all leads in pericarditis (as opposed to coronary distribution specific ST segment elevation see in acute MI).

Causes of dysrhythmias

Key facts

Over 1/3 of patients develop supraventricular arrhythmias postoperatively, as a result of changes in automaticity and conduction. A number of common factors contribute to arrhythmias. These need to be rapidly excluded when managing arrhythmias at the same time as instituting the specific treatment described on 📖 pp216–231.

Preoperative causes

- *Age*: postoperative arrhythmias are commoner in older patients.

Preoperative arrhythmias:

- Patients with pre-existing AF usually revert to AF even if their initial rhythm after bypass is sinus rhythm.
- Preoperative heart block and ventricular arrhythmias are often exacerbated postoperatively.

Etiology of heart disease:

- Mitral valve disease associated with an *enlarged left atrium*.
- Patients with *dilated left ventricles* (ischemic cardiomyopathy, decompensated valvular heart disease).
- *Hypertrophic left ventricles* (aortic stenosis, hypertensive cardiomyopathy) are more prone to ventricular arrhythmias.

Intraoperative causes

- *Poor intraoperative cardioprotection.*
- Ineffective cooling.
- Inadequate cardioplegia.
- Incomplete electromechanical arrest.
- Ventricular distension.

Surgical procedure

- Incomplete myocardial revascularization.
- Damage to the conduction pathways during decalcification and suturing of mitral or aortic valve annulus, atriotomy, or ventriculotomy.
- Ineffective de-airing and exclusion of debris from the circulation.
- There is a 1–2% incidence of complete heart block requiring permanent pacemaker insertion after aortic valve replacement (📖 p263).

Postoperative causes

Cardiovascular

- Myocardial ischemia secondary to incomplete revascularization, coronary vasopasm, conduit kinking or occlusion, hypoxia, or high myocardial oxygen demand should be excluded as a cause of arrhythmias.
- Failure of valve repair or prosthesis.
- Hypovolemia.
- Pulmonary artery catheter (particularly if placed in RV).
- Less commonly, central venous catheter insertion.
- Intracardiac pacing wires.
- Endocarditis.
- Pericarditis.

Respiratory
- Hypoxia.
- Hypercapnia.
- Acidosis.
- Endotracheal tube irritation.
- Pneumothorax.
- Atelectasis.
- Pneumonia.

Electrolyte imbalance
- Hyperkalemia.
- Hypokalemia.
- Hypomagnesemia.
- Hypocalcemia.

Drug related (withdrawal/toxicity)
- β-blockers.
- Digoxin.
- Calcium channel blockers.
- Bronchodilators.
- Tricyclic antidepressants.
- Adrenergic and dopaminergic infusions are arrhythmogenic.
- Alcohol.

Metabolic
- Hyper- and hypothyroidism.
- Hypoglycemia.

Systemic
- Fever, hypothermia.
- Anxiety and pain predispose to tachyarrhythmias.
- Bradycardias are common when the patient is asleep.

Vagal stimulation
- NGT insertion.
- Gastric dilation.
- Intubation.
- Nausea and vomiting.

Mechanical
- Pericardial collections.
- Tamponade.
- Tension pneumothorax.
- Cold fluids injected into RA via central line.
- Chest contusion.

Lethal arrhythmias

Cardiac arrest in the monitored patient

- The ALS algorithms are on the inside front cover.
- Defibrillate VF or pulseless VT.
- Secure an airway.
- Hand ventilate with 100% oxygen at a rate of 10 breaths/min.
- Start external cardiac compressions at a rate of 100/min if defibrillation not possible or not indicated (EMD, asystole).
- Listen for bilateral breath sounds. Look for symmetrical expansion.
- Get iv access, check electrolytes, blood sugars, blood gases.
- Look for reversible causes: mechanical—cardiac tamponade, tension pneumothorax, massive PE, and metabolic—hypokalemia, hyperkalemia, acidosis, hypoxia, and hypovolemia and hypothermia.
- Give 10mL 1:10 000 epinephrine if no response.

Ventricular fibrillation

Features: irregular, fine, sinusoidal trace (Fig. 5.14a). No cardiac output.

Therapy: immediate defibrillation. Do ***not*** delay defibrillation for ***any*** other procedures, e.g., intubation, cardiac massage, starting iv therapy.

- 3× shock 200J (biphasic, if monophasic defibrillator give up to 360J).
- Commence CPR.
- Give 1mg epinephrine iv (10mL of 1:10,000), repeat at 3min intervals.
- Repeat cycle.
- Give 1mg/kg lidocaine, correct acidosis and electrolytes.
- Emergency re-exploration is sometimes indicated for refractory VF in early postoperative period if surgical cause suspected (kinked conduit) or for institution CPB with a view to RVAD or LVAD support.

Pulseless ventricular tachycardia

Features: wide complex tachycardia (Fig. 5.14b). No cardiac output.

Therapy: treat as VF.

- Sometimes runs of VT are short and self-limiting. If recurrent these are best treated with lidocaine or amiodarone infusions, magnesium, and correction of acid–base and electrolyte abnormalities.
- Sustained and non-sustained VT may degenerate into VF.

Asystole

Features: flat EKG trace (Fig. 5.14c) or low amplitude fibrillation. No cardiac output.

Therapy: *pace* if possible (📖 p228). If not commence CPR immediately.

- *If epicardial pacing wires are present pace immediately* (📖 p228).
- Give atropine 1mg iv once, and epinephrine 1mg iv at 3min intervals
- Fine, low amplitude VF resembles asystole and may respond to electrical defibrillation: if in doubt defibrillate.
- An absolutely straight line suggests telemetry lead disconnection.
- Treat hyperkalemia (📖 p271) aggressively with 50mL 50% dextrose and 15 units insulin iv push.

Pulseless electrical activity (PEA)

Previously called electromechanical dissociation (EMD).
Features: no cardiac output despite EKG trace compatible with output.
Therapy: commence CPR immediately and try to pace if possible. Attempt pacing if epicardial wires are present.

Complete heart block

Features: wide complex < 35bpm (Fig. 5.14d) with no or poor cardiac output.
Therapy: pace if possible. Commence CPR if no cardiac output. Usually the patient will maintain some cardiac output and chest compressions are not necessary.

- If epicardial wires are present pace immediately (📖 p228).
- If there is an adequate cardiac output do not commence chest compressions. Treat pharmacologically (see 📖 p206).
- If no cardiac output give atropine 1mg iv once, and epinephrine 1mg iv (10mL of 1:10,000) at 3min intervals until cardiac output restored.
- Transvenous pacing wires should be placed immediately if epicardial wires are not functional/present.

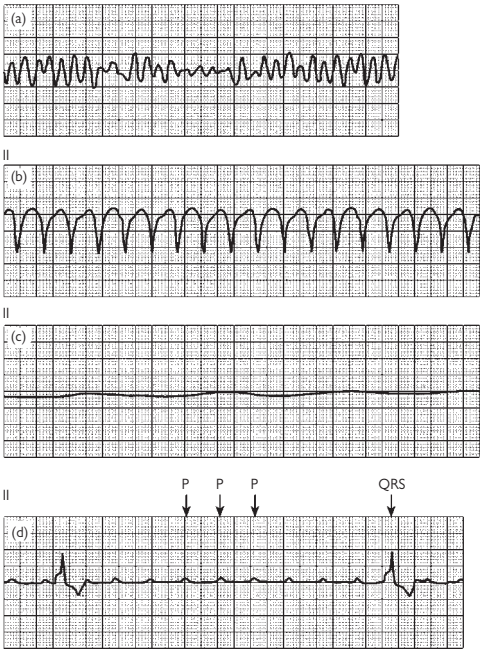


Fig. 5.14 Rhythm strips showing (a) VF, (b) VT, (c) asystole, (d) complete heart block.

Dangerous atrial arrhythmias

These arrhythmias are potentially dangerous in patients with: (1) borderline cardiac output, (2) stiff, non-compliant hypertrophic ventricles where reduced filling times lead to decreased stroke volumes, and (3) SAM (📖 p190). *Loss of synchronized atrial contraction may reduce stroke volume by up to 30%.*

Atrial fibrillation and atrial flutter

Features: irregularly irregular usually narrow complex tachycardia. Atrial rate >380 in fibrillation, atrial rate <380 in flutter (Fig. 5.15).

Therapy: there are 2 options depending on the clinical status:

- *Synchronized DC cardioversion:* This is covered on 📖 p224. It is indicated in two main groups of postoperative cardiac surgical patients:
  - Acute hemodynamic compromise
  - The stable adequately anticoagulated patient (📖 p262), who has not responded to appropriate arrhythmic and electrolyte management.
- *Pharmacological cardioversion:* This is indicated in hemodynamically stable postoperative patients.

Management of atrial fibrillation or flutter on ICU

- Correct hypokalemia: give 20mmol KCl in 50mL 5% dextrose via the central line over 10–20min, and repeat as necessary to obtain serum K^+ of 4.5–5.0mmol/L.
- Correct hypomagnesemia: give 1–2g $MgSO_4$ in 50mL 5% dextrose via the central line empirically if it has not already been given, as up to 60% of patients are hypomagnesemic postoperatively, and serum magnesium represents only 1% of total body stores.
- These two measures are usually sufficient to convert early AF.
- Correct hypoxia (📖 p160) and acidosis (📖 p170).
- Start weaning potentially arrhythmogenic infusions, e.g. isoprenaline.
- If the patient is hemodynamically compromised give 1 × 100J *synchronized* DC cardioversion (📖 p224) increased by 50–100J increments to maximum of 360J in the anesthetized patient.
- Amiodarone 150–300mg in 50mL 5% dextrose over 1h via central line, followed by amiodarone 900mg in 250mL 5% dextrose over 23h via central line (📖 p205) may achieve cardioversion in patients with good to moderate LV, and is first choice agent in most units.
- Digoxin 125 mcg in 50mL 5% dextrose over 20min via central line, repeated until rate control is achieved up to a maximum of 1250 micrograms in 24h, is preferable for rate control in patients with poor LV.
- Although β-blocker withdrawal is one of the commonest causes of postoperative AF, it is not appropriate to restart beta blockers in patients still requiring or very recently weaned from inotropic support.
- Overdrive pacing can work in atrial flutter (📖 p230).
- Consider starting a therapeutic dose of low-molecular-weight heparin e.g. enoxaparin 1mg/kg sc bid in patients who have been in AF >24h, and who are not a bleeding risk (avoid heparin for the first few days after complex aortic surgery where normal coagulation is the cornerstone of maintaining hemostasis).

- The target INR for patients in chronic AF is 2.5–3.0, and warfarin should be commenced on the 2nd or 3rd postoperative evening for patients still in AF, with no bleeding problems.

Why treat atrial fibrillation and flutter?

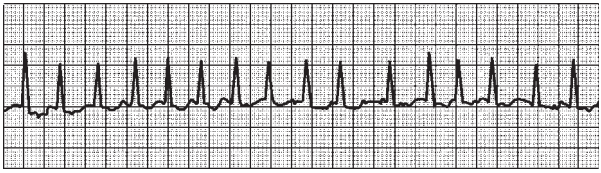
These are not benign arrhythmias. In addition to the hemodynamic compromised described earlier:

- 15% of non-anticoagulated patients have intracardiac thrombus by TEE (usually in the LA appendage) within 48h of new AF.
- 30% of patients without an embolic event can be shown to have intracardiac thrombus within 72h of onset of AF.
- TTE has a much lower sensitivity than TEE for LA thrombus.
- 5% of non-anticoagulated patients undergoing DC cardioversion for AF will have an embolic event vs. 1% of anticoagulated patients.
- The rate of stroke in patients with AF under the age of 65 is approximately 1% per year, increasing to approximately 5% per year in patients over the age of 65.
- Although the natural history of AF following CABG is not well documented, non-anticoagulated patients with a history of angina or myocardial infarction have a 6–8% risk of stroke per annum.
- The risk of intracranial hemorrhage in anticoagulated patients is about 0.5% per year, increasing to about 1% in patients > 80 years.

Supraventricular tachycardia

Features: narrow complex regular tachycardia, rate 150–250bpm. It is sometimes difficult to distinguish between an SVT and atrial flutter.
Therapy: DC cardioversion as described for AF.

- Carotid sinus massage has been recommended to slow the ventricular response rate, because re-entrant rhythms involving the AV node may be interrupted in this way, and to aid diagnosis by revealing the atrial rhythm. Bear in mind that arteriopaths are at risk of cerebral emboli from carotid plaques dislodged in this way.
- Adenosine (3mg iv bolus increased by 3 mg increments and repeated at 2min intervals) produces a transient AV block. The half-life of adenosine is 10s which is sufficient to terminate a supraventricular tachycardia but will result in up to 10s of complete heart block.
- Calcium channel blockers (diltiazem 0.25mg/kg iv over 2min repeated 15min later) convert 90% of patients to sinus rhythm.
- Digoxin helps gain rate control in refractory SVT.



II

Fig. 5.15 Rhythm strip featuring atrial fibrillation: not the irregularly irregular QRS complexes with absent p waves.

Other dangerous arrhythmias

This section describes the management of VT in *hemodynamically stable* patients. If the patient has lost cardiac output (📖 p216) get help and follow the ALS algorithm on the inside front cover.

Ventricular tachycardia (pulse present)

Features: wide complex regular tachycardia, impaired cardiac output.

Therapy:

- If at any stage the patient loses cardiac output (📖 p216) start the ALS algorithm for VF/pulseless VT.
- Correct hypokalemia: give 20mmol KCl in 50mL 5% dextrose via the central line over 10min, and repeat as necessary to obtain serum K^+ of 4.5–5.0mmol/L.
- Correct hypomagnesemia: give 1–2g $MgSO_4$ in 50mL 5% dextrose via the central line empirically if it has not already been given, as up to 60% of patients are hypomagnesemic postoperatively, and serum magnesium represents only 1% of total body stores.
- Correct hypoxia (📖 p160 and acidosis (📖 p170).
- Start weaning potentially arrhythmogenic infusions, e.g., isoprenaline.
- If the patient is hemodynamically compromised sedate with the help of an anesthesiologist and give 1 × 200J synchronized biphasic shock, and repeat if needed.
- Amiodarone 300mg in 50mL 5% dextrose over 1h via central line, followed by amiodarone 900mg in 250mL 5% dextrose over 23h via central line (📖 p205) may achieve rate control and is first choice agent in most units.
- Alternatively lidocaine 1mg/kg iv bolus, followed by an iv infusion of 4mg/min for 30min, 2mg/min for 2h, then 1 mg/min may achieve cardioversion.
- Look for and treat any evidence of myocardial ischemia (📖 p185).

Ventricular ectopy

Features: wide complexes that may occur occasionally or in couplets (ventricular bigeminy), may be unifocal or multifocal, and are usually followed by a compensatory pause.

Therapy: although ventricular ectopics occurring less than one per screen are usually benign, particularly if present preoperatively, in a small number of patients they reflect myocardial ischemia and may herald lethal arrhythmia.

- Look for signs of myocardial ischemia and treat accordingly (📖 p185).
- Correct hypokalemia: give 20mmol KCl in 50mL 5% dextrose via the central line over 10min, and repeat as necessary to obtain serum K^+ of 4.5–5.0mmol/L.
- Correct hypomagnesemia: give 1–2g $MGSO_4$ in 50mL 5% dextrose via the central line empirically if it has not already been given, as up to 60% of patients are hypomagnesemic postoperatively, and serum magnesium represents only 1% of total body stores.

- Correct hypoxia (📖 p160) and acidosis (📖 p170).
- Atrial pacing at a rate greater than the ventricular rate may abolish ectopics and improve cardiac output, but does not address any underlying pathology.

Sinus or junctional bradycardia

Features: narrow complex, rate <60bpm. Impaired cardiac output.

Therapy: if epicardial pacing wires are present, pace immediately (📖 p228).

- Stop any infusions causing bradycardia, including amiodarone, β-blockers, and digoxin.
- Give a bolus of atropine (0.3 mg iv) repeated in increments up to 1mg.
- Start isoprenaline infusion (25mg in 250 mL 5% dextrose, at a rate of 0.01–0.3 micrograms/kg/min).

Second-degree heart block

Features: either Mobitz type I—PR interval increasingly prolonged with regularly absent QRS or Mobitz type II—PR interval constant with intermittently absent QRS (Fig. 5.16).

Therapy: a permanent pacemaker may be indicated (📖 p263): discuss with cardiologists if second degree heart block persists beyond post-op day 4.

Trifascicular block

Features: wide QRS complex (>0.12s or 3 small squares) and prolonged PR interval (>0.2s or 5 small squares) (Fig. 5.16).

Therapy: do not remove temporary pacing wires until the patient has been evaluated by a cardiologist. Hold off starting β-blockers. A permanent pacemaker is indicated if trifascicular block is associated with symptomatic pauses or other sinus irregularities.

Left bundle branch block

Features: wide QRS complex (>0.12s or 3 small squares). The ST segment cannot be interpreted: elevation or depression is *not* well correlated with ischemia in left bundle branch block.

Therapy: as for trifascicular block.

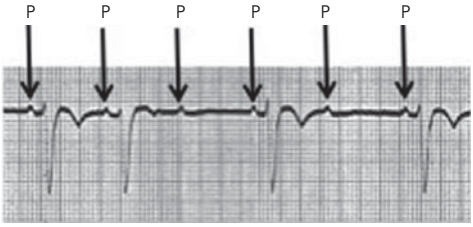


Fig. 5.16 Trifascicular block with Mobitz type II block. Note the wide QRS (left bundle block), prolonged but constant PR interval (which together indicate trifascicular block), and intermittently dropped QRS (Mobitz type II).

Benign arrhythmias

Sinus tachycardia

Features: sinus rhythm with a rate >100bpm. Rates >140 suggest SVT.

Therapy: although this is usually a benign response to sympathetic overactivity (pain, anxiety), it may be a compensatory response to poor cardiac output. Sinus tachycardia increases myocardial oxygen demand and reduces ventricular filling times. In hearts with stiff non-compliant ventricles characterizing preoperative AS, this may result in poor filling, decreased cardiac output and myocardial ischemia.

- Look for and treat the common underlying causes: pain, anxiety, pyrexia, gastric dilatation, hypercapnia, hypovolemia and anemia.
- Salbutamol nebulizers may be an unavoidable cause.
- If you are sure that the tachycardia is not compensating for poor cardiac output, and if the patient is not requiring inotropic support, and if the systolic BP is at least at pre-op values consider restarting at a low dose β- or calcium channel blocker if these were discontinued preoperatively. Short-acting drugs (metoprolol 25mg tid po) are safest.
- Inotropes, particularly epinephrine, dopamine, dopexamine, and dobutamine cause tachycardias. You may need to change or wean them.

Junctional tachycardia

Features: narrow complex tachycardia without P waves.

Therapy: junctional tachycardia is caused when junctional tissue has a faster intrinsic rate than the sinus node. Drugs slowing the sinus node should be stopped. Although it is a benign arrhythmia, loss of synchronized atrial contraction can reduce cardiac output by up to 30%.

- Stop digoxin and β-blockers.
- Attempt atrial or AV pacing.
- Consider metoprolol (25–50mg po tid) a short acting β-blocker, or diltiazem 60mg tid po to slow the junctional focus, with atrial or AV pacing if the underlying sinus rate is slow.
- Overdrive pacing may help establish synchronized AV contraction.

Atrial ectopics (Fig. 5.17a)

Features: P waves with a different morphology, associated with a longer than normal PR interval.

Therapy: premature atrial ectopics are benign but sometimes herald the onset of AF or atrial flutter, and they are worth treating.

- Correct hypokalemia: give 20mmol KCl in 50mL 5% dextrose via the central line over 10min, and repeat as necessary to obtain serum K^+ of 4.5–5.0mmol/L.
- Correct hypomagnesemia: give 1–2g $MgSO_4$ in 50mL 5% dextrose via the central line empirically if it has not already been given.
- Correct hypoxia (📖 p160) and acidosis (📖 p170).
- Give digoxin (125–250 micrograms iv over 1h, repeat up to a total of 1250 micrograms). Digoxin decreases the frequency of atrial ectopics and reduces the ventricular response rate if AF does develop.
- β-blockers decrease the frequency of ectopics.

1st-degree heart block (Fig. 5.17b)

Features: PR interval >0.2s (5 small squares).

Therapy: this is benign. In a patient that has undergone aortic valve surgery it suggests damage to the conduction pathways passing close to the aortic annulus. This often resolves over the weeks postoperatively as edema and hematoma settle. The patient warrants follow-up by a cardiologist as occasionally insertion of a permanent pacemaker is indicated, but unless first-degree heart block is accompanied by a new left bundle branch block or symptomatic bradyarrhythmias, temporary pacing wires can safely be removed on day 3/4.

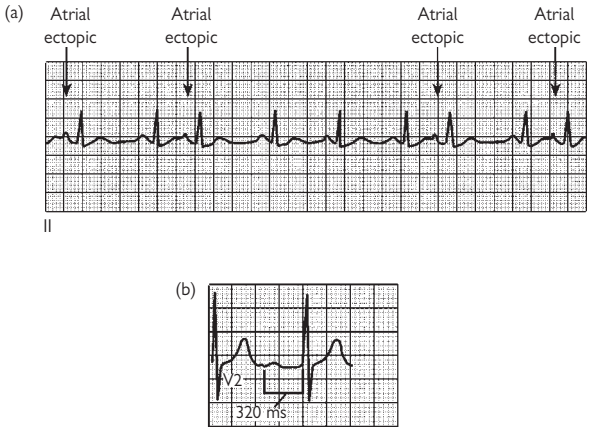


Fig. 5.17 Rhythm strips showing (a) atrial ectopics and (b) first-degree heart block.

Cardioversion

Key facts

Synchronized DC cardioversion is the treatment of choice for tachyarrhythmias compromising cardiac output, such as AF and SVT, and for AF refractory to chemical cardioversion.

Checklist for elective DC cardioversion

- Is it indicated?
  - Is the patient still in AF?
- Is it safe (📖 p262)?
  - Either AF has lasted <24h.
  - *Or* the patient must have had at least 6 weeks of formal anticoagulation.
  - *Or* a TEE excluding intracardiac thrombus.
- Is the patient ready?
  - The potassium should be 4.5–5.0.
  - The INR if anticoagulated should be >2.0.
  - The patient should have a valid consent form.
  - The patient should be starved for 6h.

DC cardioversion for AF and SVT

Patient should be anesthetized: some anesthesiologists prefer not to intubate, managing the airway with a bag and mask. You can either use adhesive external defibrillator pads which remain fixed to the patient until the procedure is completed, or hand-held paddles and gel pads. As soon as the anesthesiologist is happy:

- Expose chest.
- Place pads on chest in position shown in Fig. 5.18a: the aim is to direct as much of the current as possible through the heart.
- Place three EKG electrodes on the patient as shown, and connect to the fibrillator so that an EKG trace is visible.
- Switch defibrillator on and turn dial on to appropriate power setting (100J, 200J, 360J; Fig. 5.18c).
- ***Press the SYNC button, and ensure that each R wave is accented on the EKG: failure to do this can mean that a DC shock is delivered while the myocardium is repolarizing, resulting in VF. Check that the SYNC button is on before every shock for AF.***
- If you are using hand-held paddles hold them firmly on the gel pads.
- Perform a visual sweep to check that no one is in contact with the patient at the same time as saying clearly, 'Charging. Stand clear'.
- Press the charge button.
- Press the shock button when the machine is charged.
- If the shock has been delivered successfully the patient's muscles will contract violently: anyone in contact with the patient will experience a large electric shock.
- Check the rhythm.
- If still AF press the charge button and repeat the sequence.

Complications of DC cardioversion

- Complications of general anesthesia (📖 p256).
- Systemic embolization (📖 p275, p278).
- Failure to cardiovert.
- Burns from incorrect application of gel pads.
- Muscle pain from involuntary contraction.
- Arrhythmias including asystole and VF.

Common pitfalls

Failure to deliver a shock: check that the defibrillator is switched on and adequately charged. Check that the correct power setting has been selected. Change the machine.

Failure to cardiovert: check the latest available serum potassium was 4.5–5.0. Check that the correct power setting has been selected. Replace gel pads with fresh ones. Reposition the patient on their side and the pads as shown and try two further shocks at 200J (Fig. 5.18). Don't start at too low a power setting: each shock leaves the myocardium less sensitive to further shocks. There is some evidence that 200J as the first power setting results in less myocardial damage and a better conversion rate than multiple shocks at lower power settings.

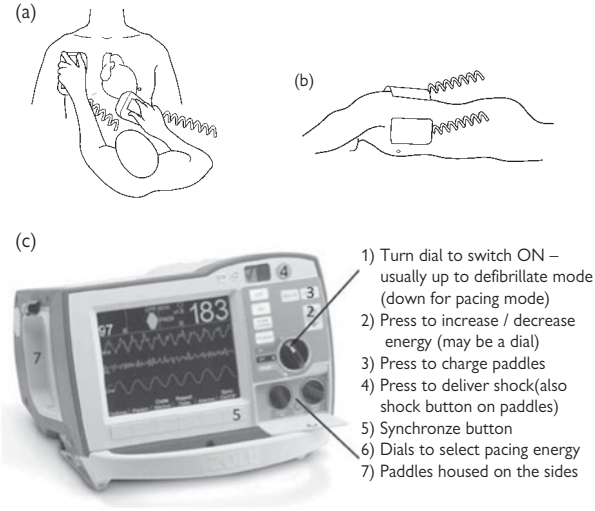


Fig. 5.18 Using defibrillators. (a) Correct positioning for defibrillation and cardioversion. (b) Alternative positioning for synchronized DC cardioversion. (c) Standard defibrillator console. 'On' dial may also be method of selecting energy level. © Zoll Medical Corporation.

Defibrillation

Key facts

- Defibrillation is the treatment of choice for VF and pulseless VT.
- *Biphasic defibrillators* cycle current direction every 10ms: the same amount of current (roughly 12Amps and 1500V) is delivered but with less energy (200J compared to 360J in older monophasic models) reducing the risk of burns and myocardial damage.

External defibrillation for VF and pulseless VT

Do not delay defibrillation for maneuvers such as intubation, massage, or administration of drugs.

- Expose chest.
- Place gel pads on chest in position shown in Fig. 5.18a: the aim is to direct as much of the current as possible through the heart.
- Switch defibrillator on (📖 Fig. 5.18c, p225) and turn dial on to appropriate power setting (200J for external defibrillation).
- Press charge button.
- If you are using hand-held paddles instead of adhesive external defibrillator pads, place them firmly on gel electrodes and hold.
- Perform a visual sweep to check that no one is in contact with the patient at the same time as saying clearly, 'Charging. Stand clear.'
- Press the red/orange shock button on the paddles.
- If the shock has been delivered successfully the patient's muscles will contract violently: personnel in contact with the patient may experience an electric shock.
- Check the rhythm: if VF charge again and repeat the sequence.
- If the rhythm changes to one compatible with an output check the pulse before proceeding further.

Internal defibrillation

- ***Familiarize yourself early on with how to connect the internal paddles to the defibrillator console: it's different in most models, ICU nurses may not be familiar with how to do this, and an emergency resternotomy is not the time to be trying to figure it out.***
- In some models one of the external paddles disconnects at the handle, and this is the part that the internal paddle lead plugs into, in other models the internal paddles plug directly into the console.
- Switch defibrillator on (📖 Fig. 5.18c, p225) and turn dial on to appropriate power setting (30–50J for internal defibrillation) before you scrub.
- Hand out lead for internal paddles and connect them to defibrillator.
- Nurse will need to press charge button.
- Place concave surface of paddles squarely and firmly over LV and RV taking real care not to dislodge coronary anastomoses (placing the anterior paddle into the left pleura reduces the risk of damaging the LIMA to LAD) or pacing wires, or lacerate the myocardium (the paddles are relatively sharp edged) and press button on paddle handles to deliver shock (Fig. 5.19).

Common pitfalls

- *Failure to deliver a shock*: check that the defibrillator is switched on and adequately charged. Check that the correct power setting has been selected. Check that the SYNC button is *off* if you are trying to defibrillate VF. Change the machine and paddles.
- *Failure to defibrillate*: exclude causes of intractable VF, e.g., coronary ischemia, hypoxia. In internal defibrillation decompress the heart using massage (with the flows down if you're on cardiopulmonary bypass). If using external paddles in OR, inflate both lungs to ensure energy reaches the heart.

Internal cardiac massage and defibrillation

If the chest is open internal defibrillator paddles can be used. Shocks are given starting at 30J to a maximum of 50J. Remove the paddles with caution as rough contact with the myocardium may be enough to cause the heart to revert into VF. Internal cardiac massage should not be performed by inexperienced hands because of the risk of AV dehiscence, RV or LV rupture, or dislodging bypass grafts. The safest method is, standing on the patient's right, to carefully place your right hand under the ventricles, your left hand over the ventricles and compress carefully with the flat of your hands, not fingertips (Fig. 5.19c). One-handed massage is less effective and more likely to rupture the RV. If the patient is in asystole or complete heart block epicardial pacing wires can easily be placed, so that pacing can commence.

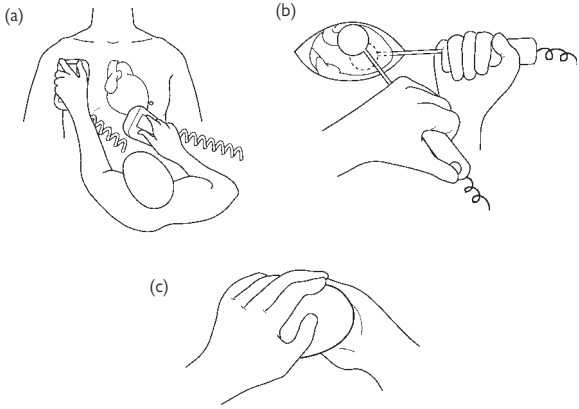


Fig. 5.19 (a) Correct positioning for defibrillation. (b) Correct positioning for internal defibrillation. (c) Correct positioning for internal cardiac massage.

Pacing overview

Key facts

- For maximum cardiac output the optimum heart rate in the immediate postoperative period is 90bpm, and the optimum rhythm is sinus.
- But arrhythmias and block are common, for the reasons described on 📖 pp214–215, and pacing is often required to optimize heart rate and rhythm.

Basics

Depending on the procedure and the patient's rhythm the surgeon may place temporary wires on the epicardium, prior to closing the chest. One or more atrial wires may be attached to the right atrial appendage, or the oversewn atrial cannulation sites. One or more ventricular wires may be attached to the epciardium of the right and sometimes left ventricle. Both sets of wires are brought out to the skin by passing them directly lateral to the xiphisternum, and normally sutured in place.

In most units the atrial wires are brought out to the right of the xiphisternum. The ventricular wires are brought out to the left of the xiphisternum. Be familiar with the convention at your unit.

Some surgeons prefer not to place any pacing wires, in order to avoid bleeding complications (📖 p259), if there is no evidence of bradycardia or heart block immediately post bypass. If the patient is bradycardic, with no evidence of abnormal AV conduction, some surgeons place atrial pacing wires, others prefer to rely on isoprenaline and atropine. Patients who have undergone valve surgery should have ventricular pacing wires inserted routinely because of the risk of developing complete heart block in the postoperative period.

Making a 'skin indifferent'

You need two pacing connections in order to pace a chamber: one indifferent and one active. If the surgeon has only placed one unipolar pacing wire (or the bipolar wire is not working) this will become the active. You need to create an indifferent. In an emergency place a green needle through the skin and attach to the pacing box. Otherwise insert an epicardial wire so that the bare wire is in contact with the skin, break off the straight needle, and attach the blunt end to the pacing box.

Pacing modes

It is possible to pace the atria, the ventricles, or both. Pacing can be *mandatory* or *asynchronous* (unaffected by any intrinsic rhythm), or demand. Demand pacing means that intrinsic action potentials are sensed and at a preset level are allowed to inhibit pacing potentials, so that the patient's own rhythm comes through. Sequential pacing means that both atria and ventricles are paced, to simulate the synchronized AV contraction of sinus rhythm. A list of modes is given as follows.

Temporary pacing modes (Table 5.3)

I Chamber paced.
II Chamber sensed.
III Response to sensing.
A Atrium.
V Ventricle.
D Dual chamber (both atrium and ventricle).
D Dual effect (inhibits and triggers pacing).
I Inhibits pacing.
O No effect on pacing.

Table 5.3 Pacing modes

I	II	III	Description	Indication
A	O	O	Asynchronous atrial pacing	Sinus bradycardia
A	A	I	Demand atrial pacing	Sinus bradycardia
V	O	O	Asynchronous ventricular pacing	No ventricular rhythm, cautery
V	V	I	Demand ventricular pacing	Heart block
D	O	O	Asynchronous dual pacing	No ventricular rhythm, cautery
D	V	I	AV sequential, ventricular demand pacing	Suitable for most bradyarrhythmias.
D	D	D	AV sequential, dual chamber demand	

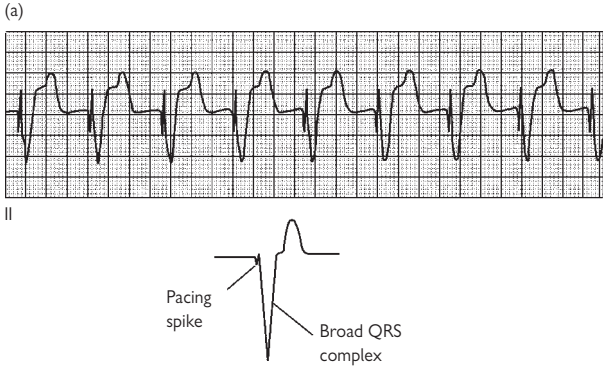


Fig. 5.20 Rhythm strip showing ventricular pacing.

Pacing

Atrial pacing

- *AOO*: asynchronous atrial pacing—useful if using electrocautery.
- *AAI* : Demand atrial pacing.

For effective atrial pacing normal AV conduction is required. It is ineffective during AF and atrial flutter. Pacing amplitude is 3–20mA. Indications include:

- Sinus bradycardia.
- Junctional bradycardia.
- Suppression of atrial and ventricular ectopics, if pacing is set at a slightly faster rate.
- Supraventricular and junctional tachycardia, and atrial flutter can be treated by overdrive pacing.

Ventricular pacing

- *VOO*: asynchronous ventricular pacing. Not often used because of the risk of VF if pacing spikes coincide with the T wave of intrinsic rhythm. Useful during surgery if patient has no underlying rhythm when electrocautery interferes with pacing.
- *VVI*: demand ventricular pacing.

For effective ventricular pacing set the pacing amplitude to 3–20mA. Ventricular pacing is much less effective than atrial and dual chamber pacing because of loss of synchronized AV contraction. Indications:

- Slow ventricular response rate to AF and atrial flutter.
- Any bradycardia with no atrial wires or failure of atrial pacing.
- Overdrive pacing of supra-ventricular tachycardia below.

Dual chamber pacing

- *DOO*: asynchronous dual chamber pacing. Caution as for VOO.
- *DVI*: dual chamber pacing, atrial asynchronous and ventricular demand.
- *DDD*: dual chamber pacing, dual chamber demand.

Effective dual chamber pacing requires atrial and ventricular pacing wires, but does not require normal AV conduction. Dual chamber pacing is more effective than ventricular pacing as synchronized AV contraction is maintained. This is particularly important in patients with hypertrophic ventricles where atrial contraction contributes up to 30% of cardiac output. It is less effective than atrial pacing at maintaining cardiac output because in ventricular pacing ventricular contraction starts from an abnormal focus and is less effective. Both pacing amplitudes should be set to 3–20mA. PR interval is set at 150ms: lengthening it can improve cardiac output by increasing ventricular filling time. Indications include:

- Complete heart block
- First- and second-degree heart block if the ventricular rate is slow.

Overdrive pacing

- This is occasionally used to treat supraventricular tachycardias. Select the maximum pacing output (20mA) and atrially pace at a rate 20bpm faster than the atrial rate (up to 600bpm!). When capture is achieved abruptly turn off the pacemaker: sinus rhythm may result.

Problems with pacing

Failure to pace—no pacing spikes: the pacing output may be too low: turn it up. The pacing wires may not be in contact with myocardium, the wires may be short-circuiting, the leads or contacts may be faulty, the box battery may be low. *Check all connections yourself.* Swap the pacing wires between active and indifferent port. Change the pacing leads, change the box. Create a skin indifferent (📖 p228) and try it with each of the epicardial wires in the active contact in turn. The EKG gain may be insufficient to show spikes: select a different lead or increase the gain. Use *transvenous pacing wires* to pace the pacing-dependent patient: in an emergency use *external pacing pads* (Fig. 5.18).

Failure to pace—pacing spikes present: pacing wires may be in contact with an area of high threshold, or out of contact altogether with the myocardium. Increase the output to maximum, swap the pacing wires or create a skin indifferent as described previously. Try an alternative mode of pacing.

Increasing threshold: in completely pacing-dependent patients it is important to check the pacing threshold daily. This is done by gradually decreasing the output until there is no longer pacing capture: the minimum output before loss of capture is the pacing threshold. Hematoma, edema, and scarring result in increased threshold. In pacing-dependent patients consider transvenous wires if threshold >10mA.

Oversensing: if the pacemaker senses T waves, or AF, demand pacing will be inhibited. Try increasing the sensing threshold or change modes.

Competition with patient's own rhythm: pacing can result in a dramatic decrease in cardiac output if it interferes with the patient's intrinsic rhythm. Use demand pacing, or try turning off the pacemaker altogether.

Pacing diaphragm: proximity of phrenic nerve causes diaphragm contraction: reduce output or use V pacing (A wires closer to phrenic nerve).

Removal of pacing wires

When the patient is no longer pacing dependent, usually by day 3 postoperatively, pacing wires are removed. It's usually safest to do this during daytime operating hours in case of bleeding complications. In anticoagulated patients the INR should be <2.5. Heparin should be discontinued until the APTT is <50. Patient should be recumbent. The wires should be untied if knotted in place, and any suture securing them removed. Apply firm pressure and pull the wires out smoothly. If much resistance is encountered pull wire taut and cut flush with the skin. Occasionally removal of pacing wires causes bleeding. Acute tamponade should be suspected in any patient who presents with hypotension following pacing wire removal (📖 p259). Some units leave one small mediastinal drain in until wires removed to avoid this problem.

Atrial electrogram

If the atrial pacing wires are attached to the two arm leads of a standard EKG (the leg leads remaining in the conventional position) a bipolar trace of atrial electrical activity (AEG) will be produced in lead I. This helps in the diagnosis of atrial arrhythmias.

Renal management

Key facts

- Cardiopulmonary bypass results in major changes in fluid distribution and renal function, moderated by pharmacological agents employed primarily for hemodynamic support (📖 p136).
- Most patients arrive in ICU 3–7kg above their preoperative weight, in a state of *total body sodium and water overload*, because of the fluid requirements driven by the changes described below.
- In a minority of patients correct fluid management is the difference between a satisfactory postoperative course and one complicated by multisystem failure. Basic fluid management is described on 📖 p238. The management of oliguria is outlined on 📖 p238. Diuretics are discussed on 📖 pp208–209. Renal pathophysiology and management is described in more depth here.

Renal effects of cardiopulmonary bypass

The period of low flow, non-pulsatile, hemodilute, hypothermic perfusion associated with bypass and the endocrine and CNS responses to major surgery have a complex series of effects on renal physiology.

- Activation of the renin–angiotensin system and aldosterone production results in sodium and water retention, and potassium excretion.
- ADH production stimulated by hypovolemia, nausea, and vomiting also causes vasoconstriction and water retention by increasing reabsorption of water in the distal collecting tubules.
- The stress response to surgery also results in elevated plasma cortisol, which contributes to sodium retention and potassium excretion.
- Complement, bradykinin, and kallikrein increase capillary permeability promoting shift of fluid into the interstitium from the vascular space.
- This is augmented by hemodilution as a result of blood loss, volume replacement given by the anesthesiologist to compensate for vasodilation at induction, and the 2L crystalloid pump prime. The reduced plasma oncotic pressure results in more movement of fluid into the comparatively high osmotic pressure interstitial space.
- Renal cortical blood flow is decreased by hypothermic vasoconstriction, but increased by hemodilution. During rewarming vasodilation of all capillary beds, not just renal, results in even greater fluid shift into the interstitium.

Etiology of renal failure

Renal dysfunction

- Creatinine >1.2mg/dL (126μmol/L), slightly lower in females.
- BUN >21mg/dL (urea >7.0 mmol/L).
- Creatinine clearance: normal is 90–130mL/min.

- *Preoperative risk factors:* age >75 years, creatinine >12.0 mg/dL, LV dysfunction, hypertension, diabetes, peripheral vascular disease, hypoperfusion as a result of diuretic therapy and vasodilators, sepsis, congestive cardiac failure, intrinsic renal damage caused by NSAIDs, contrast, aminoglycosides, diuretics, endocarditis, obstructive uropathy.

- *Intraoperative risk factors*: combined CABG and valve surgery, long bypass time, prolonged hypoperfusion, circulatory arrest.
- *Postoperative risk factors*: *pre-renal*—hypoperfusion. Many perfusionists routinely add 20g mannitol to the pump prime to protect the distal tubule and promote a diuresis. Vasoconstriction states; *renal*—sepsis, medications, hypoxia; *post-renal*—obstructive uropathy/foley catheter.

Preventing renal failure

There are a number of preoperative, intraoperative, and postoperative measures designed to reduce the risk of renal dysfunction.

- Preoperatively, ensure adequate hydration (+ acetylcysteine in patients with renal dysfunction) before undergoing procedures involving contrast for which the patient may be fasted for long periods of time
- Identify and discontinue where possible nephrotoxic medications, particularly NSAIDs and ACE inhibitors.
- Would the patient benefit from a period on the ICU preoperatively for optimizing with inotropic support and invasive monitoring? Consider stenting stenotic renal arteries (will need clopidogrel).
- Intraoperatively maintain adequate perfusion pressures (50–70mmHg).
- Postoperatively maintaining satisfactory cardiac output and optimizing intravascular volume is the most important factor in avoiding renal dysfunction (📖 pp186–187).

When should diuretics be given for oliguria?

- When you are reasonably confident that pre-renal causes, e.g., hypovolemia, hypotension, low cardiac output and post-renal causes such as Foley catheter problems or obstruction are not the cause, *and*
- Positive fluid balance will be detrimental to the patient now or in the near future, e.g., patient with borderline lung function, severe LV or RV dysfunction, obvious fluid overload.
- There are some broad rules of thumb:
  - Almost all patients need diuresis once they leave ICU.
  - Most ICU patients will do fine whether you give fluid or diuretic for low urine output, and it isn't worth a protracted debate. *But*:
  - Oliguric patients with good hemodynamic indices on minimal pressors can safely be diuresed early if extubation is a priority.
  - Oliguric patients needing large doses of pressor to maintain MAPs most likely need fluid to avoid renal malperfusion, and careful assessment of the underlying problem (📖 p234).
  - Septic (📖 p284) and vasoplegic (📖 p190) patients have high fluid requirements, and diuresis is rarely indicated in the acute phase.
  - Patients with SAM (📖 p190) or LV hypertrophy should be diuresed very, very cautiously: sometimes a little fluid overload is safer.

- Optimize preload by giving *colloid* to increase it aiming for an RAP of 12–16, a PAWP of 10–14, or an LAP of 12–16., remembering that in several conditions required filling pressures may be over 20mmHg (📖 p143).

The oliguric patient

Key facts

- Glomerular filtration rate and hence urine output is a sensitive measure of cardiac function. Most patients arrive in ICU in a state of *total body sodium and water overload*, which becomes more pronounced with the additional volume given as the patient warms and vasodilates.
- Patients with normal preoperative renal function should maintain a diuresis of at least *1mL/kg/h*: renal dysfunction is uncommon in these patients in the absence of other risk factors (📖 p233).
- If called to treat oliguria: always start by asking yourself what the cause is—once you have ruled out important causes of oliguria, then you can start to think about whether you need to treat it.

Management of oliguria

Check that the Foley catheter is not the problem

- The urine catheter may be obstructed, bypassing, or malpositioned. Is the bed wet? Flush with 60mL saline—can you draw this amount back without difficulty? If not, or if the urine is bypassing the catheter, or if the bladder is palpable, change the catheter.

Optimize cardiac function

- Patients who were markedly hypertensive preoperatively may require high mean arterial pressures to maintain a satisfactory urine output.
- Start by assessing whether cardiac output is sufficient: if the patient is warm, well-perfused, with palpable distal pulses, good hemodynamic indices (CI >2.5L/min/m^2, SVO_2 >75%, MAPs >70mmHg, HR 60–80bpm, CVP 4–16mmHg) on minimal pressors then:
  - They are unlikely to be hypovolemic, or in incipient ATN (in the absence of other risk factors), so don't give fluid.
  - If the patient is not fluid overloaded, and will tolerate positive balance, don't give a diuretic, just watch for a couple of hours.
  - If positive fluid balance will be detrimental to the patient now or in the near future, e.g., borderline lung function, severe LV or RV dysfunction, obvious fluid overload, give 20–40mg furosemide iv.
- If cardiac output is not sufficient treat this first (📖 p184), before giving diuretics, which can be harmful in low output or hypotension.
  - *Remember that RV failure (📖 p191), tamponade (📖 p144), bleeding (📖 p144), myocardial ischemia (📖 p185), mesenteric ischemia (📖 p275), ATN (📖 p232), and sepsis (📖 p239) may all present with oliguria.*
  - *Optimize preload* by giving careful fluid challenges to achieve filling pressures of 14–16mmHg (📖 p186).
  - *Optimize contractility* with inotropes: dobutamine (250mg in 50mL 5% dextrose at 3–6mcg/kg/min) is a good first-line choice: the diuresis normally seen within a few hours (📖 p188).
  - *Treat any residual hypotension* with low-dose norepinephrine) 0.03 micrograms/kg/min) if patient likely requires higher MAPs.
  - *Treat acidosis, arrhythmias, hypoxia, anemia.*

- *If you do not see a response within an hour or two there is potentially a serious problem: get senior help whatever time it is.*

Loop diuretics

- Remember diuresis will exacerbate most low cardiac output states (📖 p184).
- If the patient is adequately filled and MAPs are satisfactory *give a loop diuretic*: 20mg of furosemide iv, and if there is no response give a further 40mg furosemide iv.
- A furosemide infusion (250mg furosemide in 50mL 5% dextrose at 1–10 mg/h) can be used instead of repeat bolus doses of furosemide.
- Loop diuretics do not prevent or directly cause acute tubular necrosis but they do have a beneficial effect in converting oliguric to polyuric renal failure (📖 p208), helping to prevent or relieve pulmonary edema.
- If the urine produced in response to diuretic challenges is concentrated, the patient is probably inadequately filled: give fluid bolus.

Osmotic diuretics

- Mannitol is an *osmotic diuretic* used during bypass: some centers use 15–25g in the pump prime. It may be indicated in the fluid-overloaded patient who has a depleted intravascular space as its oncotic properties mean that fluid is drawn back into the intravascular space. If urine output does not improve quickly there is a risk of causing pulmonary edema.

Treat the important problems associated with oliguria

The management of these complications is described on 📖 p236:

- Pulmonary and cerebral edema.
- Fluid overload.
- Hyperkalemia.
- Acidosis.
- Drug toxicity.

Further assessment and management

- If these measures fail to produce a satisfactory response a more accurate assessment of cardiac indices and underlying problems is required urgently. ***In the first 48h after surgery a patient should never have several hours of very low urine output without senior assessment.***
- Most patients have *pulmonary artery catheters:* consider placing one if not already present to better evaluate left and right ventricular function, and optimize fluid status.
- A TEE will identify left and right ventricular dysfunction and tamponade.
- A CXR will show pulmonary edema, or hemothorax.
- *Hemodialysis* is indicated in the oliguric patient to avoid pulmonary edema indicated by deteriorating blood gases despite increasing respiratory support, hyperkalemia, and acidosis: it is not indicated purely for rising serum creatinine and urea in the first instance (📖 p236).

Management of renal failure

The management of oliguria is described on 📖 p234, and the pathophysiology of renal failure is described on 📖 p232. This section deals with the management of established oliguric renal failure.

Indications for renal replacement therapy

- Hyperkalemia.
- Acidosis.
- Pulmonary edema.
- Drug toxicity.
- Progressive uremia or uremia associated with pericarditis, encephalopathy, seizures, coagulopathy.

Key facts

The aim in established renal failure is firstly to avoid the potentially lethal complications of renal failure (hyperkalemia, acidosis, pulmonary and cerebral edema, severe uremia and drug toxicity), and secondly to avoid exacerbating the renal insult. Investigation of the underlying causes of renal failure is also important.

- Optimize hemodynamics to obtain higher MAPs (80–100mmHg), except in established anuria, where this is unlikely to improve function.
- Treat hypoxia aggressively.
- Aim for daily fluid balance of even to negative 500mL to avoid pulmonary edema in anuric patients.
- Remove the Foley catheter to avoid infection if the patient is anuric.
- Monitor electrolytes daily, and potassium and acid–base every few hours, avoid potassium supplements and medication that increases potassium levels (ACE inhibitors): treatment of hyperkalemia is described on 📖 p271, and acidosis on 📖 p170.
- Avoid nephrotoxic drugs (aminoglycosides, NSAIDs, ACE inhibitors) and monitor serum levels of drugs dependent on renal excretion (digoxin, antibiotics such as vancomycin and gentamycin).
- Essential amino acid diets are recommended for patients who are able to eat: patients on dialysis require high protein content (1.5g/kg/day) as dialysis results in negative nitrogen balance.
- Enteral and parenteral feeds can be similarly adjusted.
- Renal US, renal angiography may be indicated.

Hemodialysis and hemofiltration

There are two main techniques of renal replacement therapy used on cardiac ICUs: hemodialysis and hemofiltration. In both techniques access to the circulation is required and blood passes through an extracorporeal circuit that includes either a dialyser or a hemofilter.

- In hemodialysis blood flows along one side of a ***semipermeable*** membrane as a solution of ***crystalloids is pumped*** against the other side of the membrane, against the direction of blood flow.
- In hemofiltration ***blood under pressure*** passes down one side of a ***highly permeable*** membrane on the other side of which is a static crystalloid solution.

- In hemodialysis removal of solutes depends on diffusion: molecules move from high to low concentration, and smaller molecules move faster, so the amount of solute removed depends on its concentration in the dialysis fluid and on the size of the molecule.
- In hemofiltration removal of solutes with a molecular weight up to 20 000 depends on convective flow: all molecules are removed at a similar rate and virtually all ions are removed.
- In hemodialysis large molecules are not efficiently removed.
- In hemofiltration large molecules are so effectively removed that drugs such as heparin, insulin, and vancomycin may need to be replaced, and reduction in circulating inflammation mediators and pyrogens leads to a reduction in pyrexia and systemic inflammation.
- In hemodialysis controlled amounts of sodium and water are removed by creation of a transmembrane pressure gradient.
- In hemofiltration large amounts of salt and water are removed and must be replaced by an infusion of an appropriate amount of physiologic crystalloid into the distal port of the hemofilter.
- Hemofiltration has become an increasingly popular form of renalreplacement therapy for acute renal failure postoperatively, even though it is several times more expensive than dialysis. This is because the continuous nature of the process suits patients with multi-organ dysfunction. There are fewer of the dramatic swings in fluid balance and electrolytes that characterize dialysis. There are several variants.

Continuous arteriovenous hemofiltration (CAVH): this is the original and simplest form of filtration. The femoral artery and vein are cannulated, and blood passes though the hemofilter under arterial pressure alone. It is therefore less appropriate for patients with low cardiac output states. Prolonged arterial cannulation carries complications (📖 p173).

Continuous arteriovenous hemodialysis with filtration: because clearance rates are as low as 10ml/h with CAVH a dialysis circuit is added to the equipment, improving clearance rates at considerable cost.

Continuous venovenous hemofiltration (CVVH): an occlusive pump is incorporated into the circuit to drive blood so that venous cannulation is all that is required. This also allows control of blood flow and filtration rate. Clearance of up to 100mL/min can be achieved. This is the most commonly used system in intensive care units.

Continuous venovenous hemodialysis with filtration: some systems add hemodialysis to the CVVH circuit. Although there is a theoretical benefit it is rare that metabolic stability cannot be achieved with CVVH.

Management of the hemofiltered patient

- A double lumen catheter (Vascath or Mahurkar) is inserted into a central vein. The blood pump is set at 125mL/min and the hemofiltration rate to 25mL/min. The replacement fluid pump is programmed to balance the inflow and outflow of fluid to achieve a preset rate of fluid loss. The circuit is heparinized with an infusion of 200–1600 units per hour. Lactate is the commonest replacement anion, but in lactic acidosis a lactate-free replacement solution must be employed. Replacement of magnesium, calcium, phosphate, and bicarbonate is necessary. Profound hypotension is not uncommon when commencing hemofiltration.

Fluid management

Key facts

- Most patients arrive in ICU 3–7kg above their preoperative weight, in a state of *total body sodium and water overload*. The pathophysiology underlying this is discussed on 🕮 pp106–107.
- Fluid management is directed at optimizing preload (🕮 p186) without exacerbating interstitial edema, which tends to occur because of a generalized capillary leak syndrome (🕮 p137).
- Success can make the difference between a short, uncomplicated postoperative course and the patient remaining on the ICU for weeks.

Maintenance crystalloid

Regimes vary widely between units. The principle is minimizing the additional sodium and water load, while maintaining intravascular volume. Standard protocols include:

- Normal saline or 5% dextrose at a rate equal to the previous hour's urine output.
- Dextrose saline at a rate of 50mL/h.

Colloid challenges

Colloids expand the intravascular space more effectively than isotonic crystalloid. To optimize preload by maintaining preset filling pressures colloid boluses in the form of either blood, blood products, or synthetic colloids should be given. The choice depends on the patient's hematocrit and whether bleeding required correction of coagulopathy with clotting products.

Fluid replacement options

Colloids, and blood most successfully, produce a more lasting expansion of intravascular volume than crystalloid, which rapidly enters the interstium.

Myocardial perfusion, particularly of the microvasculature, is improved by maintaining lower than normal hemoglobin concentration, which also reduces transfusion minimizing exposure to the associated complications. Transfusion has associated risks. But profound anemia reduces oxygen delivery, which can only be compensated for by increased cardiac output to a certain extent (Box 5.7), so patients are generally transfused to maintain Hb>8.0g/dL if indicated by symptoms, low SVO2, hypoxia or other issues.

Sepsis, SIRS and MOD

Key facts

- Sepsis is a systemic response to infection usually characterized by fever >38°C, tachycardia, and/or WCC>12x10^9/L or <4x10^9/L.
- Systemic inflammatory response syndrome (SIRS) is a pro-inflammatory state that does not include a documented source of infection.
- Sepsis and SIRS may lead to multiple organ dysfunction (MOD), and are associated with significantly increased mortality.
- The priorities of management are volume resuscitation, maintaining oxygen deliver, and end-organ support.

Post-operative sepsis

Patients with active endocarditis may present with clinical sepsis in the early post-operative period, but more commonly sepsis does not manifest until post-operative day 3. Signs include fever, rigors, leukocytosis or neutropenia; and may be less specific e.g. vasoplegia, deteriorating RV or LV function, poor oxygenation, renal failure or confusion. Signs may be subtle in elderly patients, immunocompromised or suppressed, and those on steroids. Examine the whole patient, and culture sputum, urine, blood, Common sources include:

- Pneumonia (📖 p268).
- Wounds (📖 pp284–285).
- Indwelling lines (culture, remove and replace) (📖 p175). Groin lines highest risk.
- Urinary tract infection (including after Foley removal).
- Endocarditis.

Empiric antibiotic therapy should be started with advice from infectious disease team as soon as patient has been cultured, and switched as soon as sensitivities are available; and supportive measures as for SIRS.

Systemic inflammatory response syndrome (SIRS)

- SIRS is defined as 2 or more of tachycardia >90bpm; tachypnea >20rpm; pyrexia >38°C; WCC>12x10^9/L or <4x10^9/L; without identifiable bacteremia and in the setting of a known cause of endothelial inflammation e.g. pancreatitis, ischemia, multiple trauma, hemorrhagic shock, immune mediated organ injury, or CPB.
- The pathophysiology is mediated by activation of complement, cytokine and clotting cascades, cell and humoral mediated immunity; increased intestinal permeability and translocation of bacteria and endotoxins may be an underlying cause.
- Metabolic acidosis, low output state and MOD are common features.
- Management includes:
  - Aggressive volume resuscitation: aim is to achieve normovolemia, increase cardiac output and O2 delivery.
  - Maintenance of O2 delivery by airway protection and ventilatory support, inotropic support of cardiac function, and avoidance of pressors to reduce splanchnic malperfusion.
  - End-organ support may include enteral, renal and hepatic support.

Gastrointestinal management

GI complications including upper and lower GI bleeding, ischemia, pancreatitis cholecystitis, and paralytic ileus are described in 🕮 Chapter 8. This section deals with routine management of the GI system on ICU.

Examples of enteral feeds

- *Standard feeds*: Osmolite, Nutrison standard.
- *Fiber enriched*: Jevity, Nutrison Multi Fibre.
- *High energy*: Nutrison Energy, Ensure Plus.

Pathophysiology

Ischemia

Splanchnic hypoperfusion may be caused by caused by emboli, low perfusion pressures on bypass, and low cardiac output states postoperatively, exacerbated by sympathetic vasoconstriction mediated by the CNS response to surgery as well as exogenous catecholamine administration in the form of drugs such as norepinephrine. Mucosal ischemia and mesenteric ischemia may result, characterized in its most severe forms by focal necrosis. Salt and water retention caused by activation of the renin–angiotensin system, capillary hyperpermeability and fluid loading results in mucosal edema. Morphine reduces gastric motility and positive pressure ventilation contributes to gastric dilatation.

Peptic ulceration

The majority of cardiac surgical patients share several risk factors for peptic ulceration: age, NSAID use, and smoking.

Ileus, nausea and vomiting

Ileus, anorexia, nausea, and vomiting are caused predominantly by opiate analgesia, and lack of mobility, exacerbated by the CNS effects of bypass. They are significant causes of morbidity in the ICU.

Table 5.4 Complications of enteral and parenteral nutrition

Enteral feed complications	Parenteral feed complications
Tube malposition, complications of insertion, blockage, kinking, occlusion, loss into GI tract	Catheter malposition, complications of insertion, blocking, kinking, occlusion, embolus air/thrombus
Aspiration pneumonia, diarrhea, constipation, distension, hyperglycemia	Substrate overload, hyperglycemia, hyperosmolarity, hyperammonemia
Infection around enterostomy	Catheter-related sepsis
Fluid and electrolyte disturbances, deficiency syndromes *rare*	Fluid and electrolyte disturbances, deficiency syndromes
ARDS secondary to aspiration	Abnormal liver function, intestinal atrophy and sepsis, respiratory compromise

Basic management

- NG tube insertion maintains gastric decompression during positive pressure ventilation, removes gastric contents minimizing the risk of aspiration, decreases acidity, and allows administration of oral medication and nutrition later in the ICU.
- Patients on CPAP benefit from prophylactic NG tube insertion, otherwise most units reserve NG tubes for patients requiring enteral nutrition, or evidence of gastric dilatation.
- Patients should be routinely started on motility agents and laxatives (such as senna 1 od and lactulose 10mL bid) once extubated, as these reduce the duration of ileus, improving nausea and constipation.
- All patients should receive omeprazole 20mg po od.
- Analgesia should be downscaled from intravenous and oral opiates to non-opiate analgesia such as acetaminohen 650mg po/pr/iv as soon as possible, to avoid ileus, constipation, anorexia, nausea, and vomiting.
- Avoid NSAID analgesia in elderly patients to avoid peptic ulceration.
- Give regular iv antiemetics to treat persistent nausea and vomiting.

Enteral and parenteral nutrition

- For satisfactory wound healing and convalescence a nutritionally balanced intake of approximately 2000–3000kcal/day is required. If the patient has not been extubated within 48h, or oral calorific intake is inadequate for other reasons, NG feeding should be started. If the GI tract cannot be used, total parenteral nutrition is an alternative. It is associated with more complications than parenteral nutrition (Table 5.4) and so every effort, including nasojejunal tubes and PEG placement, should be made to feed enterally.

Most enteral tube feeds come in sterile, 500mL and 1000mL packs, are nutritionally complete and provide 1kcal/mL. They are based on whole protein. Fiber-enriched feeds also provide 1kcal/mL but are thought to be better for patients who may require long-term feeding. They contain soluble soy polysaccharides, and fructo-oligosaccharides which remain undigested passing to the colon where they are fermented by bacteria into short-chain fatty acids which promote absorption of Na^+ and water, reducing diarrhea. These molecules are also probiotic. High-energy feeds contain 1.5 kcal/mL and are used in patients requiring reduced fluid input.

- Involve the pharmacist/dietician to make patient-specific feeds.
- Renal failure patients require high energy, low volume, and electrolyte feeds: Nepro provides 2kcal/mL and high protein content.
- High-fat-low carbohydrate feeds provide 55% of the energy content as fat: this is thought to reduce CO_2 production during respiration, and is designed for patients with severe respiratory compromise: there is no clear evidence of any benefit from these regimes.
- Drugs can be given by the NG tube being used for enteral feeding. Liquid preparations designed for oral administration should ideally be used as crushed tablets may block the tube: the feed should be stopped and the NG tube flushed with saline before and after administration of drugs.
- The therapeutic effect of warfarin is reduced by the vitamin K content in feeds: it is frequently necessary to increase the dose of warfarin.
- Phenytoin interacts with enteral feeds which should be stopped for 2h before and after phenytoin administration.

Neurological management

Stroke and other neurological sequelae of cardiac surgery are described on 📖 pp278–281. This section outlines routine neurological management on the ICU.

Pathophysiology

A large proportion of patients undergoing cardiac surgery can be shown by sophisticated cognitive testing to have sustained a neurological deficit, but in the majority this is not evident clinically. A minority of patients will suffer clinically obvious CVAs (📖 p278). A number of processes contribute to neurological dysfunction. Cerebral hypoperfusion as a direct result of low perfusion pressures, embolic occlusion of vessels, hypothermia, and hypoxemia is exacerbated by inhibition of cerebral autoregulation by anesthetic drugs. Alpha-stat management during cardio-pulmonary bypass helps counter this (📖 p124). Hyperglycemia is associated with a worse outcome of anoxic brain damage, perhaps due to the generation of oxygen free radicals.

Basic management

- Anesthetic levels are decreased towards the end of the operation, commonly by discontinuing inhaled anesthetic (e.g., halothane, isoflurane, nitrous oxide) and by starting continuous infusion of propofol (0.3–3mg/kg/h), fentanyl 1–10 micrograms/kg/h) or midazolam. This provides firstly adequate anesthesia for transfer and the first few hours on the ITU, and a basis for weaning anesthesia at a controlled rate.
- Sedation is obtained by using these anesthetic drugs at lower infusion rates or dexmedetomidine (Precedex ® at 0.2–0.7 mcg/kg/hr).
- Sedation covers a wide range of conscious states from almost wide awake to unresponsive: the ideal level of sedation is one where the patient is easily rousable from light sleep, able to obey simple commands, without anxiety or agitation.
- Effective sedation minimizes distress, ensures that invasive procedures are tolerated, allows the use of artificial ventilatory patterns, and helps control fits.
- Additional neuromuscular blockade may be required once the patient has reached the ICU: the main aim is to prevent shivering which can increase oxygen consumption by over 500% and exacerbate mediastinal bleeding, and to reduce respiratory compromise in the patient who is 'fighting the ventilator', e.g., vecuronium (0.05mg/kg) iv.
- Muscle relaxants have no impact on analgesia or awareness. It is critical to ensure that the patient has an appropriate level of anesthesia: give fentanyl, propofol, or midazolam.
- As anesthesia is weaned additional analgesia may be required, and a morphine NCA which can be converted to a PCA, and regular paracetamol, are routinely started.
- Remember when bolusing narcotics that even small doses can have a major impact on ventricular function: the scenario of the agitated patient that experiences cardiovascular collapse as a direct consequence of attempts at standard sedation is well recognized.

Assessment of neurological function

The following assessments are made on admission to the ICU and then every 4h for the first 24h, and then every 12h:

- Eye opening (spontaneous, to speech, to pain, not open).
- Movement (obeys commands, purposeful, withdraws to pain, abnormal flexion, abnormal extension, no response).
- Power (all four limbs) 5 normal, 4 against resistance, 3 against gravity, 2 with gravity, 1 muscle twitch only, 0 nil.
- Comprehension and orientation (to 'yes/no' questions).
- Sensation in main dermatomes.
- Assess pupil size and reaction to light.

The ICU patient with a new neurological deficit

- Around 1% of patients will have a postoperative CVA, and a similar number of patients experience seizures (with some overlap). Open heart cases (valve surgery), deep hypothermic circulatory arrest, and severe atherosclerotic aortic disease are the main risk factors, together with a previous history of stroke or seizure.
- The rationale for making an early neurological assessment prior to resedating patients is the narrow window available to intervene on acute intracerebral pathology.
- If a patient develops a new neurological deficit they should undergo urgent CT angiography of the brain to assess both the arterial tree and perfusion, as well as identifying intracerebral hemorrhage:
  - Hemorrhage is relatively uncommon, and should be treated by correcting any coagulopathy. Mass effects may require hemicraniotomy.
  - Embolic problems are more common—intra-arterial thrombolysis or embolectomy is very occasionally possible within a window of a few hours after the ischemic event.
  - Most commonly no acute pathology is seen on the first CT scan, and the management is directed at providing MAPs of 90–100mmHg instead of 65–75mmHg, which may be associated with an immediate improvement in some neurology. This maneuver is not safe until intracranial bleeding is ruled out by CT scan.
- The rare patients who fail to regain consciousness after weaning sedation should undergo EEG: a significant proportion are in undetected status epilepticus and require anticonvulsant therapy (📖 p283).
- The intubated seizing patient may be effectively treated with propofol, but phenytoin is the first-line and mainstay of treatment.
- Management of stroke and seizures after the first postoperative day or so is covered in more detail on 📖 p283.

Placing arterial and venous lines

Arterial cannula

Set-up

- Usually prepacked kit contains: 20-gauge cannula (use a 16-gauge in the femoral artery if later IABP insertion is anticipated), connectors and three-way tap, guidewire, 2mL 1% lidocaine, 5mL syringe, blue needle, 10mL sterile saline, skin prep, gauze swabs, small drape, suture.
- Explain the procedure to the patient if appropriate.
- For radial artery cannula insertion place the forearm on a pillow so that the wrist is dorsiflexed, for femoral artery insertion lie the patient flat (if obese, tape or have someone hold stomach out of the way).
- Prep the skin and infiltrate local anesthetic with blue (22-gauge) needle.

Landmarks

- Radial artery: between tendon of flexor carpi radialis and head of radius.
- Femoral artery: lies midway between the anterior superior iliac spine and the symphisis pubis.

Technique

- Prepare and check equipment. Sterile gloves.
- Palpate pulse between 2 fingers for 2–3cm.
- Pass cannula slowly at 45° into skin watching for arterial flashback.
- Once the cannula is *in situ*, aspirate and flush via the three-way tap.
- Ask nurse to place wrist on backboard to stabilize, and connect to transducer—check the trace is arterial.
- Suture femoral arterial line in place with 3/0 silk.
- *Transfixion technique:* the cannula is passed through both artery walls, the needle completely withdrawn, and the cannula then withdrawn slowly until flashback occurs, at which point it is advanced into the artery.

Partial transfixion technique: the cannula is advanced until flashback stops, and the needle withdrawn while holding the cannula steady, which is then advanced into the artery.

Non-transfixion: the cannula is advanced carefully in 0.5mm increments until flashback is seen, at which point the catheter is carefully slid off the needle in the artery.

Guidewire

A guidewire is useful where it is possible to get flashback, but difficult to advance the catheter up the artery.

Central venous catheter

Set-up

- Usually prepacked kit contains appropriate central venous catheter, enough three-way taps for all individual lumens, 10mL 1% lidocaine, 10 mL syringe, a blue (23-gauge) needle and a green (21-gauge) needle, 20mL saline, 2 or 3/0 silk on a large handheld needle, 11 blade scalpel, skin prep, sterile drape, gauze swabs.
- You also need US probe with sterile bag if this is to be utilized.
- Sterile gloves and gown, and enough drapes to cover whole patient.

- Explain the procedure to the patient if appropriate (sign consent).
- Ask a nurse to set up pressure transducer and connection tubing.
- Patient's EKG and pulse oximetry should be continually monitored.
- Ensure that there is adequate light, a space behind the bed which you can work in, and that it is possible to place the bed in Trendelenburg.

Landmarks

- *Central approach to internal jugular vein:* apex of triangle formed by clavicular and sternal heads of sternocleidomastoid muscle, aiming the needle towards the opposite nipple.
- *Posterior approach to internal jugular vein:* point where line drawn horizontally from the cricoid cartilage to the lateral border of the clavicular head of SCM, aiming needle towards sternal notch.
- *Anterior approach to internal jugular vein:* medial border of sternal head of sternocleidomastoid, aiming needle towards ipsilateral needle.
- *Sublcavian:* advance the needle at 45° to the junction of outer and middle 1/3rd of the clavicle, then direct needle towards sternal groove.

Technique

There are numerous techniques.

- Prep and drape the patient so that all landmarks are exposed.
- Infiltrate local anesthetic around the planned puncture site.
- Spend 2–3min laying out the equipment in the order of use, secure three-way taps to central line and turn to closed position.
- Ask the nurse to place the bed in 10–20° of Trendelenburg.
- Ballot the internal jugular vein, or under US guidance using aseptic technique and a green needle on a 10mL syringe enter the skin at 45°.
- On aspirating venous blood remove the syringe but leave the green needle *in situ* as an exact marker of position, depth, and direction.
- An alternative is to use a 23G catheter over a needle: the needle is withdrawn and connection tubing attached once the catheter is in the vein, so that confirmation that this is venous not arterial can be obtained by direct manometry, prior to vessel dilatation: the guidewire is placed down the catheter once this is done.
- Take the large-bore hollow needle attached to the 10mL syringe and using the green needle as a guide cannulate the internal jugular vein.
- Aspirate 3–5mL of blood when there is flashback and then remove the syringe leaving the wide-bore needle *in situ*.
- Pass the guidewire down the wide-bore needle, watching for ectopics, and keeping hold of it at all times.
- Once an adequate length of wire is in place, remove the needle over the wire, and apply pressure to the vein.
- Make a 3mm nick in the skin over the wire with a scalpel.
- Pass the dilators over the wire into the vein.
- Remove the dilators, apply pressure, and pass the central venous pressure cannula over the wire into the vein up to about 15cm.
- The wire normally protrudes through the brown (proximal) lumen which should therefore be left open.
- Aspirate, flush, and close all lumens, suture the catheter to the skin.
- Check that there is a satisfactory venous trace.
- CXR to identify pneumothorax.

PA catheter, pericardiocentesis

PA catheter

Set-up if there is no PA catheter sheath already in situ:

- As for central line insertion (📖 p244).
- PA catheter sheath.

Set-up if there is a PA catheter sheath in situ:

- PA catheter plastic sleeve, PA catheter, 3 × three-way taps, connectors, 10mL syringe, 20mL saline flush, sterile gloves and gown, sterile drapes, skin prep, 2 or 3/0 silk suture on a large hand-held needle.
- Position patient as for central line insertion.
- Continual EKG and pulse oximetry monitoring is mandatory.
- A manometry line flushed and zeroed.
- Depending on the monitoring system being used, connection leads and box.

Technique

- Using aseptic technique insert PA catheter sheath in the manner described for central venous catheter insertion (📖 p244).
- Connect PA catheter to LED light source as per instructions and check LED functions and the balloon inflates and deflates.
- Flush all lumens with saline.
- Attach and zero the manometry line.
- Place tip of PA catheter through folded plastic sleeve (don't unravel it until the PA catheter is completely through), and into the PA sheath, connect sheath to sleeve and unroll the sleeve over the length of the PA catheter: is it now possible to continue without aseptic technique.
- Pass the PA catheter through the PA catheter sheath, into the SVC, and RA, holding the PA catheter so that it is concave to the patient's left.
- The pressure trace should show pressures for the changing vessels and chambers appropriately for every 10cm or so that the catheter is advanced.
- Once the tricuspid valve is crossed and an RV pressure trace is visible inflate the balloon.
- Pass the PA catheter further until either a wedge trace is achieved (📖 p176), or the length is >60cm.
- If no wedge trace is obtained, deflate the balloon, withdraw the catheter to about 30cm, inflate the balloon, and try again.
- Once a wedge trace is obtained, deflating the balloon should reveal the PA trace. If this is not the case deflate the balloon and withdraw the catheter a few mm and try again.
- Obtain a CXR to confirm satisfactory position.

Pericardiocentesis

Used, usually by cardiologists under fluoroscopic guidance, to relieve acute pericardial tamponade. Most surgeons favor a subxiphoid incision or even a resternotomy in early postoperative period as effusions are rarely simple and amenable to drainage through narrow-gauge catheters.

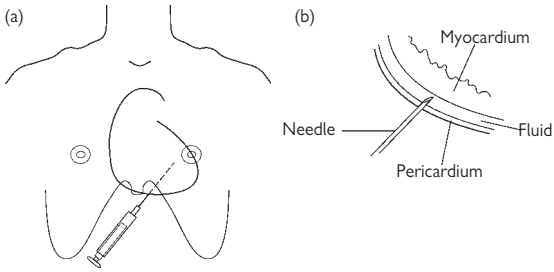


Fig. 5.21 Technique of pericardiocentesis.

Set-up

- Pericardoicentesis needle or catheter, 10mL 1% lidocaine, 10mL syringe, a blue needle and a green needle, 20mL saline, 2 or 3/0 silk on a large handheld needle, 11 blade scalpel, skin prep, sterile gauze.
- Sterile gloves, gown, and drapes.
- Explain the procedure to the patient where appropriate.
- Ensure patient has continual EKG monitoring.
- Fluoroscopy or echo guidance mandatory in elective setting.

Landmarks

1/2cm below and to the left of the xiphoid, aiming at 45° to skin, pointing at left shoulder or nipple.

Technique (Fig. 5.21)

- Prep and drape the skin, infiltrate local anesthetic and make a 3mm nick in the skin 0.5cm immediately to the left of the xiphoid tip.
- Insert the needle applying continuous aspiration aiming at 45° to skin, pointing at left shoulder or nipple, keeping an eye on echo/screening monitor (or EKG if no imaging available) as the needle is slowly advanced: if there are ectopics or changes in the ST segments stop and withdraw the needle a few mm.
- When in contact with the pericardium, 'pop' the needle a few mm into the pericardial space.
- If EKG changes, this indicates contact with the myocardium and the needle should be withdrawn slightly into the pericardial space where no ST segment elevation should be seen.
- When in the pericardial space, aspirate fluid.
- If the tamponade is successfully reduced, RAP should be decreased, cardiac output should increase and pulsus paradoxus should disappear.

Complications

- Cardiac puncture, laceration of a coronary artery, air emboli, arrhythmias, hemothorax, pneumothorax, infection.

Chest tube and cricothyroidotomy

Chest tube insertion

Set-up

- 28F intercostal drain, underwater seal containing water to up to mark, connection tubing, line clamp, Roberts or other instrument for blunt dissection, 20mL 1% lidocaine, 10mL syringe, blue (23G) and a green (21G) needle, 20mL saline, 2 or 3/0 silk on a large handheld needle, 11 blade scalpel, skin prep, sterile drape, gloves, gown, gauze.
- Explain the procedure to the patient if appropriate: recheck side on X-ray and sign consent form.
- Ensure continual monitoring of pulse oximetry.
- Position the patient at 45° with the arm abducted.

Technique

- Typical site for insertion is 5th intercostal space in the midaxillary line.
- It may extend anteriorly to the anterior axillary line.
- Prep and drape the skin. Infiltrate site for tube insertion with local anesthetic ensuring anesthesia at all layers down to and including parietal pleura, and the periosteum of the ribs posterior to the incision.
- A 2cm transverse skin incision is made and the intercostal space is dissected bluntly. Place purse-string suture now.
- The pleura is opened using blunt-tipped instrument introduced over the superior aspect of the ribs, and a finger placed into the pleural space to make sure there are no adhesions.

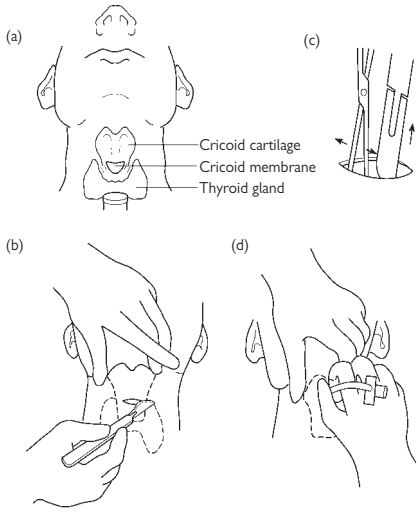


Fig. 5.22 Technique of cricothyroidomy.

- A chest tube is then passed bluntly into the pleural space, being guided superiorly for a pneumothorax and basally for a hemothorax.
- The drain is connected immediately to an underwater seal: ask the patient to cough while you hold the drain and watch for air bubbles in the underwater seal to confirm correct placement. Secure the drain.
- A pigtail can be placed using the same landmarks but employing a Seldinger technique for simple effusions and pneumothoraxes.

Complications

- Misplacement: subcutaneous, intraparenchymal.
- Trauma to other structures: diaphragm, spleen, liver, heart, aorta, lung parenchyma, intercostal arteries (entry sites too low, posterior or trocar used instead of blunt dissection).
- Surgical emphysema, wound infection, empyema, pain.

Cricothyroidotomy

Emergency need for a surgical airway

- Major maxillofacial injury.
- Oral burns.
- Fractured larynx.

Need for tracheal toilet in the extubated patient.

Set-up

- Minitracheostomy, size 6.0 ET tube or 12G cannula in emergencies.
- Artery forceps, 20mL 1% lidocaine, 10mL syringe, blue (23G) and a green (21G) needle, 20mL saline, 2 or 3/0 silk on a large handheld needle, 11 blade scalpel, skin prep, sterile drape, gloves gown, gauze.
- Explain procedure to the patient where appropriate. Sign consent.
- The trauma patient's C-spine should be immobilized.

Landmarks

- The cricoid membrane is a small diamond-shaped membrane palpable just below the prominence of the thyroid cartilage.

Technique (Fig. 5.22)

- Prep and drape.
- If the patient is conscious and maintaining their own airway, infiltrate local anesthetic using aseptic technique.
- Stabilize the thyroid cartilage with the left hand.
- With your right hand make a 2cm transverse incision (smaller for minitracheostomy) through the skin overlying the cricothyroid membrane, and then straight through the cricothyroid membrane.
- Now turn the scalpel blade 90° within the airway so that it acts as a temporary retractor.
- Place an artery forceps through the incision and open it, remove the scalpel and insert a size 6.0 ET tube.
- Suction the tube, secure, and connect to a source of oxygen.
- Some minitracheostomy kits use the Seldinger technique: aspirating air freely is a sign that the needle is in the trachea and that a guidewire can be gently passed down the lumen.

Management of chest drains

Technique of insertion is described on 📖 p248. This section describes the types of chest drainage systems available, and basic protocols for managing chest drains.

Types of drainage system

Underwater seal

Underwater seal drains used to consist of three bottles connected by tubing, the third bottle providing suction control determined by the depth the connection tube penetrated below the water level in the bottle. Now most hospital wards have reliable high-volume, low-pressure wall suction, which means that simple, lightweight, single underwater seal bottles can be used instead of the cumbersome three-bottle systems. The system has to be kept upright.

- Underwater seals are suitable for any condition requiring chest drainage.
- They can be used with or without suction.
- Suction is usually 2–5kPa.

Heimlich valves

The Heimlich valve is a one-way flutter valve within rigid tubing. It can be connected to a standard chest drain. The system allows air and fluid out of the chest cavity, but prevents both from entering. The system has to be open to air, which makes collecting liquid effluent more difficult.

- Heimlich valves are usually considered in patients with a permanent air leak for whom surgery is not appropriate, and for whom the main goal of therapy is discharge to home or palliative care.

Portex bag

- The Portex bag was designed as an ambulatory chest drainage system. It consists of a Heimlich valve within a drainage bag which has a capacity of about 1500mL, and can be emptied intermittently. This drainage system cannot be connected to suction.
- These drains are indicated in patients with chronic pleural collections, in whom surgery is not appropriate.
- As the systems are airtight, an air leak is a contraindication.

Suction

Almost all conditions can be safely managed by an underwater seal system without suction, but suction helps to reinflate the acutely collapsed lung, and improves drainage of fluid. There is a huge range in surgeons' preferences for suction protocols: the following points represent commonly used protocols.

- 1kPa = 7.5mmHg = 10cmH_2O.
- Suction should be high volume, low pressure: approximately 2–3kPa.
- Blocked suction tubing, or a blocked filter at the wall is the equivalent of clamping the drain: have a low threshold for suspecting either.
- Most surgeons put most thoracic drains on suction for 24–48h: the exceptions to this are patients with pneumonectomies who are not placed on suction.

- Ventilated patients cannot generate their own negative intrapleural pressures and therefore all chest drains in these patients, with the exception of post-pneumonectomy drains, should be placed on suction.
- Some surgeons do not routinely place their thoracic patients on suction postoperatively unless an early CXR, requested for all patients, shows that the lung has not fully re-expanded.
- Some surgeons prefer to delay suction by 6h or so in patients who have undergone chemical pleuradesis.
- It is usually safe for a patient with an underwater seal on wall suction to mobilize off-suction for brief periods.
- Discontinue suction in extubated patients after 24–48h when the lung is fully inflated on CXR, and there is no air leak (the drain does not bubble when the patient coughs).
- Suction is unlikely to secure expansion in the lung that has been collapsed chronically: it is most effective in the immediate postoperatve period.

Clamping drains

More patients have died as a result of clamped drains than unclamped drains. The practice of clamping chest drains during transfer is a dangerous one. It reveals a failure to understand how a modern underwater seal drain works, as well as reflecting outmoded practice that dates back to the time of tuberculosis, when drain bottles contained caustic sterilizing fluid that could drain back into the patient if lifted above the level of the chest during transfer. The only indications for clamping a chest drain are:

Post-pneumonectomy

The post-pneumonectomy chest drain is usually clamped for an hour at a time, and unclamped briefly to allow blood to drain. Leaving the drain unclamped risks causing mediastinal shift towards the pneumonectomy side, and cardiovascular compromise. The drain is usually removed on day 1 postoperatively.

Massive hemothorax or effusion

If more than 1500mL of fluid is drained immediately on insertion of a chest tube, and the patient appears hemodynamically compromised as a result of drainage, it is appropriate to clamp the drain for a brief period. In massive hemothorax the effect is to attempt to tamponade the bleed, buying a little time to organize surgical exploration. In a massive effusion this allows time for the lung to expand without re-expansion pulmonary edema, and to reduce mediastinal shift caused by rapid drainage.

Decision-making in long-term drains

Occasionally a surgeon may decide to see if a patient with a chronic effusion or air leak can manage without a drain, by clamping the drain. Tension pneumothorax may result from doing this in a patient with an air leak. Such patients must be observed frequently for any sign of respiratory or hemodynamic compromise, surgical emphysema, or radiological evidence of lung collapse, and if any of these occur, the drain must be unclamped. If the patient tolerates the clamp for 24h it is usually possible to remove the drain.

This page intentionally left blank